Two Answers on Heavy-Ball Dynamics

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Abstract

We determine the worst-case behavior of the heavy-ball method on smooth strongly convex functions, first for Polyak's quadratic-optimal parameters and then for arbitrary constant parameters. For Polyak's tuning, the exact dimension-uniform exponential rate is the largest growth factor of an interpolable mixture of rotating geometric sequences. Four algebraic families determine this rate at every condition number. Two frequencies suffice, and one function in dimension five attains the rate at every horizon up to constant factors. We recover the sharp convergence threshold $9 + 4\sqrt{5}$ of Badithela and Seiler. For general tunings, an exact linear feasibility criterion characterizes dimension-uniform linear convergence through the angle between the characteristic polynomials of the two extreme quadratics. This gives a continuous condition-number threshold and resolves the second heavy-ball riddle of Goujaud, Taylor and Dieuleveut. The regions between consecutive cycle tongues are Farey gaps, each containing convergent and divergent tunings. Regular portions of the threshold surface are analytic plateaus with fixed periodic frequency support. Every rational rotation number in a gap anchors a single-harmonic plateau; mixed-harmonic plateaus and an exact two-lag threshold point also occur. Along tongue edges, explicit multiplier inequalities and second-order expansions of competing Farey fans bound the transition from single-harmonic to mixed-harmonic extremals. The rate bounds and the extremal constructions follow from a common interpolation and separation argument that preserves initialization with zero momentum.

1. Introduction

Polyak's heavy-ball method accelerates strongly convex quadratic minimization by retaining the preceding displacement. On general smooth strongly convex functions, the same tuning can produce periodic or exponentially growing trajectories. We determine its exact worst-case rate under Polyak's tuning and characterize convergence for arbitrary constant parameters through spectral interpolation.

Problem setting Let $0 < \mu < L$, $\kappa = L/\mu$, and let $\mathcal{F}_{\mu,L}(\mathbb{R}^d)$ denote the class of differentiable μ-strongly convex functions on $\mathbb{R}^d$ with L-Lipschitz gradients. For a step size $\alpha > 0$ and a momentum $\beta \in [0, 1)$, the heavy-ball iteration is

$$x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta(x_k - x_{k-1}), \qquad x_{-1} = x_0. \tag{1}$$

Throughout, the initial momentum is zero. Writing $x^\star$ for the unique minimizer of f, define

$$W_N^{(d)} = \sup_{\substack{f \in \mathcal{F}_{\mu,L}(\mathbb{R}^d) \\ x_0 \neq x^\star}} \frac{\|x_N - x^\star\|}{\|x_0 - x^\star\|}, \qquad W_N = \sup_{d \geq 1} W_N^{(d)}, \qquad \rho(\alpha, \beta) = \limsup_{N \to \infty} W_N^{1/N}. \tag{2}$$

Thus ρ is the dimension-uniform worst-case exponential rate; convergence on the class means $\rho < 1$, with a linear bound uniform in the function and dimension. The quadratic subclass gives a lower bound: its rate $\rho_{\mathrm{q}}(\alpha, \beta)$ is the largest root modulus of $z^2 - (1 + \beta - \alpha\lambda)z + \beta$, $\lambda \in [\mu, L]$. Convergence on all these quadratics is equivalent to

$$0 < \alpha L < 2(1 + \beta). \tag{3}$$

Polyak's parameters [23]

$$\alpha = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \qquad \beta = q^2, \qquad q = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \tag{4}$$

balance the extreme quadratic roots and give $\rho_{\mathrm{q}} = q$. Part I determines the nonlinear rate and its extremals for every $\kappa > 1$, including $\rho \geq 1$; Part II treats all tunings satisfying (3).

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Previous results and the two heavy-ball riddles Ghadimi, Feyzmahdavian and Johansson [7] established global convergence, with linear bounds in the strongly convex case, for suitable constant parameters. Lessard, Recht and Packard [18] introduced integral quadratic constraint (IQC) bounds and exhibited nonconvergence under Polyak's tuning. Badithela and Seiler [2] identified the sharp threshold $9 + 4\sqrt{5}$: the weighted off-by-one bound reaches one and a scalar three-cycle appears. Interpolation also supports automated searches for cyclic counterexamples [8].

Goujaud, Taylor and Dieuleveut [10] proved non-acceleration in dimensions at least two using planar root-of-unity cycles, and posed two further questions [9]. Fotopoulos, Helcig, Deck and Alsbach [6] resolved the first, concerning dimension one, by rotating and projecting these cycles onto a line. They also treated coprime rotations, making j turns in k steps, whose binding interpolation lag $j^{-1} \bmod k$ is the time separation of the corresponding iterates.

The second riddle concerns gaps between a computed Lyapunov convergence region and the integer-cycle regions. Evaluating the cycle condition at fractional periods fills these gaps [9], leaving their actual convergence behavior unresolved. Wang [30], also discussed in [31, Appendix B.1], certified a convergent parameter box outside both tests. We determine the gap geometry and give an exact criterion for its convergent and divergent portions.

Liu et al. [19] quantify local attraction under Hessian regularity and give a gradient-based initialization rule. Hernanz [13] obtains the optimal asymptotic distance factor $1-8/\kappa+o(\kappa^{-1})$ after optimizing constant tunings, for arbitrary initial pairs in dimensions at least two. His extension to starts from rest in dimension three uses our additional-coordinate construction. Pokutta and Stellato [22] determine the exact asymptotic squared-distance factor of Nesterov's method through mixtures of planar spirals and linear optimization over measures.

Main results For Polyak's tuning, Theorems 2.6 and 2.8 characterize the exact rate by interpolable mixtures of rotating geometric sequences and determine it through four algebraic families: a special spiral, a single-frequency family, and two two-frequency families. At most two frequencies suffice. Theorem 2.1 gives matching bounds on W_N up to constants independent of N, attained from below at every horizon by one function in $\mathcal{F}_{\mu,L}(\mathbb{R}^5)$ and one initial point. At the critical condition number a trajectory from rest approaches a three-cycle; above it an explicit trajectory grows exponentially (Theorem 2.2).

For general tunings, let $\Theta(\omega)$ be the angle between the characteristic polynomials of the extreme quadratics on the unit circle. Theorem 3.3 gives convergence if $\Theta(\omega) > \omega/2$ for all $\omega \in (0, \pi]$, and a trajectory approaching a k-periodic orbit if $\Theta(2\pi j/k) \le \pi/k$. The rotation number $\omega/(2\pi)$ measures turns per step; fractional periods correspond to contacts with $\omega/2$. The gaps lie between resonances $1/(K+1)$ and $1/K$. One strict spectral certificate at one exponent bounds the rate (Theorem 3.2). At exponent zero, the resulting exact criterion expresses the divergence set as $\kappa \ge \kappa_\partial(\alpha L, \beta)$, with continuous $\kappa_\partial \in (1, \infty]$ (Theorems 3.7 and 3.12). Every gap contains convergent tunings near its lag-one boundary and divergent tunings meeting each rational tongue; for $K \ge 3$, divergence also reaches an arc of the junction of the bounding tongues (Theorem 3.25 and Proposition I.2).

Regular boundary points lie on analytic plateaus with fixed finite harmonic support, where finitely many lag constraints decide convergence (Theorems 3.16 and G.7). Every Farey fraction j/k in every gap anchors a single-harmonic plateau with $\kappa_\partial = \kappa_{j/k}$ (Theorem 3.19 and Corollary 3.20). Two-lag certificates and competing two-frequency Farey fans give explicit bounds on its edge away from a discrete exceptional set, the latter through a second-order inverse-period expansion (Theorems H.2 and 3.23). Exact examples exhibit convergence in a gap, mixed-harmonic extremals and a two-lag boundary point (Propositions 3.13 and 3.14, Theorems 3.17 and G.15).

Methods and relation to control theory Smooth strongly convex interpolation tests whether prescribed points, gradients and values belong to one function [1, 27]. A frequency describes a rotating geometric sequence; a positive measure combines these sequences. The interpolation inequalities are linear in this measure and indexed by time lags. For periodic trajectories, this harmonic linear program

is established in [10, Theorem 6.13] and rederived in [6, Proposition 5.1], which also gives a mixed cycle beyond the single-harmonic regions. We extend the formulation to arbitrary spectra and growth exponents. Separation gives finite dual certificates whose lag inequalities yield energy bounds for all trajectories; an additional coordinate realizes extremal spectra from rest.

These lag certificates are discrete-time O'Shea–Zames–Falb multipliers [32, 33], with a Toeplitz frequency representation [11] and energy estimates in the IQC framework [18, 20]. Related results include quadratic Lyapunov searches [28], periodic counterexamples [25], harmonic duality bounds [34] and converse multiplier theorems [26]. For slope-restricted feedback systems, Kharitenko and Scherer [16] prove exactness of the multiplier test for input-output stability uniform over the lifted plants $G \otimes I_d$ for all finite d; Kharitenko [15] shows conservatism for a scalar interconnection. Our criterion concerns convergence uniform over all finite dimensions, with zero initial momentum and explicit autonomous realizations.

For Polyak's tuning, the spectral constraints reduce to polynomials in $\cos\omega$. One family has a rational parametrization; another uses rescaled variables at the critical three-cycle and a quintic number field. Polynomial contact systems and rational interval continuation determine the four families; polynomial positivity and analytic lag tails complete the classification [14, 21, 24]. For general tunings, the harmonic minimum principle extends one-sided certificates from circle boundaries to disks or annuli, and angle convexity and Farey relations [12] organize the gaps. Strict complementarity and analytic continuation produce the plateaus; contact curvature and quadratic minimization over lag residues control their edges.

Formal verification. The mathematical results are accompanied by formal proofs in Lean. The package is available at `PolynomialRealizationsProgram/polyak`[3].

Source code. The code for the numerical experiments, figures, exact certificate generation and proof-data archives is available at `PolynomialRealizationsProgram/polyak-code`[4].

2. Polyak's tuning

We first fix Polyak's parameters (4) and determine the worst-case rate as a function of the condition number. A spectral characterization reduces this problem to four algebraic families with at most two frequencies. Their realization from rest gives matching bounds at every horizon and the sharp convergence threshold.

2.1. Exact worst-case rate under Polyak's tuning

For the parameters (4), write $\rho_{\mathrm{HB}}(\kappa) = \rho(\alpha, \beta)$, $\varphi = (1+\sqrt{5})/2$ and $\kappa_c = 9+4\sqrt{5} = \varphi^6$. The worst-case rate is determined by four algebraic families of spectra. There is an optimal spectrum with at most two frequencies; a single-frequency construction gives the sharp threshold for convergence. Let $\rho_{\mathrm{sp}}(\kappa)$ denote the unique positive root of

$$F(\rho, q) = \rho^2(\rho^2+1) - 2q\rho(\rho^2-\rho+1) - q^2(\rho^2+1) = 0. \tag{5}$$

Theorem 2.1 (Exact rate and realization)**.** *For every $\kappa > 1$ the limits*

$$\rho_{\mathrm{HB}}(\kappa) = \lim_{N\to\infty} W_N(\kappa)^{1/N} = \lim_{N\to\infty} W_N^{(5)}(\kappa)^{1/N}$$

exist, and $\rho_{\mathrm{HB}}(\kappa)$ is given by the finite algebraic prescription of Theorem 2.8. There are constants $0 < c_\kappa \le C_\kappa < \infty$ such that

$$c_\kappa\, \rho_{\mathrm{HB}}(\kappa)^N \le W_N^{(5)}(\kappa) \le W_N(\kappa) \le C_\kappa\, \rho_{\mathrm{HB}}(\kappa)^N \quad (N \ge 0). \tag{6}$$

[3] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/polyak`
[4] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/polyak-code`

The lower bound is attained by one function in $\mathcal{F}_{\mu,L}(\mathbb{R}^5)$ and one initial point, independent of N. Moreover, one function in $\mathcal{F}_{\mu,L}(\mathbb{R}^3)$ and one initial point satisfy

$$\|x_N - x^\star\| \geq \frac{\rho_{\mathrm{sp}}(\kappa)^N}{\sqrt{5}} \|x_0 - x^\star\| \quad (N \geq 0). \tag{7}$$

The proof is given in Appendix C.5; the explicit three-dimensional construction is in Appendix C.1.

Theorem 2.2 (Sharp convergence threshold)**.** *The heavy-ball method with Polyak's tuning converges on every $f \in \mathcal{F}_{\mu,L}$ from every initial point if and only if $\kappa < \kappa_c$. More precisely,*

$$\rho_{\mathrm{HB}}(\kappa) < 1 \quad (1 < \kappa < \kappa_c), \quad \rho_{\mathrm{HB}}(\kappa_c) = 1, \quad \rho_{\mathrm{HB}}(\kappa) > 1 \quad (\kappa > \kappa_c),$$

and $\rho_{\mathrm{HB}}(\kappa) \leq \sqrt{\varphi q}$ for $\kappa < \kappa_c$. At κ_c a function in $\mathcal{F}_{\mu,L}(\mathbb{R}^3)$ has a trajectory with $\|x_N - x^\star\| \geq \|x_0 - x^\star\|/\sqrt{5}$ for all N, which converges to a three-cycle; above κ_c the same construction diverges exponentially.

The proof is given in Appendix B.3.

Badithela and Seiler [2] identified κ_c and a three-cycle of a scalar piecewise quadratic function at this condition number. Their explicit solution of the weighted off-by-one linear matrix inequality also gives $\rho \leq 1$ at κ_c. Theorem 2.2 recovers their threshold and combines it with the exact rate and a realization from rest valid at every horizon, including exponential growth above the threshold.

Theorem 2.3 (Regularity and critical behavior)**.** *The function ρ_{HB} is continuous and strictly increasing on $(1, \infty)$ and real analytic outside the five switching points $\kappa_W < \kappa_1 < \kappa_a < \kappa_b < \kappa_2$ of Theorem 2.8. It is analytic at κ_c, and*

$$\rho_{\mathrm{HB}}(\kappa) = 1 + \frac{\kappa - \kappa_c}{4\kappa_c} + O\big((\kappa - \kappa_c)^2\big). \tag{8}$$

As $q \downarrow 0$,

$$\rho_{\mathrm{HB}} = (1 + \sqrt{2})\, q - \Big(2 + \frac{3\sqrt{2}}{2}\Big) q^2 + O(q^3), \quad \rho_{\mathrm{HB}} - \rho_{\mathrm{sp}} = \frac{10 + 7\sqrt{2}}{4}\, q^3 + O(q^4),$$

and as $\kappa \to \infty$, $\rho_{\mathrm{HB}}(\kappa) \to \rho_* = 1.71298\ldots$, the positive root of $\rho^4 - 2\rho^3 + 2\rho^2 - 2\rho - 1$.

Regularity is proved in Appendix B.7, and the asymptotic expansions in Appendix C.6.

Two-frequency spectra attain a strictly larger rate than the spiral on two intervals of condition numbers. The rate equals ρ_{sp} exactly on $[\kappa_W, \kappa_1] \cup \{\kappa_c\} \cup [\kappa_2, \infty)$ and exceeds q everywhere. Figure 1 displays the rate and its regimes.

2.2. Spectral characterization

The reduction applies to every quadratically stable tuning (3). Translate the minimizer to zero and normalize the function by

$$\psi(e) = \frac{2}{L - \mu}\Big(\frac{L + \mu}{4}\|e\|^2 - f(x^\star + e) + f(x^\star)\Big). \tag{9}$$

Then $\psi(0) = 0$, $\nabla\psi(0) = 0$ and $\nabla\psi$ is 1-Lipschitz; these properties define $\mathcal{S}$. For centered functions with $f(0) = 0$ this is a bijection, and with $e_k = x_k - x^\star$ and $\sigma_k = \nabla\psi(e_k)$ the iteration becomes

$$e_{k+1} = Ae_k + B\sigma_k - \beta e_{k-1}, \quad A = 1 + \beta - \frac{\alpha(L + \mu)}{2}, \quad B = \frac{\alpha(L - \mu)}{2}. \tag{10}$$

Identify the plane with $\mathbb{C}$. A wave $e_k = z^k$, $z = e^{t + i\omega}$, rotates by ω and scales by e^t per step; t is its growth exponent. The recurrence requires $\sigma_k = m(t, \omega) z^k$, where

$$m = m(t, \omega) = \frac{z + \beta/z - A}{B}. \tag{11}$$

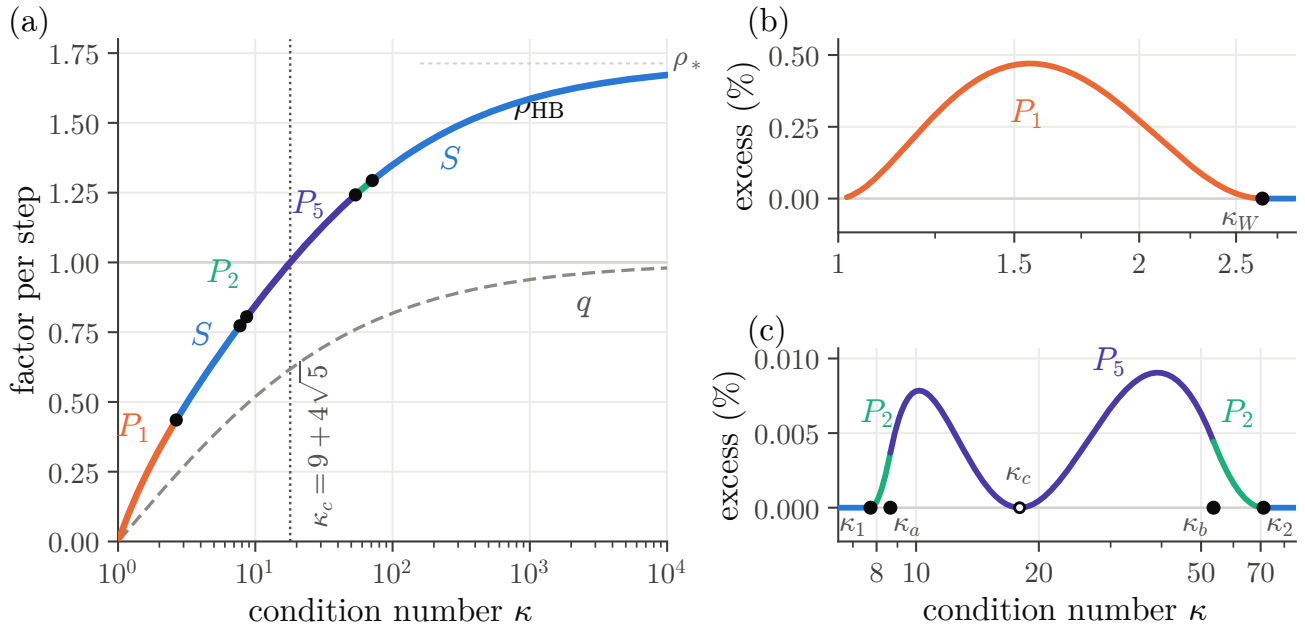


Figure 1: Exact Polyak rate and optimal regimes. (a) ρ_{HB}, the quadratic factor q, the threshold κ_c and the limit ρ_*; dots mark the five switches. (b), (c) $100(\rho_{\mathrm{HB}}/\rho_{\mathrm{sp}} - 1)$ on its two positive windows, each at its own scale. At κ_c the optimal spectrum is the three-cycle.

For an orthogonal sum of waves with common exponent t, a probability measure ν on $[0, \pi]$ records their squared initial amplitudes, normalized to sum to one. Each atom is a mass at one frequency and represents one wave. Write bars for averages with respect to ν and put $g = 1 - |m|^2$, $s = \operatorname{Im} m$ and

$$a_m = g\,(\cosh mt - \cos m\omega), \quad b_m = 2s \sin m\omega \quad (m \geq 1). \tag{12}$$

Here the index $m \geq 1$ denotes a time lag. The interpolation conditions of Taylor, Hendrickx and Glineur [27], extended to arbitrary subsets of a Hilbert space by Azagra and Mudarra [1], give a linear feasibility problem for this spectrum. The scalar C prescribes its function values; the precise interpolation statement is Theorem A.2.

Definition 2.4 (Feasible spectrum)**.** A pair (ν, C) is feasible at the exponent t if

$$\begin{gathered} |C| \leq \bar{g}, \\ \ell_m := \bar{a}_m - \bar{b}_m + C \sinh mt \geq 0, \quad \upsilon_m := \bar{a}_m + \bar{b}_m - C \sinh mt \geq 0 \quad (m \geq 1). \end{gathered} \tag{13}$$

The numbers ℓ_m and υ_m are the lower and upper lag-m slacks.

The first inequality enforces interpolation with the minimizer; a lag constraint is active when its slack is zero. A finite nonnegative combination of these inequalities gives a strict dual certificate of infeasibility when its C coefficient cancels and its remaining function of frequency, or symbol, is uniformly negative:

$$\begin{gathered} \Phi_t(\omega) = \sum_m \{\alpha_m(a_m - b_m) + \beta_m(a_m + b_m)\} + (\gamma_+ + \gamma_-)g \leq -\varepsilon \quad (0 \leq \omega \leq \pi), \\ \sum_m (\alpha_m - \beta_m) \sinh mt + \gamma_+ - \gamma_- = 0. \end{gathered} \tag{14}$$

Here $\alpha_m, \beta_m, \gamma_\pm \geq 0$ have finite support and $\varepsilon > 0$. Such a certificate gives an upper bound valid for every trajectory. Under the hypotheses below, a finitely supported feasible spectrum gives a lower bound realized by one trajectory at every horizon.

Theorem 2.5 (Spectral characterization)**.** *Assume* $0 < \alpha L < 2(1 + \beta)$.

(a) *The conditions* (13) *are infeasible at* t *if and only if a strict certificate at* t *exists.*
(b) *If strict certificates exist at every* $t \geq t_0$*, then* $\rho < e^{t_0}$*.*
(c) *If* (ν, C) *is feasible at* t*,* ν *has* r *atoms, and* $|C| < \bar{g}$ *in case* $t > 0$*, then there are* $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^{2r+1})$ *and* $x_0 \neq x^\star$ *with* $\|x_N - x^\star\| \geq c_0\, e^{Nt}\|x_0 - x^\star\|$ *for all* $N \geq 0$*, where* $c_0 > 0$*. In particular* $\rho \geq e^t$*.*

Proof. For (a), integrating the symbol of a strict certificate against a feasible measure gives $0 \leq \int \Phi_t \, d\nu \leq -\varepsilon$, a contradiction. Conversely, the minimizer inequalities restrict C to $[-1, 1]$, and the probability measures on $[0, \pi]$ form a weakly compact set. Thus infeasibility implies that a finite subsystem, including both minimizer inequalities, $\int \phi_i \, d\nu + \varsigma_i C \geq 0$, $i \in \mathcal{I}$, is infeasible (Lemma A.4). The closed convex set

$$\operatorname{conv}\{(\phi_i(\omega))_i : 0 \leq \omega \leq \pi\} + \mathbb{R}\varsigma - \mathbb{R}_+^{\mathcal{I}}$$

excludes zero; it is closed because it is the sum of a compact set and a polyhedral cone. Strict separation gives nonnegative multipliers λ_i, with $\lambda \cdot \varsigma = 0$ and $\sum_i \lambda_i \phi_i(\omega) \leq -\varepsilon$ uniformly in ω. These are the coefficients of (14).

For (b), let ρ_Γ be the asymptotic growth factor of arbitrary finite trajectory segments, defined in (38). Lemma A.5 gives $\rho \leq \rho_\Gamma$. Lemma A.6 converts a strict certificate into an energy estimate with only endpoint terms; Lemma A.7 consequently excludes a strict certificate at $\log \rho_\Gamma$. If $\rho_\Gamma \geq e^{t_0}$, the hypothesis supplies precisely such a certificate. Hence $\rho \leq \rho_\Gamma < e^{t_0}$.

For (c), Lemma A.3 gives interpolable wave data in $2r$ dimensions. Lemma A.8 appends one decaying quadratic coordinate and changes the first gradient so that $x_{-1} = x_0$. Its hypotheses are exactly feasibility and, when $t > 0$, $|C| < \bar{g}$. The resulting single interpolant satisfies $\|x_N - x^\star\| \geq c_0 e^{Nt}\|x_0 - x^\star\|$ for every N. Taking Nth roots gives $\rho \geq e^t$. □

For Polyak's tuning, $A = 0$, $B = 2q$, $\beta = q^2$ and $m = \cosh(v + i\omega)$, with

$$v = t - \log q, \quad u = \sinh v, \quad c = \cos\omega, \quad G = \cosh t. \tag{15}$$

Consequently $g = 1 - c^2 - u^2$, $s = u \sin\omega$, and

$$a_m(c) = g(c)\big(T_m(G) - T_m(c)\big), \quad b_m(c) = 2u(1 - c^2)U_{m-1}(c), \tag{16}$$

where T_m, U_m are the Chebyshev polynomials. On changing variables from ω to $c = \cos\omega$, we use ν for the resulting measure on $[-1, 1]$. The relative exponent $v = t - \log q$ measures growth against the quadratic rate q. In the auxiliary program v and t vary independently, with $u = \sinh v$; the actual tuning follows $v = t - \log q$. Set

$$V(t) = \max\{v \geq 0 : \ (13) \text{ is feasible at } (v, t)\}, \quad T(q) = \max\{t : \ |t - \log q| \leq V(t)\}. \tag{17}$$

Theorem 2.6 (Spectral rate for Polyak's tuning)**.** *For every* $q \in (0, 1)$*,* $\rho_{\mathrm{HB}}(\kappa) = e^{T(q)}$*. The function* V *is even, bounded by* $\operatorname{arsinh} 1$*, and the set of* v *feasible at a given* t *is* $[-V(t), V(t)]$*. At every optimal point with* $T(q) \neq 0$*, every optimal measure is finitely supported in* $(-1, 1)$ *and satisfies* $|C| < \bar{g}$*, and only finitely many lag constraints are active.*

The proof is given in Appendix A.4.

The feasible region is $|v| \leq V(t)$. The parameters of the method satisfy $v = t - \log q$; the largest feasible intersection gives $T = \log \rho_{\mathrm{HB}}$ (Figure 2). A limiting dual symbol is analytic for $t \neq 0$. Its finitely many interior zeros carry every optimal measure, and strict slack at the minimizer makes every sufficiently large lag inactive. These properties reduce the infinitely many interpolation inequalities to a finite algebraic classification.

2.3. Complete spectral classification

2.3.1. Single-frequency construction and convergence threshold

The three contacts $\ell_1 = \upsilon_2 = \upsilon_3 = 0$ determine an explicit single-frequency spectrum.

Proposition 2.7 (The special spiral)**.** *For every $t \neq 0$, the data*

$$u = 1 - \frac{1}{2G}, \quad c = -\frac{1}{2G}, \quad \nu = \delta_c, \quad C = g \tanh t \tag{18}$$

are feasible at $(\operatorname{arsinh} u, t)$*, and* $\ell_1 = \upsilon_2 = \upsilon_3 = 0$*. On this spectrum*

$$\frac{\ell_m}{g} = \frac{E_{m+1}(t)}{G}, \quad \frac{\upsilon_m}{g} = \frac{E_{m-1}(t)}{G}, \quad E_n(t) = \cosh nt + 2G^2 \cos n\omega, \tag{19}$$

and for every $n \geq 3$, $G^{n-2}E_n(t)$ is a polynomial in $G^2 - 1$ with nonnegative coefficients, positive for $t \neq 0$. The relation $u = \sinh(t - \log q)$ on this curve is (5) *with $\rho = e^t$.*

The proof is given in Appendix B.3.

At $t = 0$ this spectrum is the three-cycle $\omega = 2\pi/3$, $u = 1/2$. Combining the lag-one and minimizer inequalities gives the sharp strip $|v| \leq |t| + \log \varphi$, strictly for $t \neq 0$ (Proposition B.6). Intersecting it with $v = t - \log q$ proves $\rho_{\mathrm{HB}} \leq \sqrt{\varphi q} < 1$ for $q < \varphi^{-1}$; the spiral gives the matching transition to a nondecaying cycle and then exponential divergence.

2.3.2. Algebraic description of the boundary

Each family pairs a feasible measure with a nonpositive dual symbol that vanishes on its support, as required by complementary slackness. The families P_1 and S have one frequency; P_2 and P_5 have two selected frequency points, with zero masses omitted at endpoints. On the open noncritical regimes their active upper lags are respectively $\{3\}$, $\{2, 3\}$, $\{2, 3\}$ and $\{2, 3, 5\}$, together with the lower lag one. The corresponding polynomial contact systems are (48), (49) and (52); all their coefficients and branch selections are specified in Appendix B. For the two-frequency families let $x < y$ be the cosine coordinates of the support points, with masses $1 - p$ and p, and put $d = C \sinh t$.

Theorem 2.8 (Complete spectral classification)**.** *Let $G_W = (1 + \sqrt{3})/2$, let $G_1 \in (1.03329, 1.03330)$ be the root of $32G^3 - 36G^2 + 4G - 1$, and let $G_a \in (1.02358, 1.02359)$ be the root of* (62)*. Then $V(t) = \operatorname{arsinh} u(\cosh t)$, where*

$$u(G) = \begin{cases} u_5(G), & 1 \leq G \leq G_a, \\ u_2(G), & G_a \leq G \leq G_1, \\ 1 - (2G)^{-1}, & G_1 \leq G \leq G_W, \\ u_1(G), & G_W \leq G < \infty, \end{cases} \tag{20}$$

and u_1, u_2, u_5 are the values of u on the families P_1, P_2, P_5. On P_2, $G_1 = G(\zeta_1)$ and $G_a = G(\zeta_a)$, where ζ_1 is the real root of $\zeta^3 - \zeta^2 - 2$ and $\zeta_a \in (1.6989, 1.699)$ is a root of the polynomial N_{12} of degree twelve in Lemma B.11; at $G = 1$ the central family reduces to the three-cycle, $u = \frac{1}{2}$, $x = -\frac{1}{2}$, $p = d = 0$. The branches are selected by the parametrization of P_2 and the unique analytic continuations of P_1 and P_5 from the endpoint data of Appendix B; their values agree at the endpoints. The function $t \mapsto t - V(t)$ is continuous and strictly increasing, so

$$t - \operatorname{arsinh} u(\cosh t) = \log q \tag{21}$$

has exactly one solution $t = T(q)$, and $\rho_{\mathrm{HB}} = e^{T(q)}$. At most two atoms are needed.

Proof. For each $G \geq 1$ we construct a feasible spectrum and a nonpositive dual symbol $\Phi(c) = -(G - c)Q(c)$ whose zeros contain its support. The normalized upper lags and the lower lag one cancel $C \sinh t$; Lemma B.1 shows that increasing $u \geq 0$ decreases the symbol by at least $(G - c)$ times the increment of u^2. Thus such a primal–dual pair determines the maximal feasible u. For $G > 1$ the decrease is uniformly strict. At $G = 1$, adding equal positive minimizer weights makes it strict also at $c = 1$, as shown after Lemma B.1.

The factorization (55) proves this pairing for $G_1 \leq G \leq G_W$. Proposition B.7 gives the unique continued branch for $G \geq G_W$; Theorem B.13 gives the rationally parametrized branch on $[G_a, G_1]$;

Table 1: Optimal regimes for Polyak's tuning. At switching points the adjacent families agree; zero spectral weights are omitted. The critical three-cycle has the additional active constraints described below.

Condition number	Family	Zero raw lag slacks
$(1, \kappa_W)$	P_1	ℓ_1, υ_3
$[\kappa_W, \kappa_1]$	S	$\ell_1, \upsilon_2, \upsilon_3$
(κ_1, κ_a)	P_2	$\ell_1, \upsilon_2, \upsilon_3$
$[\kappa_a, \kappa_b] \setminus \{\kappa_c\}$	P_5	$\ell_1, \upsilon_2, \upsilon_3, \upsilon_5$
$\{\kappa_c\}$	Three-cycle	ℓ_m $(m \equiv 0, 1)$, υ_m $(m \equiv 0, 2)$
(κ_b, κ_2)	P_2	$\ell_1, \upsilon_2, \upsilon_3$
$[\kappa_2, \infty)$	S	$\ell_1, \upsilon_2, \upsilon_3$

In the critical row $m \geq 1$ and congruences are modulo 3; these are all zero raw slacks, so the active set is infinite. Switching values, rounded to five decimal places, are

$$\kappa_W \approx 2.65646, \quad \kappa_1 \approx 7.73195, \quad \kappa_a \approx 8.64845, \quad \kappa_b \approx 53.64303, \quad \kappa_2 \approx 71.14732.$$

and Proposition B.19 gives the central branch on $[1, G_a]$. Each result proves feasibility for every lag, dual nonnegativity, and the stated endpoint identities. In particular, the branch values agree at G_W, G_1, G_a, and Theorem B.18 identifies the critical spectrum at $G = 1$. This proves (20), including its branch selections.

The same results give $|V'| < 1$ on every open arc; on the spiral this follows from $du/dG = 1/(2G^2)$. Continuity at the junctions implies that $t - V(t)$ is strictly increasing. It tends to $-\infty$ and $+\infty$ at the two ends of the real line because V is bounded. Its unique level $\log q$ is therefore the largest feasible exponent on $v = t - \log q$, hence equals $T(q)$ by Theorem 2.6. Every constructed measure has at most two atoms. □

For conversion to condition numbers put $q_\pm(G) = (G \pm \sqrt{G^2 - 1})(\sqrt{1 + u(G)^2} - u(G))$ and $\kappa(q) = ((1 + q)/(1 - q))^2$. Then

$$(\kappa_W, \kappa_1, \kappa_a) = \big(\kappa(q_-(G_W)), \kappa(q_-(G_1)), \kappa(q_-(G_a))\big), \quad (\kappa_b, \kappa_2) = \big(\kappa(q_+(G_a)), \kappa(q_+(G_1))\big).$$

The symmetry $(t, C) \mapsto (-t, -C)$ produces the two copies of the central families in Table 1.

2.4. Finite-horizon realization

2.4.1. Initialization from rest

An atom of positive mass supplies a plane rotating by ω and scaling by e^t at each step. To initialize these waves with $x_{-1} = x_0$, append a scalar quadratic coordinate Yp_k and adjust the first gradient. For Polyak's tuning one may take $p_k = (-q^2)^{\lceil k/2 \rceil}$. The extra coordinate adds $\frac{1}{4}Y^2(p_i - p_j)^2$ to the interpolation margin. A sufficiently large Y absorbs the first-step defect uniformly over all later times; the strict minimizer slack controls the growing case. Hence r frequencies give one example in dimension $2r + 1$, with one start valid for all horizons. For the special spiral the choice $Y = 2$ gives the uniform factor $1/\sqrt{5}$.

The zeros of the dual polynomial Q select the stationary frequencies. At $G = G_1$ or $G = 1$ one selected point has zero mass. At κ_c the stationary component is periodic with period three, and the initialized trajectory approaches this orbit as the extra coordinate decays. Figure 8 illustrates the selected stationary spectra.

2.4.2. Uniform bound at the exact rate

The spectral characterization first yields $W_N \leq C_\delta e^{N(T+\delta)}$. The boundary certificate has a strict negativity margin of order δ at exponent $T + \delta$. Real linear combinations of finitely many consecutive positions turn it into a local window functional; these functionals remain bounded as $\delta \downarrow 0$. Their limit is

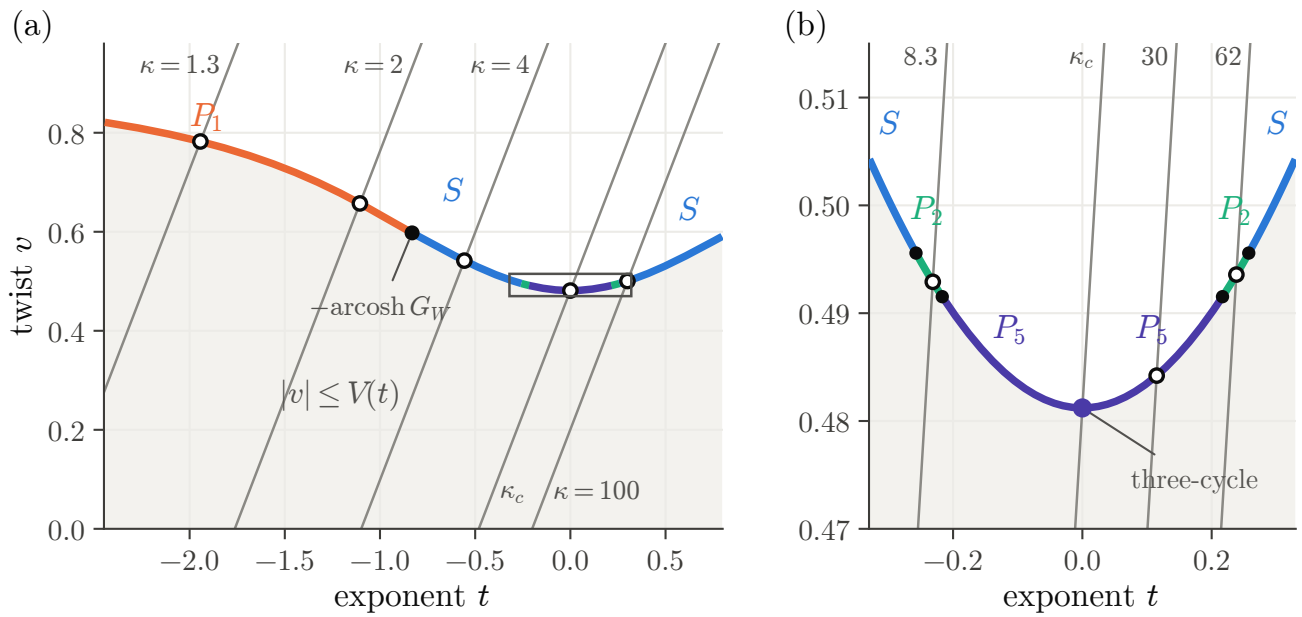


Figure 2: The upper spectral boundary $v = V(t)$; the shaded region is $0 \le v \le V(t)$. A line $v = t - \log q$ meets the boundary at the method's exponent $T = \log \rho_{\mathrm{HB}}$. Panel (b) magnifies the three-cycle and the switches $\pm \operatorname{arcosh} G_a$, $\pm \operatorname{arcosh} G_1$.

nonnegative on every admissible window, grows by at most e^{2T} per step, and is strictly positive on every stationary optimal spectrum (Lemma C.8). Separation and the energy estimate give a strictly smaller growth factor where this functional vanishes (Lemma C.9). Compactness extends this contraction to a neighborhood of that set (Lemma C.10). Outside that neighborhood the limiting functional controls the window norm, which gives $W_N \le C_\kappa e^{NT}$. Together with the two-frequency realization this proves Theorem 2.1. Appendix C gives this argument and the asymptotic expansions of Theorem 2.3.

3. General tunings

We now allow the step size and momentum to vary within the region of quadratic stability. The spectral characterization gives an exact criterion for uniform convergence and describes the boundary of its complement. Angle inequalities, Farey geometry and continuation of finite spectra determine the global organization and local structure of this boundary. We first give explicit phase tests, then the exact measure criterion of Theorem 3.7, and finally the global threshold and its local plateau structure.

3.1. Spectral characterization for general tunings

In this part q denotes the parameter triple (κ, y, β). Here $\kappa > 1$, $0 \le \beta < 1$ and $0 < y < 2(1+\beta)$; $x = y/\kappa = \alpha\mu$ and $y = \alpha L$. This is precisely the range (3) of stability on quadratics. Spectra in this part are measures in the frequency variable $\omega \in [0, \pi]$. Write $\rho(q)$ for (2) and call $\mathcal{D} = \{q : \rho(q) \ge 1\}$ the divergence set, the complement of uniform exponential convergence in (2). Its boundary can support nondecaying periodic behavior; exponential growth requires $\rho > 1$.

The endpoint characteristic polynomials are

$$a(z) = z^2 - (1 + \beta - x)z + \beta, \qquad b(z) = z^2 - (1 + \beta - y)z + \beta. \tag{22}$$

Their largest root moduli ρ_a, ρ_b give $\rho_{\mathrm{q}} = \max(\rho_a, \rho_b) \le \rho$. Between the damping curves

$$x = (1 - \sqrt{\beta})^2, \qquad y = (1 + \sqrt{\beta})^2, \tag{23}$$

every curvature is underdamped or critically damped and $\rho_{\mathrm{q}} = \sqrt{\beta}$. Outside this region an endpoint has a dominant real root. Polyak's tuning is the intersection of the curves.

The link with the spectral program is the identity

$$g - 2is = (1-m)(1+\overline{m}) = -\frac{a(z)\overline{b(z)}}{B^2|z|^2}, \qquad z = e^{t+i\omega}. \tag{24}$$

At $t = 0$, $s = (1-\beta)\sin\omega/B > 0$ on $(0,\pi)$. The argument of $g - 2is$ compares the phases of the endpoint characteristic polynomials and determines the threshold geometry.

Definition 3.1 (Angle function)**.** For $\omega \in (0,\pi]$ let $\Theta(\omega) \in (0,\pi]$ be the angle between $b(e^{i\omega})$ and $-a(e^{i\omega})$, so that $g - 2is = |g - 2is|\, e^{-i\Theta}$ at $t = 0$.

The curve is determined by a concave polynomial Ξ of degree at most two: $\cot\Theta(\omega) = \Xi(\cos\omega)/\sin\omega$. Its two endpoint values are negative; the coefficients and endpoint formulas are in Lemma D.2.

Extend Θ continuously by $\Theta(0) = \pi$. The spectral upper bound becomes particularly useful because one exponent already determines a strict rate bound. Let $r_0 < 1$ be the largest root modulus of $z^2 - Az + \beta$.

Theorem 3.2 (One exponent suffices)**.** *Assume* (3)*. If a strict certificate exists at some t with $e^t > r_0$, then $\rho < e^t$. In particular, infeasibility of* (13) *at $t = 0$ implies $\rho < 1$.*

Proof. Put $\varrho = e^t$ and rescale a trajectory of (10) by $\tilde{e}_k = \varrho^{-k} e_k$. Lemma D.4 converts the strict certificate into a quadratic constraint satisfied by the scaled gradient of every $\tau\psi$, $0 \le \tau \le 1$. The corresponding linear filter has transfer function

$$G_t(z) = \frac{B\varrho z}{\varrho^2 z^2 - A\varrho z + \beta}.$$

Its poles lie inside the unit circle because $\varrho > r_0$, and it vanishes at infinity. Moreover, $G_t(e^{i\omega}) = 1/m(t,\omega)$ is nonzero on the unit circle, and

$$[G_t; 1]^* \Pi [G_t; 1] = |G_t|^2 \Phi_t \le -\varepsilon \min_{|z|=1} |G_t(z)|^2 < 0.$$

The scaled gradient acts pointwise in time, is 1-Lipschitz and fixes zero; scaling ψ by $\tau \in [0,1]$ preserves $\mathcal{S}$. Thus every hypothesis of Lemma D.5 is satisfied. Its energy bound is uniform in the function, dimension and initial pair:

$$\sum_{k \ge 0} \varrho^{-2k} \|e_k\|^2 \le C_0(\|e_0\|^2 + \|e_{-1}\|^2).$$

Lemma D.6 therefore yields $\rho \le \varrho\theta^{1/4} < \varrho$ for one $\theta < 1$. At $t = 0$ the filter is stable by (3), and the spectral alternative supplies a strict certificate whenever (13) is infeasible. □

3.2. Convergence and periodic orbits

Two elementary sufficient conditions determine large subsets of the parameter region: a lag-one multiplier proves convergence, while a single-frequency feasible measure gives a periodic obstruction.

Theorem 3.3 (Phase locking)**.** *Assume* (3)*.*

(i) *If $\Theta(\omega) > \omega/2$ for every $\omega \in (0,\pi]$, then the method converges.*

(ii) *If $\Theta(2\pi j/k) \le \pi/k$ for coprime integers $1 \le j < k/2$, then there are $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^3)$ and x_0 with $\|x_N - x^\star\| \ge c\|x_0 - x^\star\| > 0$ for all N; the trajectory converges to a k-periodic orbit that rotates by $2\pi j/k$ in a plane. If the inequality is strict, then $\rho > 1$.*

(iii) *Let Ω_1 be the set of tunings in* (i)*. On its closure relative to* (3)*, $\rho \le 1$. At every boundary point of Ω_1 at which $\min_\omega(\Theta(\omega) - \omega/2) = 0$ is attained at $\omega = 2\pi/K$, the hypothesis of* (ii) *holds with $j = 1$ and $k = K$; there $\rho = 1$, and the point lies on the boundary of the convergence region.*

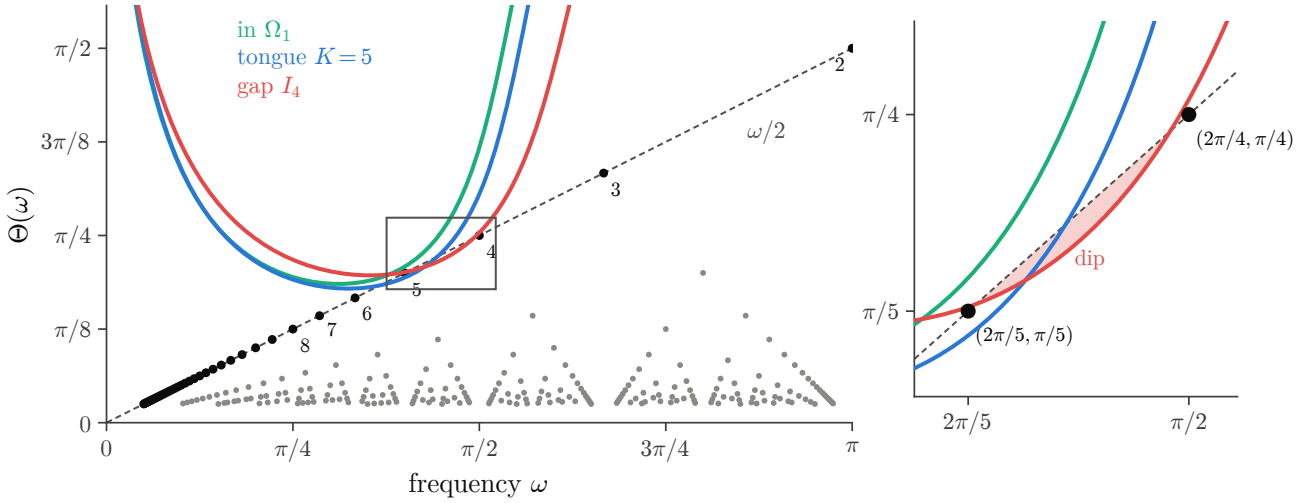


Figure 3: The rational points $(2\pi j/k, \pi/k)$ and three angle curves at $\kappa = 100$. The points $j = 1$ lie on $\omega/2$. A curve above the line gives convergence; meeting a rational point gives a periodic obstruction. The magnified dip between $2\pi/5$ and $2\pi/4$ contains the two-harmonic example of Proposition 3.13.

The proof is given in Appendix D.2.

For fixed j/k, the tunings satisfying the periodic criterion form the j/k-tongue; the K-tongue corresponds to $j/k = 1/K$. The two conditions compare the angle curve with the line $\omega/2$ and with rational phase values (Figure 3). At a phase-locked boundary point, the tangency frequency equals $2\pi/K$. The cycle criterion agrees with the roots-of-unity regions of [6, 10]; its algebraic form is given in Corollary D.11.

For $c = \cos\omega$, let $H_r(c) = \mathrm{Re}[(1-z)a(z)\overline{b(z)}]$, $z = re^{i\omega}$. It has degree at most three in c (Lemma D.3). Inside the lag-one region Ω_1, it bounds the rate and identifies a region $\mathcal{Q}$ of exact agreement with the quadratic worst case.

Theorem 3.4 (Rates)**.** *Assume* (3).

(a) *If* $\Theta(\omega) > \omega/2$ *on* $(0, \pi]$*, then the set* $\{r \in (\rho_b, 1) : H_r(c) \geq 0$ *for all* $c \in [-1, 1]\}$ *is a nonempty interval with right endpoint* 1*. With* r_1 *its infimum,* $\rho_{\mathrm{q}} \leq \rho \leq r_1 < 1$*.*

(b) *Suppose* $\rho_{\mathrm{q}} > \sqrt{\beta}$ *and* $\Theta(\omega) > \omega/2$ *on* $(0, \pi]$*. Then* $r_1 = \rho_{\mathrm{q}}$ *if and only if* $H_{\rho_{\mathrm{q}}}(c) \geq 0$ *for all* $c \in [-1, 1]$*. On the set* $\mathcal{Q}$ *of such tunings,* $\rho = \rho_{\mathrm{q}}$*.*

(c) *If* $\rho_{\mathrm{q}} = \sqrt{\beta}$*, then* $\rho > \rho_{\mathrm{q}}$*.*

The proof is given in Appendix D.2.

An interior underdamped quadratic mode has interpolation slack and can be perturbed to grow faster; this explains the strict gap in part (c). An overdamped endpoint mode is tight, and the sign of $H_{\rho_{\mathrm{q}}}$ decides whether the lag-one bound is sharp. The signs reduce to quadratics (Corollary D.11); Figure 4 shows the resulting regions.

Corollary 3.5 (Polyak's tuning and gradient descent)**.** (a) *For Polyak's tuning, the method lies in* Ω_1 *if and only if* $\kappa < \kappa_c$*; at* κ_c *the tuning is the phase-locked point* $K = 3$ *of* $\partial\Omega_1$*, and for* $\kappa > \kappa_c$ *it lies in the three-tongue.*

(b) *For* $\beta = 0$ *and* $0 < \alpha L < 2$*, both* Ω_1 *and* $\mathcal{Q}$ *are the whole range, and* $\rho = \max(|1 - \alpha\mu|, |1 - \alpha L|)$*.*

The proof is given in Appendix D.2.

The complement of Ω_1 is the union of the fractional-cycle regions $\Theta(\omega) \leq \omega/2$, the continuation of the integer tongues to real cycle lengths [9]. The gaps between consecutive integer tongues contain both convergent tunings and multi-frequency periodic obstructions. The full spectral program distinguishes these behaviors.

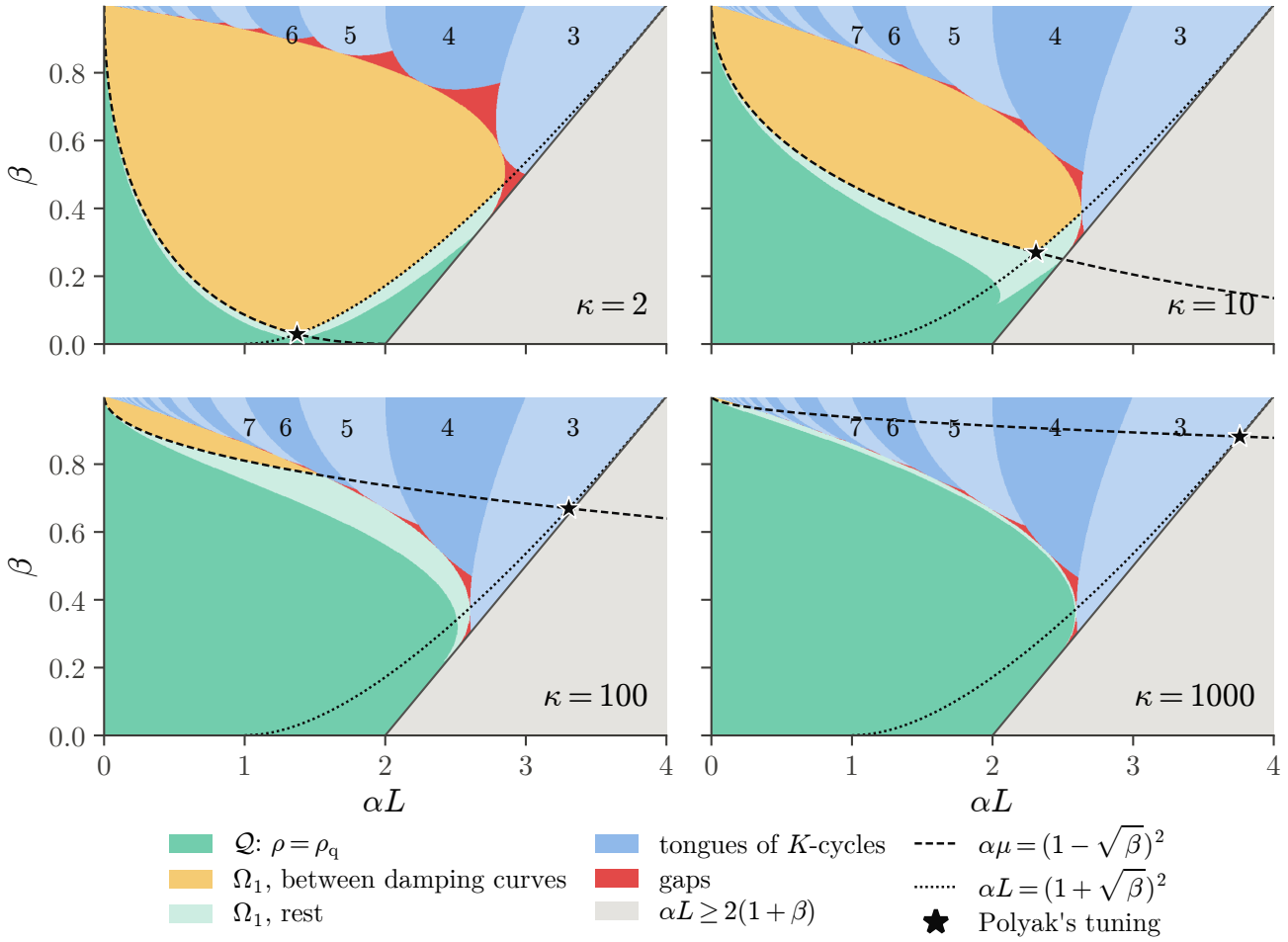


Figure 4: Phase diagrams for $\kappa = 2, 10, 100, 1000$. Green: $\mathcal{Q}$, where $\rho = \rho_q$. Yellow and pale green: the remaining lag-one convergence region Ω_1. Blue: integer-cycle tongues. Violet: additional rational-cycle tongues. Red: the remaining displayed gap regions. Grey: quadratic instability. The damping curves are dashed and dotted; the star is Polyak's tuning.

3.3. The threshold function and the divergence set

At exponent zero all lag inequalities can be written using the angle alone:

$$\Lambda_n(\omega) = \cos\Theta - \cos(n\omega - \Theta) = 2\sin\frac{n\omega}{2}\sin\Big(\frac{n\omega}{2} - \Theta\Big), \qquad n \in \mathbb{Z}\setminus\{0\}. \tag{25}$$

The magnitude $|n|$ is the temporal separation in an interpolation constraint; positive and negative indices denote the lower and upper inequalities, respectively. A curve is a continuous map $\Theta : [0,\pi] \to (0,\pi]$ with $\Theta(0) = \Theta(\pi) = \pi$.

Definition 3.6 (Full program and certificate symbols)**.** Let Θ be a curve. The full program $\mathrm{FP}(\Theta)$ asks for a nonzero positive Borel measure w of finite total mass on $[0,\pi]$ with

$$\int \Lambda_n\, dw \geq 0 \quad (n \in \mathbb{Z}\setminus\{0\}), \qquad \int \cos\Theta\, dw \geq 0.$$

The last condition is the minimizer constraint. For $\mathcal{N} \subset \mathbb{Z}\setminus\{0\}$, the program $\mathrm{FP}[\mathcal{N}](\Theta)$ keeps the lag constraints with $n \in \mathcal{N}$. A certificate symbol with lags in $\mathcal{N}$ is a function $S = \sum_{n\in N}\lambda_n\Lambda_n + \gamma\cos\Theta$ with $N \subset \mathcal{N}$ finite, $\lambda_n \geq 0$ and $\gamma \geq 0$; it is strict if $S \leq -\varepsilon$ on $[0,\pi]$ for some $\varepsilon > 0$.

This is a linear feasibility problem in w. The symbol combines its constraint functions with nonnegative dual weights: its integral against a feasible measure is nonnegative, so strict negativity certifies

infeasibility.

Theorem 3.7 (Exact threshold criterion)**.** *Let q be a tuning.*

(a) *The conditions* (13) *are feasible at $t = 0$ if and only if* $\mathrm{FP}(\Theta_q)$ *is feasible; up to normalization, $\nu \mapsto |g - 2is|\,\nu$ is a bijection between the spectra feasible at $t = 0$ with $C = 0$ and the solutions of* $\mathrm{FP}(\Theta_q)$*. Feasibility at $t = 0$ depends on the tuning only through the curve Θ_q, that is, through the three coefficients of Ξ.*

(b) *If* $\mathrm{FP}(\Theta_q)$ *is feasible, then $W_N \geq c$ for all N, with $c > 0$ independent of N; in particular $\rho(q) \geq 1$.*

(c) *$\rho(q) < 1$ if and only if* $\mathrm{FP}(\Theta_q)$ *is infeasible, if and only if a strict certificate symbol exists. Consequently $\mathcal{D} = \{q : \ \mathrm{FP}(\Theta_q)$ is feasible$\}$.*

Proof. At exponent zero, feasibility with any admissible C implies $\bar{g} \geq |C| \geq 0$, so $C = 0$ may be used. With $d(\omega) = |g - 2is| = |a(e^{i\omega})b(e^{i\omega})|/B^2$, the lag and minimizer functions satisfy

$$g(1 - \cos n\omega) - 2s \sin n\omega = d\Lambda_n, \qquad g = d \cos \Theta_q.$$

The function d is continuous and strictly positive on $[0, \pi]$ by quadratic stability. Consequently $w = d\nu$ and $\nu = d^{-1}w$ preserve finiteness and positivity and give exactly the correspondence in (a). Lemma E.2 gives the equivalence between infeasibility and a strict finite symbol. Such a symbol is a strict spectral certificate at $t = 0$, hence implies $\rho < 1$ by Theorem 3.2. Conversely, a feasible measure admits the start-up realization of Appendix E.1: multiplication by $e^{i\omega}$ in $L^2(\nu)$ gives a stationary trajectory, and one decaying coordinate enforces the initial condition with a fixed constant Y. Restricting its interpolation data to the finite span used before a horizon N gives a finite-dimensional objective with $\|e_N\| \geq 1$ and $\|e_0\| \leq \sqrt{1 + Y^2}$. Thus $W_N \geq (1 + Y^2)^{-1/2}$ for every N, proving (b) and the remaining implication in (c). □

3.3.1. Tangencies and gaps between periodicity regions

The cotangent $h_q = \Xi(\cos\omega)/\sin\omega$ is strictly concave. Consequently the closed dip, the set of frequencies where the curve meets or falls below the lag-one convergence line, $\bar{\Delta} = \{\omega \in (0, \pi] : \Theta_q(\omega) \leq \omega/2\}$ is a compact interval in $(0, \pi)$, possibly empty or a point (Theorem E.4 and Corollary E.5). Every feasible measure charges this interval (Lemma E.6).

Theorem 3.8 (The boundary of Ω_1)**.** *Let q satisfy $\Theta_q \geq \omega/2$ on $(0, \pi]$ with equality at some ω_*. If $\omega_* \neq 2\pi/K'$ for every $K' \geq 3$, then a strict certificate symbol with the lags 1 and $-K$ exists, where $2\pi/(K + 1) < \omega_* < 2\pi/K$, and $\rho(q) < 1$. Consequently $\partial\Omega_1 \cap \mathcal{D}$ consists exactly of the phase-locked points of Theorem 3.3*(iii)*, where $\rho = 1$.*

The proof is given in Appendix E.2.

Definition 3.9 (Gaps)**.** A tuning lies in the gap I_K of level one, $K \geq 2$, if the dip $\{\Theta < \omega/2\}$ is nonempty with rotation numbers in $(1/(K + 1), 1/K)$, while $\Theta(2\pi/K) > \pi/K$ and $\Theta(2\pi/(K + 1)) > \pi/(K + 1)$.

Theorem 3.10 (Level zero)**.** *Every tuning q falls into exactly one of the following cases.*

(I) *$\bar{\Delta} = \emptyset$: $q \in \Omega_1$, and $\rho(q) < 1$.*

(II) *$2\pi/K' \in \bar{\Delta}$ for some $K' \geq 3$: $\Theta(2\pi/K') \leq \pi/K'$, the tuning lies in the closed K'-tongue, $\delta_{2\pi/K'}$ solves* $\mathrm{FP}(\Theta_q)$*, and $\rho(q) \geq 1$.*

(III) *$\emptyset \neq \bar{\Delta} \subset (2\pi/(K + 1), 2\pi/K)$ for a unique $K \geq 2$. Either $\bar{\Delta}$ is a point, q is a point of $\partial\Omega_1$ that is not phase-locked, and $\rho(q) < 1$; or $\bar{\Delta}$ has interior, and $q \in I_K$.*

The proof is given in Appendix E.2.

Thus only the open gaps remain after lag-one convergence and integer cycles have been classified. For $c_K = \cos(2\pi/K)$, each gap has the coordinates

$$1 + c - \Xi_q(c) = a(c - r_1)(c - r_2), \qquad c_K < r_1 < r_2 < c_{K+1}, \qquad a > a_0(r_1, r_2) := \frac{2}{(1 - r_1)(1 - r_2)}.$$

Proposition E.7 proves that this parametrization is a homeomorphism and, in particular, that I_K is connected.

3.3.2. Continuity and periodic approximation

At exponent $t = 0$, introduce a separate parameter s by shifting the cotangent vertically:

$$\Theta_{q,s} = \operatorname{arccot}(h_q + s) \quad (0 < \omega < \pi), \qquad \Theta_{q,s}(0) = \Theta_{q,s}(\pi) = \pi. \tag{26}$$

Increasing s lowers the curve and preserves feasibility. In the monotone form of Proposition E.9, this is the identity

$$\ell_n[h + s] = \ell_n[h] + s(1 - \cos n\omega), \qquad \ell_n[h] = h(1 - \cos n\omega) - \sin n\omega.$$

There is a finite attained threshold $\sigma(q)$: $\mathrm{FP}(\Theta_{q,s})$ is feasible exactly for $s \geq \sigma(q)$, and $\mathcal{D} = \{\sigma \leq 0\}$ (Proposition E.11).

Feasible measures at this smallest shift are called threshold extremals; extremality here refers to the attained shift. For $L \geq 3$, let $\Omega_L = \{2\pi j/L : 0 < 2\pi j/L \leq \pi\}$. The L-periodic program P_L restricts the measure to Ω_L, whose waves all repeat after L iterations; lag constraints then depend only on $n \bmod L$. The harmonic reduction for cycles with free initial states is given in [10, Theorem 6.13]; Proposition E.12 enforces the initialization in (1). Write $\mathcal{T}_L$ for its feasible tunings, σ_L for its shift threshold, and $\sigma_{\mathcal{N}}$ for the threshold using lags in $\mathcal{N}$. Then $\sigma_{\mathcal{N}} \leq \sigma \leq \sigma_L$.

Theorem 3.11 (Periodic sampling)**.** *For every tuning q, $\sigma(q) = \inf_{L \geq 3} \sigma_L(q)$. If $q \in \mathcal{D}$, then either $q \in \mathcal{T}_n$ for some n, and then $(\kappa', \alpha L, \beta) \in \mathcal{T}_n$ for all $\kappa' \geq \kappa$, or for every $\kappa' > \kappa$ there is L_0 with $(\kappa', \alpha L, \beta) \in \mathcal{T}_L$ for all $L \geq L_0$. In particular $\mathcal{D} = \operatorname{cl} \bigcup_L \mathcal{T}_L$.*

The proof is given in Appendix E.4.

Theorem 3.12 (Global structure)**.** (a) *For every $L \geq 3$ the periodic threshold σ_L is finite and continuous on the set of tunings, and the threshold σ is continuous.*

(b) *σ is strictly decreasing in κ at fixed $(\alpha L, \beta)$, and $\sigma \to +\infty$ as $\kappa \downarrow 1$.*

(c) *Put*

$$\kappa_\partial(\alpha L, \beta) = \inf\{\kappa > 1 : \ \sigma(\kappa, \alpha L, \beta) \leq 0\} \in (1, \infty]$$

on the strip $\{\beta \in [0, 1), \ 0 < \alpha L < 2(1 + \beta)\}$. Then

$$\mathcal{D} = \{(\kappa, \alpha L, \beta) : \ \kappa \geq \kappa_\partial(\alpha L, \beta)\},$$

and κ_∂ is continuous as a map into $(1, \infty]$.

The proof is given in Appendix E.4. Writing κ_L for the condition-number threshold of P_L, we also have $\kappa_\partial = \inf_{L \geq 3} \kappa_L$ (Corollary E.18).

Here $(1, \infty]$ carries its order topology; an infinite boundary means convergence for every finite condition number. This occurs throughout the zero-momentum face.

The essential approximation step is to preserve all signed lag constraints simultaneously. A strict cotangent shift gives positive slack unless the measure is already supported on a periodic frequency grid. Lemmas E.14 and E.15 turn that slack into a uniform margin after smoothing; Lemma E.16 then preserves it on every sufficiently fine periodic grid. Compactness and continuity of the finite-grid thresholds give continuity of σ. Strict decrease in κ follows from Lemma E.10 and yields the boundary graph, including its possible value $+\infty$.

On $\partial\mathcal{D}$, continuity gives $\sigma = 0$, so threshold extremals solve the original program. Their supports are constrained simultaneously by one limiting dual symbol. By complementary slackness, the symbol vanishes on every threshold-extremal support, and each positive dual coefficient forces its corresponding constraint to hold with equality (Theorem E.19). Its zeros will determine the periodic structure of the boundary.

3.4. Farey geometry and finite spectral measures

The gap I_4 already exhibits the distinction between one harmonic and the full program.

Proposition 3.13 (A two-harmonic thirteen-cycle)**.** *Let $\kappa = 100$, $\alpha L = 70471/31800$ and $\beta = 6169/9900$. Then the tuning lies in the gap I_4, $\Theta > \pi/6$ on $(0, \pi]$, and no one-atom measure solves* FP(Θ)*. The spectrum ν with the weights $149/797$ at $2 \cdot 2\pi/13$ and $648/797$ at $3 \cdot 2\pi/13$, with $C = 0$, is feasible at $t = 0$; equivalently, $w = |g - 2i \operatorname{Im} m|\, \nu$ satisfies the hypothesis of Theorem F.10(a) with $p = 3$, $k = 13$. Hence a function in $\mathcal{F}_{\mu,L}(\mathbb{R}^5)$ has a trajectory that converges to a thirteen-periodic orbit with the two harmonics $2/13$ and $3/13$, and $\rho \geq 1$.*

The proof is given in Appendix F.3.

Proposition 3.14 (Convergence in a gap)**.** *Let $\kappa = 100$, $\alpha L = 111/50$ and $\beta = 31/50$. Then the tuning lies in the gap I_4, and the two-lag inequality* (30) *holds with $K = 4$ and the rational multipliers listed in Appendix F. Hence $\rho < 1$.*

The proof is given in Appendix F.3.

Thus I_4 contains both a two-frequency periodic obstruction and a tuning certified to converge by the lags 1 and -4.

3.4.1. Farey parents and binding constraints

Introduce

$$\xi = \frac{\omega}{2\Theta}, \qquad \eta = \frac{\pi}{\Theta}. \tag{27}$$

The image of the angle curve is $\mathcal{C}$, and its radial direction $\xi/\eta = \omega/(2\pi)$ is the rotation number, the fraction of a full turn per iteration. A lag function is negative exactly in the open strips $0 < n\xi - \ell\eta < 1$, $\ell \in \mathbb{Z}$; the rational contact points become the lattice points (j, k) (Theorem F.2).

For $0 < j/k < 1/2$ in lowest terms, write $a/b < j/k < c/d$ for its Farey parents: $j = a + c$, $k = b + d$ and $bc - ad = 1$ [12, Chapter III]. A single atom at $2\pi j/k$ is feasible exactly when its phase is at most π/k. As the phase crosses π/k, its first violated lags are b and $-d$, the denominators of the two Farey parents; they form one residue class modulo k (Theorem F.3).

The parent denominators identify the binding lags. Each periodic obstruction in I_K has an anchor, an atom in the closed dip, at a reduced fraction j/k of its window; necessarily $k \geq 2K + 1$ (Proposition F.5). For later use put

$$\omega_0 = 2\pi j/k, \quad c_0 = \cos\omega_0, \quad s_0 = \sin\omega_0, \qquad c' = \pi/k, \quad h_0 = s_0 \cot c', \quad b \equiv j^{-1} \pmod{k}.$$

The single-harmonic tongue is the explicit graph region $\kappa \geq \kappa_{j/k}(y, \beta)$ of Lemma F.4. Hence $\kappa_\partial \leq \inf_{j/k} \kappa_{j/k}$: the single-harmonic envelope bounds the full threshold from above, and periodic mixtures may lower it (Proposition 3.13). By Theorem 3.19, each Farey fraction carries an open plateau of equality. Figure 5 shows the competing tongue graphs across I_4 and their lower envelope. The fraction attaining this envelope changes in steps; the undamped quadratic rotation number of Proposition 3.21 gives the limiting locations of the single-harmonic plateaus.

3.4.2. Compensation and pinning

The sequences of neighboring fractions $(a + mj)/(b + mk)$ and $(c + mj)/(d + mk)$, called Farey fans, approach j/k from its two sides. For the endpoint $1/K$ the fan has

$$k_p = pK + 1, \qquad p/k_p, \quad p \geq 1. \tag{28}$$

Its common upper parent lag is $-K$; the associated lag family is

$$\mathcal{N}_{\text{fan}}(K) = \{mK + 1 : m \geq 0\} \cup \{-mK : m \geq 1\}. \tag{29}$$

Under the phase bounds of Theorem F.10, the lag functions at an anchor p/k_p and a second frequency, its companion $(p - 1)/k_p$, have opposite signs for the lags 1 and $-K$; positive masses can therefore

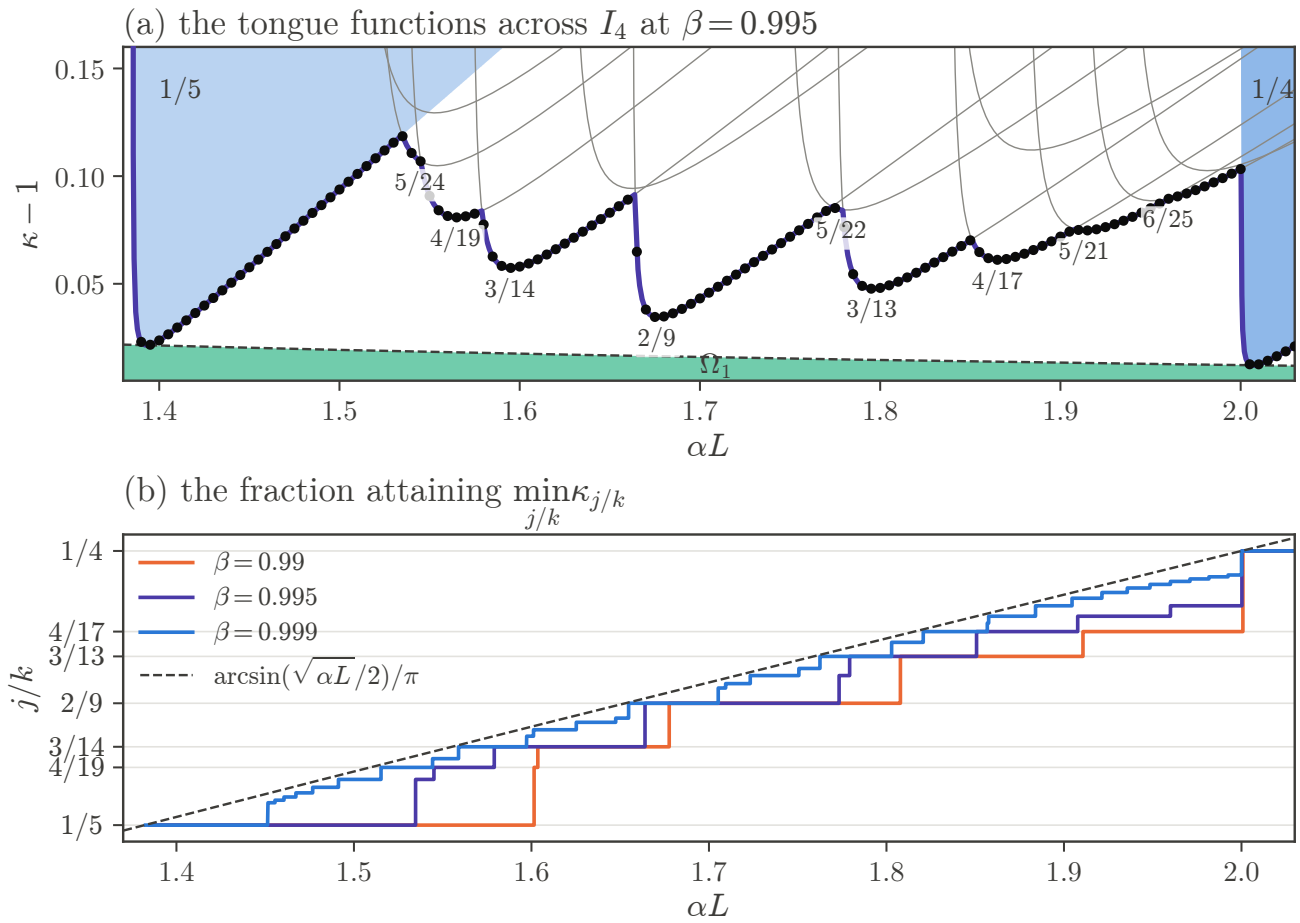


Figure 5: Tongue graphs and their lower envelope across I_4. (a) At $\beta = 0.995$: individual graphs $\kappa_{j/k} - 1$ with $k \le 30$ (grey), their envelope over $k \le 60$ (violet), and periodic upper bounds on the full threshold, using periods at most 60 (dots). The dashed curve bounds Ω_1; blue regions are the tongues of 1/5 and 1/4. (b) The fraction attaining the envelope over $k < 80$, at three values of β. The dashed curve is the undamped quadratic rotation number $\arcsin(\sqrt{\alpha L}/2)/\pi$ (Proposition 3.21).

compensate the violated constraints. Theorem F.10 gives the exact determinant criterion; Theorem F.9 gives the complete ratio interval for every anchor and companion, selected by Farey parents one level below the anchor. The underlying condition excludes lattice points from a specified box: every lag function negative at the anchor must be positive at its companion.

The dual construction uses the same lags. With Chebyshev polynomials T_n, U_n, the simplest certificate is

$$\Xi(c)\big[(1-c) + r(1 - T_K(c)) + \gamma\big] < (1-c^2)(1 - rU_{K-1}(c)), \qquad -1 \le c \le 1, \tag{30}$$

where $r \ge 0$ and $\gamma > 0$. It proves $\rho < 1$ by Theorem F.11. Allowing the other fan lags gives the general polynomial criterion (86).

Finally, lags in one residue class agree on Ω_k but have different derivatives there. Their exact difference is the pinning identity of Lemma E.13; for the fan, $-pK \equiv 1$ and $(p-1)K + 1 \equiv -K \pmod{k_p}$ (Lemma F.12). Redistributing dual weight within these classes preserves contact values and controls contact slopes. This permits continuation of a threshold measure with fixed periodic support, as established in Subsection 3.5.

3.5. Regularity of the divergence boundary

In a gap, convergence and divergence meet on the surface

$$\mathcal{B}_K = \partial\mathcal{D} \cap I_K = \{q \in I_K : \sigma(q) = 0\}.$$

The surface is the graph of κ_∂ over its projection to the $(\alpha L, \beta)$ plane. Off this surface, either a finite-lag certificate or a periodic measure determines the behavior. On the surface, the support and regularity of a threshold measure determine the local geometry. Call a solution of $\mathrm{FP}(\Theta_q)$ at $q \in \mathcal{B}_K$ an extremal. Write $\mathcal{P} = \{q \in \mathcal{B}_K : \sigma_L(q) = 0 \text{ for some } L\}$, and let $\mathcal{R}$ be its regular points, in the sense of Definition G.1.

Theorem 3.15 (Classification)**.** *Let $q \in I_K$. Exactly one of the following holds.*

(i) $\kappa < \kappa_\partial(\alpha L, \beta)$*: $\sigma(q) > 0$, a strict certificate symbol with finitely many lags exists, and $\rho(q) < 1$.*

(ii) $\kappa > \kappa_\partial(\alpha L, \beta)$*: $\sigma(q) < 0$, there is L with $q \in \mathcal{T}_{mL}$ for all $m \geq 1$, and $\rho(q) \geq 1$.*

(iii) $\kappa = \kappa_\partial(\alpha L, \beta)$*: $q \in \mathcal{B}_K$, $\rho(q) \geq 1$, and q lies in exactly one of $\mathcal{R}$, $\mathcal{P} \setminus \mathcal{R}$ and $\mathcal{B}_K \setminus \mathcal{P}$.*

In (ii) and (iii), every solution of every feasible $\mathrm{P}_L(\Theta_q)$ has an anchor at a Farey fraction $j/k \in (1/(K+1), 1/K)$ with $k \mid L$ and $k \geq 2K+1$.

The proof is given in Appendix G.2.

Thus convergence off the boundary admits a finite-lag certificate, and divergence off the boundary admits a periodic spectral measure. Boundary points divide into regular periodic points, singular periodic points, and points with no threshold extremal supported on a finite periodic grid. The last class is approached by the thresholds of every sufficiently large period; plateau endpoints belong to the second class (Theorem G.4).

3.5.1. Analytic continuation of regular spectra

At an L-regular point, $S \subset \Omega_L \cap (0, \pi)$ is the finite frequency support, N is a finite set of certificate lags, and w^*, λ^* are their positive normalized weights. Write $\mathcal{A} = (N \setminus L\mathbb{Z}) \bmod L$ for the active nonzero residue classes, whose lag constraints hold with equality at w^*, and $\Lambda_n^{q,s} = \Lambda_n[\Theta_{q,s}]$. Definition G.1 requires a one-dimensional kernel for the active constraint matrix, strict inactive and minimizer inequalities, negative second derivatives of the dual symbol at its zeros, a positive endpoint moment and positive pinning factors, and full rank of the pinning equations. These conditions permit the masses and the certificate to vary while the frequencies stay fixed.

Theorem 3.16 (Mode locking)**.** *Let (q_*, s_*) be L-regular with data S, N, w^*, λ^*. Then $s_* = \sigma(q_*) = \sigma_L(q_*)$, and there are a neighborhood U of q_* and real analytic maps $s : U \to \mathbb{R}$, $w : U \to \mathbb{R}^S_{>0}$ with $s(q_*) = s_*$, $w(q_*) = w^*$, such that for every $q \in U$:*

(a) $\sigma(q) = \sigma_L(q) = s(q)$*, and* $\det\big(\Lambda_\varrho^{q,s(q)}(\omega)\big)_{\varrho \in \mathcal{A}, \omega \in S} = 0$*;*

(b) $w(q)$ *solves* $\mathrm{FP}(\Theta_{q,s(q)})$*;*

(c) *the solutions of* $\mathrm{FP}(\Theta_{q,\sigma(q)})$ *are exactly the measures* $c\,w(q) + a\,\delta_0$ *with* $c > 0$ *and* $0 \leq a \leq c \int \cos\Theta_{q,\sigma(q)}\, dw(q)$*; the threshold extremal on* $(0, \pi]$ *is unique up to scaling and carried by* S*;*

(d) $\nabla s(q) = -\sum_{\omega \in S} \varpi_\omega(q)\, \nabla_q h_q(\omega)$ *with weights* $\varpi_\omega(q) > 0$*,* $\sum_\omega \varpi_\omega = 1$*; in particular* $\partial_\kappa s < 0$*;*

(e) $(q, s(q))$ *is L-regular with the data S, N.*

Consequently, if $s_* = 0$, then near q_* the boundary $\partial\mathcal{D}$ is the analytic graph $\kappa = \kappa_\partial(\alpha L, \beta)$ of the function defined by $s = 0$, and along it the threshold extremal keeps the support S.

The proof is given in Appendix G.1.

The determinant equation determines the shift and the positive primal masses by the analytic implicit function theorem. The independent slope equations then determine positive dual weights whose symbol has the same contacts. The strict curvature and complementary slackness conditions ensure that these contacts remain its only nonzero zeros. They give both optimality of the continued shift and the stated classification of every threshold measure.

Table 2: Mixed-spectrum boundary points and rational enclosures.

anchor	$(\alpha L, \beta)$	interval for $\tilde{\kappa}$	$S \cdot L/(2\pi)$
2/9	$(2.201327, 0.6268)$	$(100.5508210, 100.5508214)$	$\{3,4\}$, $L = 18$
5/21	$(2.240775, 0.6152)$	$(106.8055118, 106.8055121)$	$\{3,4,5\}$, $L = 21$

The regular boundary is relatively open. On each connected component its support, least period, and tight residue classes are constant (Corollary G.2); these components are the plateaus. Thus mode locking fixes the frequencies, while their masses and the boundary height κ_∂ may vary analytically. The finite lag set of a regular certificate gives an exact local decision procedure, and competing grids incur a quadratic loss (Theorems G.7 and G.8).

The least period determines all feasible threshold periods. The endpoint fan realizes a plateau precisely when its least period is coprime to the gap index (Corollary G.3).

3.5.2. Mixed-spectrum plateaus

Regular threshold measures may require more than one nonzero frequency. The following two boundary points in I_4 have supports of sizes two and three, respectively.

Theorem 3.17 (Two mixed-spectrum plateaus)**.** *For each row of Table 2, take the displayed decimal values of $(\alpha L, \beta)$ exactly. There is a unique boundary value $\tilde{\kappa}$ in the stated open interval. The tuning $(\tilde{\kappa}, \alpha L, \beta)$ lies in I_4 and is L-regular with support S. It lies on an open analytic plateau whose extremals are exactly the positive multiples of a measure with positive mass at every point of S, together with the admissible atom at 0 of Theorem 3.16(c). Throughout its intersection with I_4, its unique anchor is the displayed fraction. The least period of the first plateau is 18, and none of its threshold periods has the form $4p + 1$. Every extremal on the second plateau has three nonzero frequencies.*

The proof is given in Appendix G.4.

At some points a two-lag relaxation already has the threshold of the full program. Such a point occurs when its two contact frequencies lie on a common grid and the remaining residue constraints are strict (Theorem G.12).

Theorem G.15 gives such an exact point with anchor 3/13 at $\kappa \approx 83.5343$. Its threshold spectrum is $\{2,3\} \cdot 2\pi/13$, up to scaling and an admissible atom at 0; the lags $1, -4$ certify the threshold at this point, and regularity places it on an analytic plateau.

The geometry behind this example is a double tangency between the quadratic Ξ_q and a rational Chebyshev profile (Proposition G.14). It reduces the contact equations to one scalar equation and the global symbol inequality to the sign of a linear polynomial; see Appendix G.

Figure 6 illustrates a finite-period approximation to $\mathcal{B}_4$ at $\kappa = 150$, across the gap between the tongues of 1/5 and 1/4. The displayed frequency intervals are indexed by Farey fractions, with the widest interior interval at 3/13. These intervals illustrate the mode locking of Theorem 3.16; the mixed examples above describe possible spectral supports within plateaus.

3.6. Single-frequency plateaus

Every reduced fraction j/k in the window of a gap also supports a single-harmonic plateau. Its threshold lies on the edge of its tongue, and its entire regular component admits an exact description. The edge and fan criteria apply to every reduced $0 < j/k < 1/2$, $k \geq 3$. Use the notation $\omega_0 = 2\pi j/k$, $c_0 = \cos\omega_0$, $s_0 = \sin\omega_0$, $c' = \pi/k$, $h_0 = s_0 \cot c'$ of Subsection 3.4. Let b be the inverse of j modulo k and set $\mathcal{N}_{j/k} = (b + k\mathbb{Z}) \cup (k\mathbb{Z} \setminus \{0\})$. For every tuning define

$$s_*(q) = \cot c' - h_q(\omega_0), \qquad \Theta_* = \Theta_{q,s_*(q)}, \qquad \bar{n}(q) = 2\Theta'_*(\omega_0).$$

The single atom δ_{ω_0} is feasible at this shift, so

$$\sigma(q) \leq \sigma_k(q) \leq s_*(q). \tag{31}$$

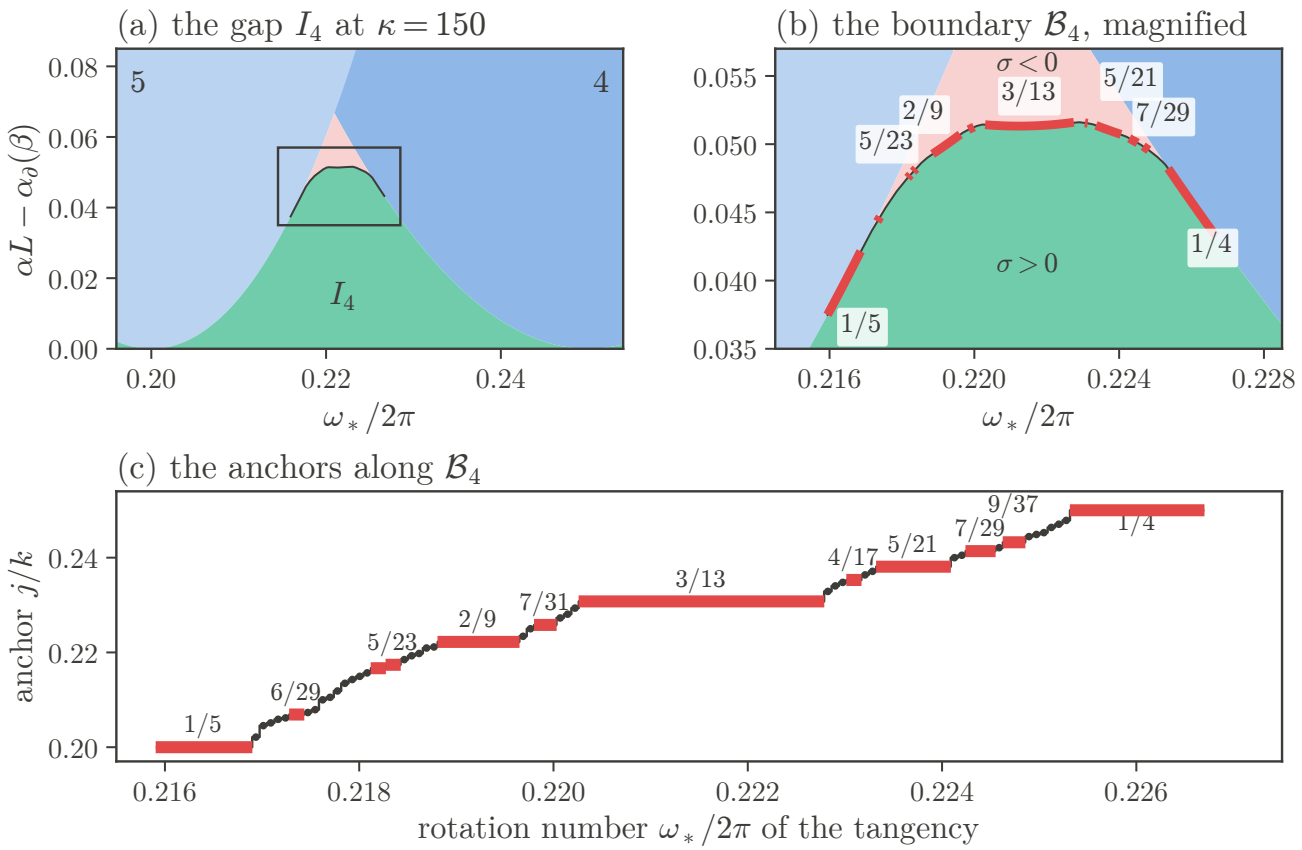


Figure 6: A finite-period approximation to the section at $\kappa = 150$, using programs with $9 \le L \le 120$. The horizontal coordinate is the rotation number $\omega_*/2\pi$ of the tangency on $\partial\Omega_1$; the vertical coordinate is $\alpha L - \alpha_\partial(\beta)$, where $(150, \alpha_\partial(\beta), \beta) \in \partial\Omega_1$. (a) The gap between the tongues of 1/5 and 1/4, with green and pale red below and above the computed curve. (b) The boxed region magnified; red segments mark constant selected frequencies. (c) The rotation number of the largest atom of the minimizing periodic measure in cotangent form, $\nu = \sin\Theta\, w$ (Proposition E.9). The constant-frequency intervals illustrate periodic mode locking.

On the tongue edge $s_* = 0$. For a fraction $j/k \in (1/(K+1), 1/K)$, parametrize that edge by

$$\Xi^{a,D}(c) = h_0 + (1+D)(c - c_0) - a(c - c_0)^2, \qquad a > 0.$$

Whenever $\Xi^{a,D}(\pm 1) < 0$ it defines a unique tuning $q(a, D)$. The slope index records twice the derivative of the angle curve at the contact:

$$\bar{n}(D) = 2\sin^2 c'(1 + D + \cot c' \cot \omega_0). \tag{32}$$

Let $a_E(D)$ be the explicit piecewise affine threshold for membership in I_K (Lemma H.6). Let $a_F(D)$ be the least curvature required by a finite single-harmonic certificate, defined by the one-variable minimax problem in Definition H.7. Its admissible finite nonnegative lag weights cancel the contact slope; a_F minimizes the global maximum of their curvature profiles. When the slope index lies strictly between two consecutive lags in $b + k\mathbb{Z}$, the adjacent-lag symbol is the convex combination with zero derivative at ω_0.

Definition 3.18 (Edge symbol)**.** Let q be a tuning with $\bar{n} = \bar{n}(q) \notin b + k\mathbb{Z}$. Let $n_1 < \bar{n} < n_2 = n_1 + k$ be the members of $b + k\mathbb{Z}$ adjacent to $\bar{n}$, put $\vartheta = (\bar{n} - n_1)/k \in (0, 1)$, and, with the lag functions of Θ_*,

$$T_q = (1 - \vartheta)\Lambda_{n_1} + \vartheta\Lambda_{n_2}, \qquad F_q = -\tfrac{1}{2}T_q.$$

Write $\lambda^{\mathrm{e}}_{n_1} = 1 - \vartheta$ and $\lambda^{\mathrm{e}}_{n_2} = \vartheta$. The associated profile is

$$\Psi_{\mathrm{e}}(c) = \frac{\sin\omega \sum_{i=1}^{2} \lambda^{\mathrm{e}}_{n_i} \sin(n_i\omega)}{\sum_{i=1}^{2} \lambda^{\mathrm{e}}_{n_i}(1-\cos(n_i\omega))}, \qquad c = \cos\omega,$$

$$R_{\lambda^{\mathrm{e}}}(c) = \frac{h_0 + (1+D)(c-c_0) - \Psi_{\mathrm{e}}(c)}{(c-c_0)^2}.$$

Values at the endpoints and at c_0 are understood by continuous extension. The curvature threshold $a_2(D) = R_{\lambda^{\mathrm{e}}}(c_0)$ is where the contact curvature vanishes (Lemma H.8); Definition H.7 gives the full class of profiles defining a_F.

Theorem 3.19 (The single-harmonic plateau of j/k)**.** *Let $K \geq 2$ and let $j/k \in (1/(K+1), 1/K)$ be a Farey fraction. Put*

$$\Pi_{j/k} = \{q(a,D) : \ D \in \mathbb{R}, \ a > \max\{a_E(D), a_F(D)\}\}.$$

(a) *$\Pi_{j/k}$ is a plateau contained in $\mathcal{B}_K$, and it consists of all regular points of $\partial\mathcal{D}$ whose threshold extremal is carried by $\{\omega_0\}$.*

(b) *On $\Pi_{j/k}$ the threshold extremal is δ_{ω_0}, up to scaling and an atom at 0, its unique anchor is j/k, and $\kappa_\partial = \kappa_{j/k}$ on the projection of $\Pi_{j/k}$ to the $(\alpha L, \beta)$ plane.*

(c) *If $\bar{n}(D) \notin b + k\mathbb{Z}$, then $a_2(D) \leq a_F(D) \leq \max_{[-1,1]} R_{\lambda^{\mathrm{e}}}$, and every defined $q(a,D)$ with $a < a_2(D)$ lies in the interior of $\mathcal{D}$, where the fan pairs of j/k have a smaller threshold than the single harmonic (Theorem 3.23).*

The proof is given in Appendix H.3.

Lemma H.8 identifies strict regular certificates with $a > \max R_\lambda$, and Theorem H.9 supplies such a certificate at every sufficiently large curvature. A common continuous upper bound for these sufficient curvatures makes the parameter region connected. The converse uses the uniqueness of a regular extremal to exclude contact with an integer tongue. The curvature comparison in (c) then follows from the exact second-order fan asymptotics below.

Corollary 3.20 (A plateau at every Farey fraction)**.** *For every $K \geq 2$, every Farey fraction $j/k \in (1/(K+1), 1/K)$ is the unique anchor of the plateau $\Pi_{j/k} \neq \emptyset$ of $\mathcal{B}_K$, whose threshold extremal is the single harmonic at $2\pi j/k$ and on which κ_∂ is the rational function $\kappa_{j/k}$. In particular $\mathcal{B}_K$ contains regular points for every $K \geq 2$.*

The proof is given in Appendix H.3.

Proposition 3.21 (Location of $\Pi_{j/k}$)**.** *For fixed D, as $a \to \infty$,*

$$\beta \to 1, \qquad a(1-\beta)^2 \to \frac{2\tan(\pi/(2k))}{\sin\omega_0}, \qquad \alpha L \to 4\sin^2\frac{\pi j}{k}, \qquad \kappa \to 1$$

along $q(a,D)$. Thus the projection of $\Pi_{j/k}$ to the $(\alpha L, \beta)$ plane accumulates at the point $(4\sin^2(\pi j/k), 1)$, at which the characteristic polynomial $z^2 - (2-\alpha L)z + 1$ of the undamped quadratic has the roots $e^{\pm 2\pi i j/k}$.

The proof is given in Appendix H.3.

3.6.1. Second-order asymptotics of Farey fan pairs

The same curvature that certifies the single harmonic controls the nearest rational competitors. For the edge symbol of Definition 3.18,

$$F_q''(\omega_0) = \sin c'\, \Theta_*''(\omega_0) - \tfrac{1}{2}\cos c'\, k^2\vartheta(1-\vartheta).$$

Positive curvature, together with the global sign and endpoint conditions, gives regularity. Nonpositive curvature obstructs every regular single-harmonic certificate at that frequency (Theorem H.2).

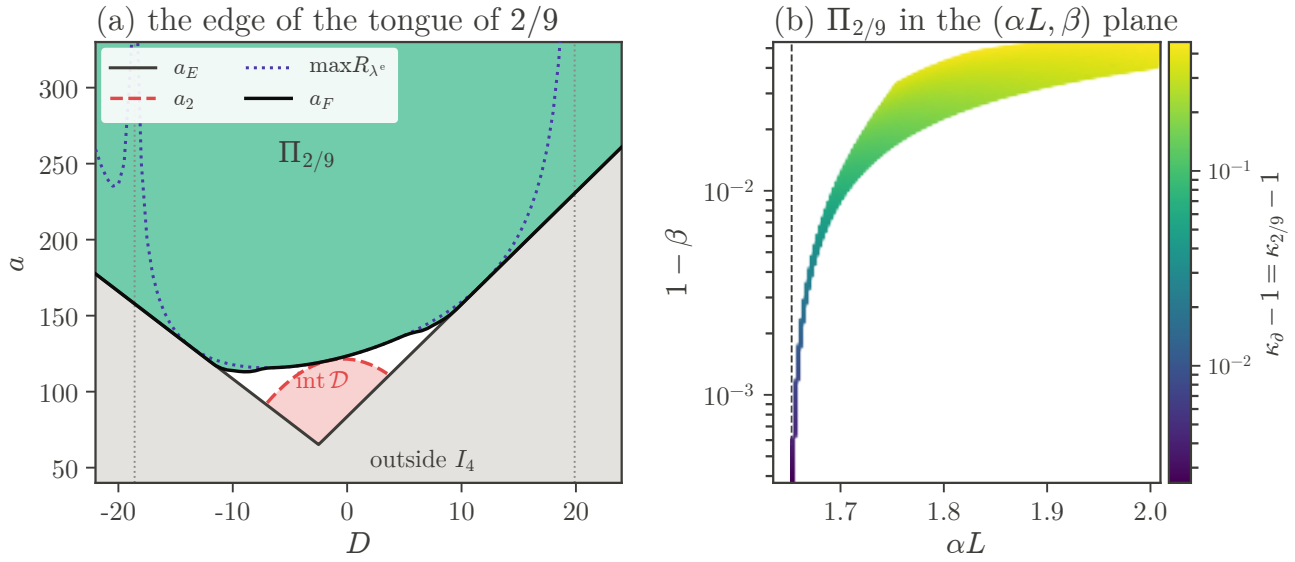


Figure 7: The single-harmonic plateau of 2/9. (a) In edge coordinates: the gap boundary a_E, the curvature obstruction a_2 (dashed), and a numerical evaluation of the adjacent-lag bound $\max R_{\lambda^e}$ (dotted). The solid curve approximates a_F by finite-lag programs. (b) The image in $(\alpha L, 1-\beta)$, with logarithmic vertical scale; its limiting abscissa is $4\sin^2(2\pi/9)$. Color indicates $\kappa_\partial - 1$.

Definition 3.22 (Fan pairs of j/k)**.** Let $\varsigma \in \{\pm 1\}$ and let $L > k$ satisfy $jL \equiv \varsigma \pmod k$. Put $J = (jL - \varsigma)/k$,

$$\omega_a = \frac{2\pi J}{L}, \qquad \omega_c = \frac{2\pi(J+\varsigma)}{L}, \qquad \delta = \omega_a - \omega_0 = -\frac{2\pi\varsigma}{kL}.$$

Then $\gcd(J, L) = 1$, $\omega_c = \omega_0 - (k-1)\delta$ and $(k-1)\omega_a + \omega_c = 2\pi j$. The fan pair of order L is $\{\omega_a, \omega_c\}$; ω_a is its point nearer to ω_0.

Theorem 3.23 (Fan pairs at a Farey fraction)**.** *Let q be a tuning with $\bar{n}(q) \notin b + k\mathbb{Z}$, let $\varsigma \in \{\pm 1\}$, and put $e_*(q) = -(k-1)F_q''(\omega_0)/(2\sin^3 c')$.*

(a) *For every $e < e_*(q)$ there are $z \in \mathbb{R}$ and L_0 such that for every $L \geq L_0$ with $jL \equiv \varsigma \pmod k$ the measure with mass $k - 1 + z\delta$ at ω_a and mass 1 at ω_c solves $\mathrm{FP}(\Theta_{q,s})$ at the shift $s = s_*(q) - e\delta^2$.*

(b) *For every $e > e_*(q)$ there is L_0 such that for every $L \geq L_0$ with $jL \equiv \varsigma \pmod k$ no nonzero measure carried by $\{\omega_a, \omega_c\}$ solves $\mathrm{FP}(\Theta_{q,s})$ at $s = s_*(q) - e\delta^2$.*

(c) *The threshold $s_L^{\mathrm{pair}}(q)$ of* FP *restricted to measures carried by the fan pair of order L satisfies*

$$s_L^{\mathrm{pair}}(q) = s_*(q) + \frac{(k-1)F_q''(\omega_0)}{2\sin^3 c'}\,\delta^2 + o(\delta^2) \qquad (L \to \infty,\ jL \equiv \varsigma \pmod k).$$

(d) *If $F_q''(\omega_0) < 0$, then $\sigma(q) < s_*(q)$. In particular a point of the edge with $F_q''(\omega_0) < 0$ lies in the interior of $\mathcal{D}$.*

(e) *If $F_q > 0$ on $(0, \pi] \setminus \{\omega_0\}$, $F_q''(0) > 0$ and $F_q''(\omega_0) > 0$, then for $L \to \infty$ with $k \nmid L$*

$$\sigma_L(q) - s_*(q) \;\geq\; \frac{F_q''(\omega_0)}{2\sin^3 c'}\Big(\frac{2\pi}{kL}\Big)^2(1 + o(1)),$$

and for $jL \equiv \pm 1 \pmod k$ also $\sigma_L(q) - s_*(q) \leq s_L^{\mathrm{pair}}(q) - s_*(q)$.

The constants are locally uniform: on every compact set of tunings on which $\bar{n}$ stays in one interval $(n_1, n_1 + k)$, and for every $\eta_0 > 0$, (a) *holds with $e = e_*(q) - \eta_0$ and* (b) *with $e = e_*(q) + \eta_0$ for all $L \geq L_0$, with L_0 independent of q.*

The proof is given in Appendix H.2.

The proof reduces all lag constraints to a trigonometric polynomial on the finite grid of phases attained by the periodic frequencies. The two zeros of its limiting polynomial require separate local estimates. At one zero the grid spacing controls the linear perturbation; at the other the leading constraints form a convex quadratic on $b + k\mathbb{Z}$. Balancing its two adjacent values gives e_*. Uniform Taylor bounds near the zeros and a strict minimum on their compact complement control every lag, including those increasing with L.

Negative contact curvature therefore gives a two-frequency threshold strictly smaller than the one-frequency threshold. Under the stated global sign conditions, positive curvature yields a quantitative L^{-2} lower bound for grids that omit the threshold frequency.

3.7. Gap boundaries and tongue junctions

The edges of I_K are the closing dip on $\partial\Omega_1$ and the adjacent tongue edges at $2\pi/(K+1)$ and $2\pi/K$; for $K = 2$ the latter endpoint is $\alpha L = 2(1+\beta)$. Near a tongue tip the single-harmonic spectrum persists even at a resonant slope index. Fix $K' \geq 3$, put $\omega_* = 2\pi/K'$, $c' = \pi/K'$, and use $s_*(q) = \cot c' - h_q(\omega_*)$. A tip satisfies $\Theta_q \geq \omega/2$ with equality only at ω_*.

Theorem 3.24 (Tips of the tongues)**.** *Let $K' \geq 3$ and let q_0 be a tip of the K'-tongue. There is a neighborhood V of q_0 such that for every $q \in V$:*

(a) *$\sigma(q) = \sigma_{K'}(q) = \cot(\pi/K') - h_q(2\pi/K')$; in particular σ is real analytic on V;*
(b) *the threshold extremal is $\delta_{2\pi/K'}$, up to scaling and an atom at 0;*
(c) *$\mathcal{D} \cap V = \{q \in V : \ \Theta_q(2\pi/K') \leq \pi/K'\}$, the closed tongue, and $(I_{K'-1} \cup I_{K'}) \cap V \subset \{\sigma > 0\}$;*
(d) *$(q, s_*(q))$ is K'-regular with support $\{2\pi/K'\}$ and the lag set $\{1, 1+K', 1-K'\}$.*

The proof is given in Appendix I.1.

At the other points of $\partial\Omega_1$, a strict finite certificate persists. Consequently $\operatorname{cl}\mathcal{B}_K$ stays away from $\partial\Omega_1$ (Corollary I.1). In the dip coordinates, convergence occupies a neighborhood of the diagonal side, while every curvature ray eventually becomes divergent.

Theorem 3.25 (The boundary in every gap)**.** *Let $K \geq 2$.*

(a) *The sets $I_K \cap \{\sigma > 0\}$ and $I_K \cap \{\sigma < 0\}$ are nonempty and open, and $\mathcal{B}_K = I_K \cap \{\sigma = 0\}$ separates them: every path in I_K from one to the other meets $\mathcal{B}_K$. In particular $\mathcal{B}_K \neq \emptyset$.*
(b) *In the coordinates (a, r_1, r_2) of Proposition E.7, every point (a, r, r) with $c_K < r < c_{K+1}$ and $a > a_0(r, r)$ has a neighborhood in which $\sigma > 0$; near the corners $r_1 = r_2 = c_{K+1}$ and, for $K \geq 3$, $r_1 = r_2 = c_K$, $\sigma > 0$ on I_K as well (Theorem 3.24(c)); and every ray $\{(a, r_1, r_2) : a > a_0(r_1, r_2)\}$ over a point of the triangle contains points with $\sigma < 0$.*
(c) *For every r with $c_K < r \leq c_{K+1}$, and also for $r = c_K$ if $K \geq 3$, there is $\delta > 0$ such that for all $c_K < r_1 < r_2 < c_{K+1}$ with $|r_1 - r|, |r_2 - r| < \delta$ the ray over (r_1, r_2) meets $\mathcal{B}_K$. Thus the projection of $\mathcal{B}_K$ to the triangle $c_K < r_1 < r_2 < c_{K+1}$ contains a relative neighborhood of its diagonal side, including the corners for $K \geq 3$ and the corner $r_1 = r_2 = c_3$ for $K = 2$.*

Proof. Fix $c_K < r < c_{K+1}$ and $a > a_0(r, r)$. By Proposition E.7, the point (a, r, r) is a nonlocked point of $\partial\Omega_1$. Corollary I.1 gives a neighborhood with $\sigma > 0$. Theorem 3.24(c) gives the same conclusion at the stated corners, with the stability endpoint excluded when $K = 2$. This proves the convergent-neighborhood assertions.

On a ray with $r_1 < r_2$, choose a reduced rational frequency $\omega_0 = 2\pi j/k$ with $r_1 < \cos\omega_0 < r_2$. Then

$$\Xi^{a,r_1,r_2}(\cos\omega_0) = 1 + \cos\omega_0 + a(\cos\omega_0 - r_1)(r_2 - \cos\omega_0) \longrightarrow +\infty.$$

For all sufficiently large a, the single-frequency condition $\Theta_{q,s}(\omega_0) \leq \pi/k$ therefore holds at some $s < 0$. Theorem F.3 gives $\sigma(q) \leq s < 0$. Both signs consequently occur in I_K, and both sign sets are open by Theorem 3.12. Continuity forces a zero on every path joining them, proving (a) and the remaining part of (b).

Finally fix a point of the diagonal side or one of the allowed corners, and choose a finite a strictly above its lower boundary. For all nearby ordered roots (r_1, r_2), the same a is admissible and belongs to the convergent neighborhood. Increasing a along each such ray eventually gives the negative sign just proved. The intermediate value theorem therefore gives a point of $\mathcal{B}_K$ on every such ray, which is (c). □

For $K \geq 3$, divergence also enters the gap from a nonempty arc where the two adjacent tongues meet. Proposition I.2 locates this arc explicitly by the negative curvature of the edge symbol. Together, these results describe the boundary in each gap: the closing-dip side has a convergent neighborhood, some tongue-junction arcs have divergent neighborhoods, and the separating surface contains an analytic single-frequency plateau for every Farey fraction in the gap window.

4. Conclusion

A common spectral interpolation problem gives both universal rate bounds and realizable obstructions. For Polyak's tuning it yields the exact rate and matching finite-horizon bounds, attained by one five-dimensional function initialized from rest. For general tunings it gives a continuous condition-number threshold, while Farey geometry and analytic continuation describe its regular plateaus. Single-harmonic edge certificates and competing fan spectra bound their extent; the mixed-harmonic examples exhibit the additional supports that a complete boundary classification must accommodate.

Several questions remain. Are the regular points dense in each threshold surface $\mathcal{B}_K$? Which mixed supports occur, and how do the plateaus accumulate along their relative boundaries? A more explicit description of the variational function a_F in Theorem 3.19 would sharpen the two available edge bounds. The exact point of $3/13$ also raises a classification problem: which Farey fractions admit a threshold extremal with a two-lag certificate? Proposition G.14 reduces this question to an explicit tangency equation and the remaining global sign conditions.

The boundary may also contain regimes outside periodic mode locking. Does some point of $\mathcal{B}_K$ have no periodic extremal? By Proposition G.5, a connected boundary set contains uncountably many such points if, after removing the zero atom and normalizing, every extremal has an atom of mass at least one fixed $m_0 > 1/2$ and the common location of these atoms is nonconstant. Further questions concern the gap geometry after imposing additional lag constraints, the smallest dimension of mixed-harmonic obstructions, and the number of frequencies needed to attain the rate inside the lag-one convergence region. In particular, does the tuning of Proposition 3.13 admit a nonconvergent scalar trajectory, and is the rate in $\Omega_1 \setminus \mathcal{Q}$ always attained by at most two frequencies?

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A. Interpolation and spectral characterization

A.1. Normalization and smooth interpolation

Throughout, $\alpha > 0$, $\beta \in [0, 1)$ and (3) hold, A and B are as in (10), and all Hilbert spaces are real. Since $B > 0$ and $A \pm B = 1 + \beta - \alpha\mu,\ 1 + \beta - \alpha L$, condition (3) gives

$$|A| + B = \max(|1 + \beta - \alpha\mu|, |1 + \beta - \alpha L|) < 1 + \beta. \tag{33}$$

Lemma A.1 (Normalization)**.** *The map $f \mapsto \psi$ is a bijection between functions in $\mathcal{F}_{\mu,L}$ with minimizer 0 and value 0 there, and $\mathcal{S}$. With $e_k = x_k - x^\star$ and $\sigma_k = \nabla\psi(e_k)$, the iteration* (1) *becomes* (10)*. For Polyak's tuning $A = 0$, $B = 2q$ and $\beta = q^2$.*

Proof of Lemma A.1. Put $M = L - \mu$ and $f_0 = f - \mu\|\cdot\|^2/2$. The convex M-smooth function f_0 has a $1/M$-cocoercive gradient [27]; hence $\nabla\psi = \mathrm{id} - 2\nabla f_0/M$ is nonexpansive. Conversely, $\nabla f = (L + \mu)\mathrm{id}/2 - M\nabla\psi/2$ is μ-strongly monotone and L-Lipschitz by Cauchy–Schwarz, so it defines a member of $\mathcal{F}_{\mu,L}$. The two transformations are inverse. Substitution gives (10); (4) gives $A = 0$, $B = 2q$ and $\beta = q^2$. □

Theorem A.2 (Interpolation on arbitrary sets)**.** *Let $\{(x_i, \sigma_i, \psi_i) : i \in I\}$ be data in a real Hilbert space containing the datum $(x_\star, \sigma_\star, \psi_\star) = (0, 0, 0)$ of an index $\star \in I$. A function $\psi \in \mathcal{S}$ with $\nabla\psi(x_i) = \sigma_i$ and $\psi(x_i) = \psi_i$ exists if and only if, for all $i, j \in I$,*

$$|T_{ij}| \le R_{ij}, \quad T_{ij} = \psi_j - \psi_i - \tfrac{1}{2}\langle\sigma_i + \sigma_j, x_j - x_i\rangle, \quad R_{ij} = \tfrac{1}{4}(\|x_j - x_i\|^2 - \|\sigma_j - \sigma_i\|^2). \tag{34}$$

Proof of Theorem A.2. Set $F_i = \psi_i + \|x_i\|^2/2$ and $G_i = \sigma_i + x_i$. Expanding (34) gives precisely

$$F_i \ge F_j + \langle G_j, x_i - x_j\rangle + \tfrac{1}{4}\|G_i - G_j\|^2 \quad (i, j \in I). \tag{35}$$

This is the convex $C^{1,1}$ extension criterion of Azagra and Mudarra [1, Theorem 1.3], with gradient Lipschitz constant two. It applies to arbitrary subsets of a real Hilbert space. Repeated sites give identical values and gradients by the same inequality. Thus there is a convex 2-smooth interpolant F; cocoercivity makes $\nabla F - \mathrm{id}$ nonexpansive, and $\psi = F - \|\cdot\|^2/2$ is the required interpolant. Necessity follows by applying the same convex smoothness inequality to $F = \psi + \|\cdot\|^2/2$. An explicit conjugate construction is

$$H(y) = \sup_{i\in I}\{\langle x_i, y\rangle - F_i + \tfrac{1}{4}\|y - G_i\|^2\}, \qquad \psi = H^* - \tfrac{1}{2}\|\cdot\|^2. \tag{36}$$

Indeed (35) makes the jth quadratic attain the supremum at G_j, with subgradient x_j; strong convexity of H gives the claimed smooth conjugate. □

Lemma A.3 (Exact spectral interpolation)**.** *For the orthogonal sum of waves with spectrum ν at the exponent t and function values $\psi_k = \frac{1}{4}(C + 2\int \mathrm{Re}\, m\, d\nu)e^{2kt}$,*

$$R_{j,j+m} \pm T_{j,j+m} = \tfrac{1}{2}e^{(2j+m)t}(\bar a_m \mp \bar b_m \pm C\sinh mt), \quad R_{j\star} \pm T_{\star j} = \tfrac{1}{4}e^{2jt}(\bar g \pm C). \tag{37}$$

Consequently the data are interpolated by a function in $\mathcal{S}$ if and only if (ν, C) is feasible at t.

Proof of Lemma A.3. For one wave and $j = 0$, the defining expressions reduce to $T_{0n} = \frac{C}{4}(e^{2nt} - 1) - s\ \mathrm{Im}\, z^n$ and $R_{0n} = \frac{g}{4}|z^n - 1|^2$. Insert $e^{2nt} - 1 = 2e^{nt}\sinh nt$ and $|z^n - 1|^2 = 2e^{nt}(\cosh nt - \cos n\omega)$. Shifting by j multiplies both expressions by e^{2jt}, and a minimizer pair gives $T_{\star j} = Ce^{2jt}/4$, $R_{j\star} = ge^{2jt}/4$. Orthogonal sums replace these expressions by their averages. For a general measure use the real Hilbert space $L^2([0, \pi], \nu; \mathbb{C})$ and $e_k(\omega) = e^{k(t+i\omega)}$. Theorem A.2 now gives both directions. □

A.2. Separation and trajectory growth

The next lemma collects the topological facts used below.

Lemma A.4 (Compactness)**.** *For fixed t, the set of pairs (ν, C) feasible at t is compact in the product of the weak topology on probability measures on $[0,\pi]$ and the real line. For Polyak's tuning the set of (v,t,ν,C) feasible in the sense of* (17) *with (v,t) in a compact set is compact, and feasibility implies $|\sinh v| \le 1$.*

Proof. Probability measures on $[0,\pi]$ are weakly compact. The minimizer constraint gives $|C| \le \bar{g} \le 1$, and every fixed-lag constraint is closed because its integrand is continuous, jointly with the parameters on compact sets. In the Polyak case, $0 \le \bar{g} = 1 - \int c^2 d\nu - \sinh^2 v \le 1 - \sinh^2 v$. □

For the upper bound we allow arbitrary initial states. A segment consists of positions $e_0, \dots, e_n$, gradients and values at $e_1, \dots, e_{n-1}$, and the minimizer datum; it satisfies all interpolation inequalities among these data and the recurrence (10) for $1 \le k < n$. Its state is $S_k = (e_{k-1}, e_k)$. Let

$$\Gamma(p) = \sup \frac{\|S_{k+p}\|}{\|S_k\|}, \quad \Gamma(0) = 1, \tag{38}$$

over all segments, admissible k and nonzero S_k.

Lemma A.5 (Growth envelope)**.** Γ *is finite and submultiplicative, the limit $\rho_\Gamma = \lim_p \Gamma(p)^{1/p} = \inf_{p\ge1} \Gamma(p)^{1/p}$ exists, $\rho_\Gamma \ge \rho_{\mathrm{q}} > 0$, and $W_N \le \sqrt{2}\,\Gamma(N)$ for $N \ge 0$. If (ν, C) is feasible at t, then $\Gamma(p) \ge e^{pt}$ for all p.*

Proof. The minimizer inequality gives $\|\sigma_k\| \le \|e_k\|$, hence

$$\begin{pmatrix} \|e_{k+p-1}\| \\ \|e_{k+p}\| \end{pmatrix} \le \begin{pmatrix} 0 & 1 \\ \beta & |A| + B \end{pmatrix}^p \begin{pmatrix} \|e_{k-1}\| \\ \|e_k\| \end{pmatrix}.$$

This proves finiteness. Restricting and concatenating ratios proves submultiplicativity; a zero state remains zero. The submultiplicative lemma gives the limit and infimum. Quadratic trajectories give $\rho_\Gamma \ge \rho_{\mathrm{q}} > 0$, and the prescribed initial state has norm $\sqrt{2}\|e_0\|$. A feasible spectrum has $\|S_k\|^2 = e^{2kt}(1 + e^{-2t})$, proving the last assertion. □

Lemma A.6 (Energy estimate)**.** *Suppose* (14) *holds at the exponent t with lags at most M'. There are $K > 0$ and an integer $w \ge M' + 2$ such that for every segment and $w \le i \le j \le n - w$,*

$$\sum_{k=i}^{j} \mathsf{b}_k \le K(E_i + E_j), \quad \mathsf{b}_k = e^{-2kt}\|e_k\|^2, \quad E_k = \sum_{|l-k|\le w} \mathsf{b}_l. \tag{39}$$

Proof. Let $I^{\pm}_{k,n} = R_{k,k+n} \pm T_{k,k+n}$, $I^{\pm}_{k\star} = R_{k\star} \pm T_{\star k}$ and form

$$Z_k = \sum_{n \le M'} 2e^{-nt}(\alpha_n I^{+}_{k,n} + \beta_n I^{-}_{k,n}) + 4\gamma_+ I^{+}_{k\star} + 4\gamma_- I^{-}_{k\star} \ge 0.$$

After $f_k = e^{-kt}e_k$, $h_k = e^{-2kt}\psi_k$ and the recurrence, $e^{-2kt}Z_k = \mathcal{Q}_k(f) + \mathcal{H}_k(h)$. The quadratic form couples positions at most $M' + 2$ steps apart and is invariant under time shifts; its value on $f_k = e^{ik\omega}$ is $\Phi_t(\omega)$. The value coefficients sum to zero by the second line of (14), so their sum has only endpoint terms. The minimizer inequality gives $|h_k| \le \|f_k\|^2/2$. For every finitely supported Hilbert-valued sequence F, Parseval yields

$$\sum_k \mathcal{Q}_k(F) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi_t(|\theta|) \|\widehat{F}(\theta)\|^2 \, d\theta \le -\varepsilon \sum_k \|F_k\|^2.$$

Apply this to $F = f\mathbf{1}_{[i-1, j+M'+1]}$. Every truncation error lies in fixed-width endpoint windows; $|\langle f_a, f_b\rangle| \le (\|f_a\|^2 + \|f_b\|^2)/2$ bounds their sum by $K_1(E_i + E_j)$. The value terms have the same bound, giving (39). □

Lemma A.7 (No strict certificate at the growth exponent)**.** *There is no strict certificate at $t = \log \rho_\Gamma$.*

Proof. Suppose there is one and put $r = \rho_\Gamma$, $\gamma(p) = \Gamma(p) r^{-p}$, $\bar\gamma(p) = \max_{0\le l\le p} \gamma(l)$. By Lemma A.5, $\gamma(p) \ge 1$ and $\bar\gamma(p)^{1/p} \to 1$. For $l \ge k \ge 1$, $\|e_l\| \le \|S_l\| \le \Gamma(l-k)\|S_k\|$, hence

$$\mathsf{b}_l \le c_r \gamma(l-k)^2 (\mathsf{b}_{k-1} + \mathsf{b}_k), \quad c_r = \max(1, r^{-2}). \tag{40}$$

Let K, w be as in Lemma A.6, $\theta = 1 + 1/(2K)$, $a = 2w+1$, and choose an integer p_0 with

$$\theta^{p_0} > 1 + p_0 a c_r \bar\gamma(p_0 a + w)^2, \tag{41}$$

which is possible by subexponential growth of $\bar\gamma$. Write $L' = p_0 a$.

First, if $i \ge w$, $j \ge i + 2w$ and $j \le n - L'$, then

$$B_i(j) := \sum_{k=i}^{j} \mathsf{b}_k \le 2K E_i. \tag{42}$$

Otherwise (39) gives $E_l > B_i(l)/(2K)$ for $j \le l \le n-w$. Since $E_l = B_i(l+w) - B_i(l-w-1)$, this gives $B_i(l+w) \ge \theta B_i(l-w-1)$, and iterating at $l = j + w + 1 + pa$, $0 \le p < p_0$, yields $B_i(j+L') \ge \theta^{p_0} B_i(j)$. On the other hand (40) with $k = j - w$ gives $B_i(j+L') \le B_i(j)\{1 + L' c_r \bar\gamma(L'+w)^2\}$, which contradicts (41).

Second, put $n' = n - L'$ and $\mathsf{T}_i = \sum_{k=i}^{n'} \mathsf{b}_k$. For $w \le i \le n' - 2w$, (42) gives $\mathsf{T}_i \le 2KE_i \le 2K(\mathsf{T}_{i-w} - \mathsf{T}_{i+w+1})$, hence $\mathsf{T}_{i+w+1} \le \theta^{-1}\mathsf{T}_{i-w}$, and $\mathsf{T}_{pa} \le \theta^{-p}\mathsf{T}_0$ as long as the windows fit. Moreover $\mathsf{T}_0 \le (1+2K)E_w \le C_w \|S_1\|^2$, with C_w depending only on $\Gamma(l)$, $l \le 2w$, and r. The last such window ends at distance at most $L' + 4w + 2$ from n; applying (40) over these finitely many steps gives $\|S_n\|^2 \le C r^{2n} \theta^{-n/a} \|S_1\|^2$ with C independent of the segment and of n. Hence $\rho_\Gamma \le r\theta^{-1/(2a)} < r$, a contradiction. □

A.3. Initialization from rest

Lemma A.8 (One-coordinate initialization)**.** *Let (ν, C) be feasible at t, with atoms $\omega_1, \dots, \omega_r$ and weights $\nu_j > 0$, and assume $|C| < \bar g$ if $t > 0$. Then there are $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^{2r+1})$ and x_0 with $x_{-1} = x_0$ such that $\|x_k - x^\star\| \ge c\, e^{kt} \|x_0 - x^\star\|$ for all $k \ge 0$, with $c > 0$ independent of k. The trajectory is the orthogonal sum of the r waves in $2r$ coordinates and a sequence $y_k \to 0$ in the last coordinate.*

Proof. Choose $\tau \in (-1, 1)$ and let $p_{-1} = p_0 = 1$, $p_{k+1} = (A + B\tau)p_k - \beta p_{k-1}$. By (33), $p_k \to 0$. Each p_k is a nonconstant polynomial in τ with leading coefficient B^k; avoid its countably many roots of $p_k = 1$. Then $\delta_0 = \inf_{k\ge1}(1 - p_k)^2 > 0$. In $\mathbb{C}^r \oplus \mathbb{R}$ prescribe

$$\begin{aligned} e_0 &= ((\sqrt{\nu_j})_j, Y), & \sigma_0 &= ((\sqrt{\nu_j}(z_j - A + \beta)/B)_j, \tau Y), \\ e_k &= ((\sqrt{\nu_j} z_j^k)_j, p_k Y), & \sigma_k &= ((\sqrt{\nu_j} m_j z_j^k)_j, \tau p_k Y) \quad (k \ge 1), \end{aligned}$$

with $\psi_0 = \langle \sigma_0, e_0 \rangle/2$ and the stationary wave values plus $\tau p_k^2 Y^2/2$ for $k \ge 1$. These data satisfy the recurrence from rest. The scalar quadratic contributes $T = 0$ and $R = (1-\tau^2)(y_i - y_j)^2/4$ to every pair. All pairs excluding zero already interpolate. The pair $(\star, 0)$ gains $(1-\tau^2)Y^2/4$; every $(0, k)$ gains at least $(1-\tau^2)\delta_0 Y^2/4$. For $t \le 0$ their wave deficits are uniformly bounded. For $t > 0$, the stationary margin is at least $(\bar g - |C|)e^{2kt}/4 - O(e^{kt})$, while altering the initial gradient changes it by only $O(e^{kt}) + O(1)$. Thus their negative parts remain uniformly bounded in both cases. A single sufficiently large Y makes every pair interpolate. Theorems A.2 and A.1 yield one f and one trajectory, with $\|e_k\| \ge e^{kt}$ and $\|e_0\| = \sqrt{1 + Y^2}$. □

A.4. Feasibility for Polyak's tuning

We turn to Polyak's tuning, treating the relative exponent v, also called the twist, and the exponent t as independent parameters. Here (ν, C) is feasible at (v, t) if (13) holds with the data (16).

Lemma A.9 (Symmetries and monotonicity in the twist)**.** *For Polyak's tuning:*

(i) $(t, C) \mapsto (-t, -C)$ *at fixed* (v, ν) *preserves feasibility;*

(ii) $(v, C) \mapsto (-v, -C)$ *at fixed* (t, ν) *interchanges the lower and upper constraints and preserves feasibility;*

(iii) *if* (ν, C) *is feasible at* (v, t) *and* $0 \le v' \le v$*, then* $(\nu, \theta C)$ *with* $\theta = \sinh v'/\sinh v$ *(and* $\theta = 1$ *if* $v = 0$*) is feasible at* (v', t)*.*

Consequently the set of v *feasible at* t *is* $[-V(t), V(t)]$*,* V *is even, and* $V \le \operatorname{arsinh} 1$*.*

Proof. (i) a_m and g are even in t, b_m does not depend on t, and $\sinh mt$ is odd. (ii) a_m and g are even in $u = \sinh v$ and b_m is odd. (iii) The case $v = 0$ is immediate. Otherwise, with $u' = \sinh v' = \theta u$, $b_m(v') = \theta b_m(v)$ and

$$a_m(v') = a_m(v) + (u^2 - u'^2)(T_m(G) - T_m(c)) \ge a_m(v),$$

because $T_m(G) \ge 1 \ge T_m(c)$. Hence $\ell_m(v') = \theta \ell_m(v) + (1-\theta)\bar{a}_m(v) + (\bar{a}_m(v') - \bar{a}_m(v))$, and $\bar{a}_m(v) = \frac{1}{2}(\ell_m + \upsilon_m) \ge 0$; the same holds for υ_m. The minimizer slack satisfies $\bar{g}(v') \ge \bar{g}(v) \ge |C| \ge \theta|C|$. The consequences follow from (ii), (iii), closedness, and Lemma A.4. □

Lemma A.10 (Finite support and inactive tails)**.** *For Polyak's tuning let* $T = T(q) \ne 0$ *and* $v = T - \log q$*. Every optimal pair* (ν, C) *at* (v, T) *has* ν *finitely supported in* $(-1, 1)$ *and* $|C| < \bar{g}$*, and only finitely many lag constraints are active.*

Proof. The feasible spiral satisfies $\rho_{\mathrm{sp}} > q$ by Proposition 2.7 and the root calculation following its proof. Consequently $T \ge \log \rho_{\mathrm{sp}} > \log q$, so $v = T - \log q > 0$. Divide the lag functions $a_m \mp b_m$ by $U_{m-1}(\cosh t)$, so that C enters every normalized lag constraint with the coefficient $\pm \sinh t$, and represent the minimizer constraints, multiplied by $\sinh |t|$, as the limits $m = \infty$: for $t \ne 0$ the normalized lag functions converge uniformly on $[-1, 1]$, as $m \to \infty$, to $g \sinh |t|$. For $t > T$ the points $(t - \log q, t)$ are infeasible by the definition of $T(q)$, and Theorem 2.5(a) gives strict certificates. Normalized to total weight one, the certificates at a sequence $t_n \downarrow T$ have a weakly convergent subsequence on the compact index set $\{\mathsf{L}_m, \mathsf{U}_m : m \in \mathbb{N} \cup \{\infty\}\}$. The limit is a nonpositive symbol Φ_T, with equal total lower and upper weight, and complementary slackness places every optimal measure in the zero set of Φ_T.

In the complex strip $|\operatorname{Im} \omega| < |T|$ the trigonometric parts of the normalized lag functions decay geometrically in m, so Φ_T is analytic there. At $c = \pm 1$ every normalized lag functional is at most $-\sinh^2 v \sinh |T| \tanh(|T|/2) < 0$, so Φ_T is not identically zero; it has finitely many zeros in $[-1, 1]$, all interior. Thus ν is finitely supported in $(-1, 1)$.

Suppose $|C| = \bar{g}$. Part (i) of Lemma A.9 maps (T, C) to $(-T, -C)$, so we may take $T > 0$. Put $u = \sinh v$ and $z_j = -\nu_j(g_j + 2iu \sin \omega_j)$, so that

$$f(m) := \operatorname{Re} \sum_j z_j e^{im\omega_j} = \sum_j \nu_j \{2u \sin \omega_j \sin m\omega_j - g_j \cos m\omega_j\}.$$

If $C = \bar{g}$, the upper constraints give $f(m) \ge -\bar{g} e^{-mT}$; if $C = -\bar{g}$, the lower constraints give the same bound for the function obtained by negating the sine term, that is, with z_j replaced by its conjugate, and the argument below applies verbatim. The atoms are distinct points of $(0, \pi)$, so the Cesàro means satisfy $N^{-1} \sum_{m \le N} f(m) \to 0$ and $N^{-1} \sum_{m \le N} f(m)^2 \to \frac{1}{2} \sum_j |z_j|^2$. Here $|z_j|^2 = \nu_j^2(g_j^2 + 4u^2 \sin^2 \omega_j) > 0$: for $u \ne 0$ the second term is positive, and for $u = 0$ one has $g_j = \sin^2 \omega_j > 0$. The lower bound makes the Cesàro mean of the negative part f^- tend to zero; together with $N^{-1} \sum f(m) \to 0$, the means of $|f|$ tend to zero, and since f is bounded the means of f^2 tend to zero, a contradiction. Hence $|C| < \bar{g}$.

Finally, with $\eta = \bar{g} - |C| > 0$ every raw lag slack is at least

$$\eta \cosh(m|T|) - (1 + u)^2, \tag{43}$$

since $|g(c)| \le 1 + u^2$ and $|2u \sin \omega \sin m\omega| \le 2u$; so only finitely many lags are active. □

Proof of Theorem 2.6. Lemma A.9 gives the description of the feasible v and the bound on V. Lemma A.4 shows that the maximum defining $T(q)$ is attained; the spiral of Proposition 2.7 is feasible, so the set is nonempty. For $t > T(q)$, $|t - \log q| > V(t)$, so the actual point $(t - \log q, t)$ is infeasible and a strict certificate exists by Theorem 2.5(a); Theorem 2.5(b) gives $\rho_\Gamma < e^{t_0}$ and $\rho_{\mathrm{HB}} < e^{t_0}$ for every $t_0 > T(q)$, hence $\rho_{\mathrm{HB}} \le e^{T(q)}$; together with Lemma A.5,

$$\rho_\Gamma = e^{T(q)}. \tag{44}$$

At $t = T(q)$ the actual point is feasible. If $T(q) \neq 0$, Lemma A.10 gives a finitely supported optimal spectrum with $|C| < \bar{g}$, and Lemma A.8 gives $\rho_{\mathrm{HB}} \ge e^{T(q)}$. If $T(q) = 0$, then $q = \varphi^{-1}$ by Propositions B.6 and 2.7, and the three-cycle of Proposition 2.7 is a one-atom feasible spectrum at $t = 0$, so Lemma A.8 applies again. The remaining assertions are Lemma A.10. □

B. The spectral classification for Polyak's tuning

Throughout, the parameters are those of Polyak's tuning, T_m and U_m denote Chebyshev polynomials, U_m denotes the normalized upper functional of (45), and $G = \cosh t \ge 1$, $u = \sinh v$, $c = \cos\omega$. The raw slacks of a spectrum are $\ell_m = \bar{a}_m - \bar{b}_m + d\,U_{m-1}(G)$ and $\upsilon_m = \bar{a}_m + \bar{b}_m - d\,U_{m-1}(G)$, where $d = C\sinh t$, so that $d\,U_{m-1}(G) = C\sinh mt$. All finite decimals in this appendix denote exact rationals. The computer-assisted steps are exact polynomial identities, Sturm counts and signs of rational numbers, and, for two arcs, interval inclusions in outward-rounded arithmetic [14, 21].

B.1. Duality and the four contact systems

Normalize the lag functionals and the constant C by

$$\mathsf{L}_m(c) = \frac{a_m(c) - b_m(c)}{U_{m-1}(G)}, \quad \mathsf{U}_m(c) = \frac{a_m(c) + b_m(c)}{U_{m-1}(G)}, \quad d = C\sinh t, \tag{45}$$

so that the constraints read $\bar{\mathsf{L}}_m + d \ge 0$ and $\bar{\mathsf{U}}_m - d \ge 0$. For nonnegative weights w_m, $m \in J$, with $\sum_{m\in J} w_m = 1$, the combination

$$\Phi(c) = \mathsf{L}_1(c) + \sum_{m\in J} w_m \mathsf{U}_m(c), \quad Q(c) = -\frac{\Phi(c)}{G - c}, \tag{46}$$

cancels d, and Q is a polynomial in c. Its structure is the basis of the classification.

Lemma B.1 (Dual factorization)**.** *For $G \ge 1$ and $-1 \le c \le 1$,*

$$\Phi(c) = (G - c)\{(1 - c^2 - u^2)\,\alpha_J - 2u(1 - c^2)\,\beta_J\}, \tag{47}$$

where $\alpha_J \ge 1$ and $\beta_J \ge 0$ are divided differences of Chebyshev polynomials that depend on G, c and the weights and are independent of u. In particular, for $u' \ge u \ge 0$,

$$\Phi_{G,u'}(c) \le \Phi_{G,u}(c) - (G - c)(u'^2 - u^2).$$

The proof is given in Appendix B.3.

Thus the dual symbol decreases strictly with $u = \sinh v$. A feasible measure supported on the zeros of a nonnegative Q at the same point (G, u) is therefore optimal: increasing u makes the certificate strict. This primal–dual pairing reduces the boundary $u = \sinh V(t)$ to finitely many polynomial equations once the active lags are known.

B.2. Polynomial contact systems

Each family is described by polynomial contact equations: active constraints are equalities, and interior zeros of the nonnegative polynomial Q also satisfy $Q' = 0$.

(i) The spiral family S, given by (18). Its active constraints are $\mathsf{L}_1, \mathsf{U}_2, \mathsf{U}_3$ when $t \neq 0$.

(ii) The single-frequency family P_1. Take $J=\{3\}$ in (46) and solve

$$Q(c)=0, \quad Q'(c)=0, \tag{48}$$

on the analytic branch with $u\to 1$ and $c\to 0$ as $G\to\infty$. Set $\nu=\delta_c$ and $d=-\mathsf{L}_1(c)$.

(iii) The two-frequency family P_2. Take $J=\{2,3\}$ with weights $(W,1-W)$ and $\nu=(1-p)\delta_x+p\delta_y$, $x<y$, and solve

$$\bar{\mathsf{L}}_1+d=0, \quad \bar{\mathsf{U}}_2-d=0, \quad \bar{\mathsf{U}}_3-d=0, \quad Q(y)=Q'(x)=Q'(y)=0. \tag{49}$$

Its values of (G,u) lie on the curve

$$\begin{aligned} P_2(G,u) = {} & 256G^4(u^2-1)-512G^3u^3+768G^3u \\ & -32G^2u^4-480G^2u^2-16Gu^3-u^4-16u^2=0, \end{aligned} \tag{50}$$

which is rational:

$$G=\frac{\zeta^4+4}{8\zeta(\zeta^2-2)}, \quad u=\frac{(\zeta^2-2\zeta+2)^2}{\zeta^4-4}. \tag{51}$$

(iv) The central family P_5. Take $J=\{2,3,5\}$ with weights $(w_2,w_3,1-w_2-w_3)$ and solve

$$\bar{\mathsf{L}}_1+d=0, \quad \bar{\mathsf{U}}_m-d=0 \ (m=2,3,5), \quad Q(y)=Q'(x)=Q'(y)=0. \tag{52}$$

In the two-frequency systems the condition $Q(x)=0$ follows from the primal equations and $p<1$. Every coefficient is obtained from (16)–(46).

B.3. The spiral and the sharp threshold

For one frequency the equations $\ell_1=\upsilon_2=\upsilon_3=0$ are homogeneous and linear in $(g,C,2u(1-c^2))$. With $\varrho=e^t$, their determinant factors as

$$\varrho^{-4}(\varrho^2-1)(\varrho^2-2c\varrho+1)^2\{c(\varrho^2+1)+\varrho\}.$$

For $t\neq 0$ it follows that $c=-1/(2G)$. The first two equations give $C=g\tanh t$ and $g=u/G$. Substitution into $g=1-c^2-u^2$ selects the positive root $u=1-1/(2G)$:

$$u=1-\frac{1}{2G}, \quad c=-\frac{1}{2G}, \quad g=\frac{2G-1}{2G^2}, \quad C=g\tanh t. \tag{53}$$

Lemma B.2 (Lag identities on the spiral)**.** *On the data* (53)*, for every $m\geq 1$, $\ell_m=gE_{m+1}(t)/G$ and $\upsilon_m=gE_{m-1}(t)/G$, where $E_n(t)=\cosh nt+2G^2\cos n\omega$. Moreover $E_0=1+2G^2$ and $E_1=E_2=0$.*

Proof. The relations $g=u/G$, $c=-1/(2G)$ give $b_m/g=\sqrt{4G^2-1}\sin m\omega$. Apply the addition formulas to $\cosh mt\pm\tanh t\sinh mt=\cosh(m\pm 1)t/G$ and $-\cos m\omega\mp\sqrt{4G^2-1}\sin m\omega=2G\cos(m\pm 1)\omega$. The values at $n=0,1,2$ follow immediately. □

Lemma B.3 (Positivity of the spiral kernel)**.** *Let $s=G^2-1$ and $F_n=G^{n-2}E_n$. For every $n\geq 3$, F_n is a polynomial in s with nonnegative coefficients, $F_n(0)=3$ if $3\mid n$ and $F_n(0)=0$ otherwise, and $F_n(s)>0$ for $s>0$. In particular $E_n(t)>0$ for $n\geq 3$ and $t\neq 0$.*

Proof. Put $X=G^2=1+s$. The generating function $\sum_{n\geq 0}T_n(x)r^n=(1-xr)/(1-2xr+r^2)$, applied with $x=G$ and $x=-1/(2G)$ and the variable Gr, gives

$$\sum_{n\geq 0}F_nr^n=\frac{1-Xr}{X(1-2Xr+Xr^2)}+\frac{2+r}{1+r+Xr^2}.$$

Since $F_1=F_2=0$ and $F_0=(1+2X)/X$, a direct computation gives

$$\sum_{n\geq 3}F_nr^n=\frac{(2X+1)\,r^3\,(2X-1-Xr)}{(1+r+Xr^2)(1-2Xr+Xr^2)}=\frac{(2s+3)\,r^3\,\{(1-r)+s(2-r)\}}{D},$$

where $D = (1 + r + Xr^2)(1 - 2Xr + Xr^2) = D_0 - sD_1 - s^2 D_2$ with

$$D_0 = (1-r)(1-r^3), \quad D_1 = r(2 + 3r^2 - 2r^3), \quad D_2 = r^3(2-r).$$

As formal power series in r with coefficients in $\mathbb{Q}[s]$,

$$\frac{(1-r) + s(2-r)}{D} = \sum_{k \geq 0} \Big(s\frac{D_1}{1-r} + s^2 \frac{D_2}{1-r} \Big)^k \frac{1}{(1-r^3)^{k+1}} \Big(1 + s\frac{2-r}{1-r} \Big),$$

and the series converges coefficientwise because D_1 and D_2 have no constant term. Each factor has nonnegative coefficients: $D_1/(1-r) = 2r + 2r^2 + 5r^3 + 3\sum_{j \geq 4} r^j$, $D_2/(1-r) = 2r^3 + \sum_{j \geq 4} r^j$, $(2-r)/(1-r) = 2 + \sum_{j \geq 1} r^j$, and $1/(1-r^3) = \sum_j r^{3j}$. Hence every F_n, $n \geq 3$, has nonnegative coefficients in s. The terms with $k \geq 1$ are divisible by s. The term $k = 0$ contributes to F_n the polynomial $(2s+3)\{[3 \mid n] + s\, c_n\}$, where c_n is the coefficient of r^{n-3} in $(2-r)/((1-r)(1-r^3))$, and $c_n \geq 1$. This gives the value at $s = 0$ and strict positivity for $s > 0$. Finally $s = \sinh^2 t$. □

Proof of Proposition 2.7. By Lemmas B.2 and B.3, $\ell_1 = \upsilon_2 = \upsilon_3 = 0$, $\upsilon_1 = g(1 + 2G^2)/G > 0$, and all other raw slacks are positive multiples of E_n, $n \geq 3$, hence positive for $t \neq 0$. Moreover $g > 0$ and $|C| = g|\tanh t| < \bar{g} = g$. This is (19) and the feasibility. On the spiral, $1 - \frac{1}{2G} = (\varrho^2 - \varrho + 1)/(\varrho^2 + 1)$ with $\varrho = e^t$, and $\sinh(t - \log q) = \frac{1}{2}(\varrho/q - q/\varrho)$; equating the two and multiplying by $2q\varrho(\varrho^2 + 1)$ gives $F(\varrho, q) = 0$. □

The positive root of (5) is unique. Write $F(\rho, q) = (\rho - q)^2(\rho^2 + 1) - 2q\{q - (1-q)\rho^2\}$. For $\rho > q$ the first term increases and the subtracted expression does not increase; at $\rho = q$ the value is negative and at infinity it is positive. For $0 < \rho \leq q$ the first term is smaller than $q^2(1 + q^2) \leq 2q^2(1 - q + q^2)$, which does not exceed the subtracted expression. Thus the positive root ρ_{sp} is unique, exceeds q, and satisfies $F_\rho > 0$ and $F_q < 0$, so ρ_{sp} is analytic and strictly increasing in q. At $\rho = 1$ the equation reads $2(1 - q - q^2) = 0$, so $\rho_{\mathrm{sp}} = 1$ exactly at $q = \varphi^{-1}$, that is, at $\kappa = \kappa_c$.

The next lemma controls all large lags at once.

Lemma B.4 (Amplitude identity and tail bound)**.** *For $c = \cos\omega \in [-1, 1]$ and $u \in \mathbb{R}$, $g(c)^2 + (2u \sin\omega)^2 = (1 - c^2 + u^2)^2$. Consequently, for every spectrum and every $m \geq 1$,*

$$\ell_m, \ \upsilon_m \ \geq \ (\bar{g} - |C|)\cosh mt - (\bar{g} + 2u^2). \tag{54}$$

Proof. With $k = 1 - c^2$, $(k - u^2)^2 + 4u^2 k = (k + u^2)^2$. For one frequency, $a_m \mp b_m = g\cosh mt - \{g\cos m\omega \pm 2u \sin\omega \sin m\omega\}$, and by the identity and the Cauchy–Schwarz inequality the braces are at most $1 - c^2 + u^2$ in absolute value. Integrating gives $\bar{a}_m \mp \bar{b}_m \geq \bar{g}\cosh mt - (\bar{g} + 2u^2)$, and $|C \sinh mt| \leq |C|\cosh mt$. □

Proof of Lemma B.1. Divided differences give

$$\alpha_J = 1 + \sum_m w_m \frac{T_m(G) - T_m(c)}{(G-c)U_{m-1}(G)}, \qquad \beta_J = \sum_m w_m \frac{U_{m-1}(G) - U_{m-1}(c)}{(G-c)U_{m-1}(G)}.$$

The quotients by $G - c$ are polynomials. Since $T_m(G) \geq 1 \geq T_m(c)$ and $U_{m-1}(G) \geq m \geq U_{m-1}(c)$, they have the required signs on $G \geq 1$, $c \in [-1, 1]$. Subtracting the two symbols proves the monotonicity inequality. □

A feasible measure supported on the zeros of $Q \geq 0$ is optimal. Indeed increasing u gives a strictly negative symbol when $G > 1$. At $G = 1$ its upper bound is $-(1-c)S(c)$ with $\min S > 0$. Adding equal minimizer weights ϵ yields

$$-(1-c)S + 2\epsilon g \leq -(1-c)(\min S - 4\epsilon) - 2\epsilon u'^2 < 0 \quad (0 < 4\epsilon < \min S),$$

which gives a strict certificate there as well.

For the spiral and $J = \{2, 3\}$, the weight that makes the zero of the symbol at $c = -1/(2G)$ double is

$$W(G) = \frac{2(1 + 2G - 2G^2)}{4G + 1}, \quad 1 - W(G) = \frac{4G^2 - 1}{4G + 1},$$

and an exact factorization gives

$$\Phi(c) = -\frac{(G - c)(2Gc + 1)^2}{2G^3(4G + 1)}\Big\{2G\Big(c - \frac{2G - 1}{4G}\Big)^2 + \frac{32G^3 - 36G^2 + 4G - 1}{8G}\Big\}. \tag{55}$$

Equivalently,

$$\Phi(c) = -\frac{(G - c)(2Gc + 1)^2}{2G^3(4G + 1)}\{2Gc^2 - (2G - 1)c + 4G(G - 1)\},$$

with $2Gc^2 - (2G - 1)c + 4G(G - 1) = 2G\big(c - \frac{2G-1}{4G}\big)^2 + \frac{32G^3-36G^2+4G-1}{8G}$. The weight is nonnegative exactly for $G \le G_W = (1 + \sqrt{3})/2$, and the residual quadratic is nonnegative on $[-1, 1]$ exactly for $G \ge G_1$, the root of $32G^3 - 36G^2 + 4G - 1$ in $(1.03329, 1.03330)$; its vertex $(2G - 1)/(4G)$ lies in $(0, 1)$. Hence the spiral is optimal at fixed G for $G_1 \le G \le G_W$. At G_1 the certificate acquires a second double zero, at G_W the weight of U_2 vanishes; these are the two transitions continued below.

Proposition B.5 (The spiral outside its regime)**.** *Let $q \in (0, 1)$, let $t_{\mathrm{sp}} = \log \rho_{\mathrm{sp}}(\kappa) \neq 0$, and let $G = \cosh t_{\mathrm{sp}}$. If $G > G_W$ or $1 < G < G_1$, then $\rho_{\mathrm{HB}} > \rho_{\mathrm{sp}}$.*

Proof. Let $f = (\ell_1, \upsilon_2, \upsilon_3)$ at the spiral and $\mathsf{J} = [\partial_\omega f, \partial_C f]$. The vector $y = (1, W/U_1(G), (1 - W)/U_2(G))$ annihilates both columns. Their independence follows from

$$\partial_\omega(\upsilon_2/U_1 - \upsilon_3/U_2) = -\sin\omega\,\frac{(4G + 1)(2G^2 + 1)}{2G^3(2G + 1)} \neq 0;$$

thus $\operatorname{im}\mathsf{J} = y^\perp$. For $G > G_W$, $W < 0$, so the entries of y have both signs. Choose $r > 0$ with $y^T r = y^T \partial_t f$ and solve $\mathsf{J}\Delta = r - \partial_t f$. Increasing t and changing (ω, C) in direction Δ makes all three contacts positive. For $1 < G < G_1$, $y > 0$, but (55) is positive at $c_0 = (2G - 1)/(4G)$. Adding mass $\epsilon\tau$ at c_0 gives a direction d_0 with $y^T d_0 = \Phi(c_0) > 0$. Choose ϵ so that $y^T(\partial_t f + \epsilon d_0) > 0$ and solve the same equation with a positive r and right side $r - \partial_t f - \epsilon d_0$. All other fixed-lag and minimizer slacks stay positive. Since $t \neq 0$, (54) handles the remaining lags uniformly in a neighborhood. The perturbed measure is feasible at a larger exponent; Theorem 2.6 gives the strict inequality. □

The threshold is located by an inequality that combines two constraints.

Proposition B.6 (Threshold strip)**.** *Every point (v, t) at which (13) is feasible satisfies*

$$|v| \le |t| + \log\varphi, \tag{56}$$

strictly if $t \neq 0$. At $t = 0$ the three-cycle $\nu = \delta_{-1/2}$, $C = 0$, is feasible at $v = \log\varphi$.

Proof. By symmetry take $v \ge 0$ and put $\varrho = e^{|t|}$. Add $|\sinh t|$ times the appropriate minimizer inequality to ℓ_1. The result is $\int h_\varrho\, d\nu \ge 0$, where

$$h_\varrho(c) = (1 - c^2 - u^2)(\varrho - c) - 2u(1 - c^2).$$

It decreases with $u \ge 0$. At $u_\varrho = \sinh(|t| + \log\varphi)$, substitute $c = 1 - 2X$, $\varrho = 1 + Y$. Write $-4\varphi^2\varrho^2 h_\varrho = A_3X^3 + A_2X^2 + A_1X + A_0$. Expansion gives $A_3 = 16(3 + \sqrt{5})(1 + Y)^2 > 0$; A_0 and $4A_1A_3 - A_2^2$ are polynomials in Y with zero constant term and positive remaining coefficients. Therefore

$$A_3X^3 + A_2X^2 + A_1X + A_0 = A_3X\left(X + \frac{A_2}{2A_3}\right)^2 + \frac{4A_1A_3 - A_2^2}{4A_3}X + A_0$$

is positive for $Y > 0$, $X \in [0, 1]$. This excludes $u \ge u_\varrho$ when $t \neq 0$. For $t = 0$, $u = 1/2$ gives $h_1(c) = -(1 - c)(c + 1/2)^2$; increasing u makes it negative on $[-1, 1)$, forcing $\nu = \delta_1$, contrary to $\bar{g} \ge 0$. Finally $\delta_{-1/2}$, $C = 0$ is feasible because its slacks are gE_n, with $g = 1/2$ and $E_n = 1 + 2\cos(2\pi n/3) \in \{0, 3\}$. □

Proof of Theorem 2.2. If $q < \varphi^{-1}$, then $\log q < -\log\varphi$, and every $t \ge 0$ has $t - \log q > t + \log\varphi$, so by Proposition B.6 the actual point $(t - \log q, t)$ is infeasible. Hence $T(q) < 0$. At $t = T(q)$, $v = T - \log q \le |T| + \log\varphi = -T + \log\varphi$ gives $T \le \frac{1}{2}\log(\varphi q)$, so $\rho_{\mathrm{HB}} = e^{T(q)} \le \sqrt{\varphi q} < 1$ by Theorem 2.6. If $q = \varphi^{-1}$, the three-cycle is feasible at the actual point $(\log\varphi, 0)$, and for $t > 0$ the actual point has $v = t + \log\varphi$, which Proposition B.6 excludes; thus $T(q) = 0$ and $\rho_{\mathrm{HB}} = 1$. If $q > \varphi^{-1}$, then $\rho_{\mathrm{sp}} > 1$, and the spiral is feasible at the actual point with $t = \log\rho_{\mathrm{sp}} > 0$, so $\rho_{\mathrm{HB}} \ge \rho_{\mathrm{sp}} > 1$. The trajectory statements are the case $\rho_{\mathrm{sp}} \ge 1$ of (7), proved in Appendix C: at κ_c the spiral is the three-cycle, and the realized trajectory consists of the cycle and a coordinate $2(-q^2)^{\lceil k/2\rceil} \to 0$. □

B.4. Continuation of the algebraic families

The remaining three families have algebraic descriptions. For P_1 write c_1 for the atom, $\nu = \delta_{c_1}$, and use the coordinates $z = G^{-1}$ and

$$u = 1 - \frac{z}{2} + z^2\mathcal{A}, \quad c_1 = z^2\mathcal{B}. \tag{57}$$

For $J = \{3\}$, multiplying Q by its positive parameter denominator gives the quartic

$$Q_1(c) = -\{(1 - c^2 - u^2)[2 + zc + z^2(c^2 - 1)] - 2uz(1 - c^2)(1 + zc)\}, \tag{58}$$

and (48) states that c_1 is a double root of Q_1. After (57), both $Q_1(c_1)/z^2$ and $Q_1'(c_1)/z^2$ are polynomials in $(z, \mathcal{A}, \mathcal{B})$. At $z = 0$ their solution is $(\mathcal{A}, \mathcal{B}) = (\frac{1}{8}, -\frac{1}{4})$, and the Jacobian in $(\mathcal{A}, \mathcal{B})$ is nonsingular there. For P_5 put $h = G - 1$ and

$$u = \tfrac{1}{2} + h\mathcal{A}, \quad x = -\tfrac{1}{2} + h\mathcal{B}, \quad p = hP, \quad d = hD, \tag{59}$$

with unknowns $(\mathcal{A}, \mathcal{B}, y, P, D, w_2, w_3)$. Dividing each of the four primal equations of (52) by h before clearing denominators, and keeping the three dual equations, gives a polynomial system $F(h, Z) = 0$, $Z = (\mathcal{A}, \mathcal{B}, y, P, D, w_2, w_3)$, which is regular at $h = 0$.

For the two continuations use the parametric Krawczyk operator [21, §8.2]. For a polynomial system $F(s, Z) = 0$, parameter interval I, centered box X, and nonsingular preconditioner R, check

$$Z_0 - RF(I, Z_0) + (\mathrm{Id} - RF_Z(I, X))(X - Z_0) \subset \operatorname{int} X, \qquad \|\mathrm{Id} - RF_Z(I, X)\| < 1. \tag{60}$$

The norm is a weighted infinity norm. The preconditioned fixed-point map is a contraction on X for every $s \in I$; its unique root is analytic. At a shared endpoint, inclusion of the left root enclosure in the right box identifies the branches. Derivatives follow from $Z' = (\mathrm{Id} - \mathcal{M})^{-1}(-RF_s)$, with $\mathcal{M} = \mathrm{Id} - RF_Z$; the Neumann remainder is bounded by $\|\mathcal{M}\|\|RF_s\|/(1 - \|\mathcal{M}\|)$.

Table 3: Domains of the verified continuations. Each row uses 4000 adjacent parameter slabs; the small extension beyond G_a allows exact endpoint identification.

Family	Parameter interval	Unknowns	Equations
P_1	$0 \le z \le 733/1000$	$(\mathcal{A}, \mathcal{B})$	(48), (57)
P_5	$0 \le h \le 0.023582$	$(\mathcal{A}, \mathcal{B}, y, P, D, w_2, w_3)$	(52), (59)

Proposition B.7 (The family P_1)**.** *The continued branch of P_1 on $0 < z \le z_W = \sqrt{3} - 1 = G_W^{-1}$ consists of feasible and optimal spectra: $\nu = \delta_{c_1}$ with $-z/2 \le c_1 < 0$, $\eta = g - |C| \ge z^3/4$, $Q_1(c) = (c - c_1)^2(q_2c^2 + q_1c + q_0)$ with $q_2 = z^2$ and $q_0 - |q_1| > 0$, and all lag slacks nonnegative. At z_W the branch equals the spiral, $\mathcal{A} = 0$, $\mathcal{B} = -1/(2z_W)$. Moreover u increases in G, $\frac{1}{2} < u < 1$, and $|V'| < 1$ along the branch.*

Proof. The continuation verifies, on every slab, $\mathcal{B} < 0$, $g/z > 1/2$, $u < 1$, $u_z < 0$, $(1/2 + z\mathcal{B})' < 0$, $q_0 - |q_1| > 0$ and $u_z^2 z^2(1 - z^2) < 1 + u^2$. Exact substitution at z_W identifies the spiral endpoint; the derivative signs give $-z/2 \le c_1 < 0$ and $1/2 < u < 1$. Put $\mathsf{D} = 2 + zc_1 + z^2(c_1^2 - 1)$, $\mathsf{E} = 1 + zc_1 + z^2(2c_1^2 - 1)$.

The contact equation gives $g = 2uz(1-c_1^2)(1+zc_1)/\mathsf{D}$ and $d = 2u(1-c_1^2)\mathsf{E}/\mathsf{D}$. Thus $|C| \le g\sqrt{1-z^2}$ and $\eta \ge gz^2/2 \ge z^3/4$. After multiplication by positive factors the raw slacks become

$$z^{m-1}\{z(1+zc_1)[T_m(1/z) - T_m(c_1)] \pm \mathsf{D}U_{m-1}(c_1) \mp \mathsf{E}U_{m-1}(1/z)\}. \tag{61}$$

The contacts vanish identically. For $m = 2$ the upper expression is $-2z(2c_1+z)(c_1z-1) \ge 0$. For the other slacks with $m \le 12$, substitute $c_1 = -\theta z/2$ and remove z^2 from upper slacks with $m \ge 4$; their tensor Bernstein coefficients on $[0, 3/4] \times [0, 1]$ are positive. The basis functions are nonnegative and sum to one, so this certifies the sign throughout the box. For $m \ge 13$, (43) gives the lower bound $z^{3-m}/4 - 4 > 0$. The nonnegative dual factor proves optimality, and $V' = (-z^2u_z)\sinh t/\sqrt{1+u^2}$ gives $|V'| < 1$. □

B.5. The rational two-frequency family

The family P_2 is treated in closed form: its curve is rational, and all primal and dual data are rational functions of one parameter.

Lemma B.8 (The rational curve)**.** *Let $G(\zeta)$ and $u(\zeta)$ be given by* (51)*. Then $P_2(G(\zeta), u(\zeta)) \equiv 0$, where P_2 is the polynomial* (50)*, and $u/G = 8\zeta(\zeta^2 - 2\zeta + 2)/\{(\zeta^2+2)(\zeta^2+2\zeta+2)\}$.*

Proof. Substitution and clearing denominators prove both identities. One way to recover the parametrization is to put $u = \sigma G$ in $P_2 = 0$. The remaining equation is quadratic in G^2; its discriminant has square-free part $\sigma^2 - 24\sigma + 16$. Parametrize this conic, then the resulting conic for G by a line through a rational point. Simplification gives (51). □

We use $\zeta^4 + 4 = (\zeta^2 - 2\zeta + 2)(\zeta^2 + 2\zeta + 2)$ and the abbreviations

$$q_\pm = \zeta^2 \pm 2\zeta + 2, \quad r_\pm = \zeta^2 \pm 2, \quad e_\pm = \zeta^2 \pm 2\zeta - 2, \quad K_6 = \zeta^6 - 4\zeta^5 + 4\zeta^4 + 8\zeta^2 - 16\zeta + 8.$$

Definition B.9 (Data of the arc)**.** For $\zeta \in \mathcal{J} = [1.6955, 1.699]$ use (51) and put

$$W = \frac{q_-^2 q_+(7\zeta^2 - 18\zeta + 14)}{8r_+K_6}, \quad 1 - W = \frac{e_-^2 e_+^2}{8r_+K_6},$$
$$\mathsf{a} = \frac{q_-^3}{2\zeta r_- r_+}, \qquad \mathsf{b} = -\frac{2\zeta(3\zeta^4 - 10\zeta^3 + 16\zeta^2 - 20\zeta + 12)}{r_-^2 r_+}.$$

Let $x < y$ be the roots of $H(c) = c^2 + \mathsf{a}c + \mathsf{b}$. Write $L_0 + L_1c$ and $U_0 + U_1c$ for the remainders of $\mathsf{L}_1(c)$ and $\mathsf{U}_2(c)$ modulo H. Set

$$\bar{c} = -\frac{L_0 + U_0}{L_1 + U_1}, \qquad d = -L_0 - L_1\bar{c}, \qquad p = \frac{\bar{c} - x}{y - x}, \qquad \nu = (1-p)\delta_x + p\delta_y.$$

The denominator $L_1 + U_1$ stays nonzero on $\mathcal{J}$. For $t > 0$ put $C = d/\sqrt{G^2 - 1}$; the reflected branch has $-C$.

For a polynomial $\phi(c)$, reduction modulo $c^2 + \mathsf{a}c + \mathsf{b}$ gives $\phi \equiv \phi_0 + \phi_1 c$ and $\int \phi\, d\nu = \phi_0 + \phi_1\bar{c}$. Hence every raw slack of ν is a rational function of ζ.

Lemma B.10 (Algebraic identities on the arc)**.** *Where the rational expressions of Definition B.9 are defined:*

(a) *$K_6 = \zeta^4(\zeta-2)^2 + 8(\zeta-1)^2 > 0$; hence $0 < W < 1$ whenever $e_-e_+ \ne 0$.*

(b) *Let $M = G(4G^2-1)\alpha_J$ and $N = G(4G^2-1)\beta_J$ for $J = \{2, 3\}$ and weights $(W, 1-W)$, that is, $M = 8G^3 + 4G^2c + 4Gc^2 - 4G - W(4Gc^2 - 2G + c)$ and $N = 4G^2 + 4Gc - W(4Gc + 1)$. Then*

$$-\{(1-c^2-u^2)M - 2u(1-c^2)N\} = 4G(1-W)\,(c^2 + \mathsf{a}c + \mathsf{b})^2,$$

and consequently $Q(c) = \frac{4(1-W)}{4G^2-1}(c-x)^2(c-y)^2 \ge 0$.

(c) $\ell_1 = \upsilon_2 = \upsilon_3 = 0$ *identically.*

Proof. Expand $\zeta^4(\zeta-2)^2 + 8(\zeta-1)^2$ to obtain $K_6 > 0$. All remaining factors of W are positive, and the formula for $1 - W$ gives its sign. Clearing the parameter denominators in (b) gives a polynomial identity in (ζ, c). For (c), reduce each slack modulo $c^2 + \mathsf{a}c + \mathsf{b}$ and replace c by $\bar{c}$ in the affine remainder. The three resulting rational functions vanish identically. □

The formulas of Definition B.9 are obtained as follows. The coefficient identities expressing that the quartic $G(4G^2-1)Q$ is a constant times a square are two equations, linear in W, with the common root W on the curve of Lemma B.8. Then a, b follow from the cubic and quadratic coefficients, and $\bar{c}, d$ from the linear equations $\ell_1 = \upsilon_2 = 0$. The third primal equation holds automatically, by complementary slackness: $\int \Phi \, d\nu$ equals $\ell_1 + W\upsilon_2/U_1(G) + (1-W)\upsilon_3/U_2(G)$.

Lemma B.11 (Endpoints)**.**

$$\begin{aligned} H_a(G) = {} & 162529280000G^{12} - 238685519872G^{10} + 77668708352G^8 \\ & - 6246545408G^6 - 58821872G^4 + 12933112G^2 + 229441. \end{aligned} \tag{62}$$

Let ζ_1 be the real root of $\zeta^3 - \zeta^2 - 2$, and let

$$\begin{aligned} N_{12} = {} & 25\zeta^{12} - 314\zeta^{11} + 1764\zeta^{10} - 6092\zeta^9 + 14988\zeta^8 - 28736\zeta^7 \\ & + 44832\zeta^6 - 57472\zeta^5 + 59952\zeta^4 - 48736\zeta^3 + 28224\zeta^2 - 10048\zeta + 1600. \end{aligned}$$

Then $1.6956 < \zeta_1 < 1.6957$, and N_{12} has exactly one root ζ_a in $\mathcal{J} = [1.6955, 1.699]$, with $1.6989 < \zeta_a < 1.699$; both roots are simple. Moreover:

(a) *$G(\zeta_1) = G_1$ and $u(\zeta_1) = 1 - 1/(2G_1)$. With $G = G(\zeta)$,*

$$32G^3 - 36G^2 + 4G - 1 = \frac{(\zeta^3 + 2\zeta + 4)(\zeta^3 - \zeta^2 - 2)\, S_6}{16\zeta^3 r_-^3},$$

$$u - 1 + \frac{1}{2G} = \frac{8(\zeta^3 - \zeta^2 - 2)^2}{r_- r_+ q_- q_+} \geq 0,$$

with $S_6 = \zeta^6 - 8\zeta^5 - 2\zeta^4 + 16\zeta^3 + 4\zeta^2 - 32\zeta - 8$; the curve P_2 touches the spiral curve $u = 1 - 1/(2G)$ at ζ_1 and lies above it.

(b) *$G_a := G(\zeta_a)$ is the root of H_a in $(1.02358, 1.02359)$: N_{12} divides the numerator of $H_a(G(\zeta))$.*

Proof. Rational endpoint evaluations and monotonicity isolate ζ_1. For N_{12}, its values at 1.6989 and 1.699 have opposite signs, and Lemma B.12 gives $N_{12}' < 0$ on $\mathcal{J}$. The displayed factorizations prove (a). For (b), polynomial division gives $N_{12} \mid \operatorname{num} H_a(G(\zeta))$; Sturm's theorem isolates the corresponding root in $(1.02358, 1.02359)$. □

Lemma B.12 (Sign certificates)**.** *Put $\zeta_c = 1.69725$, $\delta = 0.00175$. The rational tests of Table 4 have their indicated strict signs on $\mathcal{J} = [\zeta_c - \delta, \zeta_c + \delta]$. After cancellation of the displayed endpoint factors, each numerator and denominator polynomial R satisfies the exact criterion*

$$|R(\zeta_c)| > \sum_{k \geq 1} |R^{(k)}(\zeta_c)/k!| \delta^k.$$

Proof. Taylor's formula bounds $|R(\zeta) - R(\zeta_c)|$ by the displayed sum. The criterion is a finite rational comparison for each numerator and denominator obtained from Definition B.9; no approximation to an algebraic root enters it. □

Theorem B.13 (The arc P_2)**.** *For $\zeta \in [\zeta_1, \zeta_a]$ the data of Definition B.9 have the following properties.*

(i) *G decreases from G_1 to G_a; $1.0233 < G < 1.0337$ and $0.511 < u < 0.517$.*

Table 4: Rational tests for P_2. All quantities in the second column are positive on $\mathcal{J}$. Moments are obtained by reducing modulo $H(c)$ and replacing its affine remainder by its value at $\bar{c}$.

Property	Rational tests
Parameter bounds	$-G_\zeta$, $G-1.0233$, $1.0337-G$, $u-0.511$, $0.517-u$
Root bounds	$H(-0.505)$, $-H(-0.48)$, $-H(0.25)$, $H(0.285)$
Positive mass	$-(\bar{c}+\mathsf{a}/2)$, $-R_p$
Dual weights	$W-0.362$, $0.366-W$
Minimizer margin	d, $\bar{g}-0.28$, $(\bar{g}-0.28)^2(G^2-1)-d^2$
Slope and endpoint	du/dG, $1-du/dG$, $-N_{12}'$
Finite lag slacks	The residual quantities of Table 5

Table 5: Finite-lag sign reduction on P_2. Generate every expression from (16), reduce modulo H, and apply Lemma B.12 after canceling the indicated factor.

Quantity	Factor removed	Sign of remainder on $\mathcal{J}$
$\bar{c}^2+\mathsf{a}\bar{c}+\mathsf{b}$	$\zeta^3-\zeta^2-2$	$R_p<0$
$\ell_1,\upsilon_2,\upsilon_3$	—	Identically zero
υ_5	N_{12}	Positive
Other slacks, $1\le m\le 9$	1	Positive

(ii) $-0.505<x<-0.48$ *and* $0.25<y<0.285$.
(iii) $0\le p<1$, *with* $p=0$ *exactly at* ζ_1.
(iv) $0.362<W<0.366$, *and* $Q\ge 0$ *on* $[-1,1]$ *with zeros exactly at* x *and* y.
(v) $\bar{g}-|C|>0.28$.
(vi) $\ell_1=\upsilon_2=\upsilon_3=0$; $\upsilon_5>0$ *for* $\zeta<\zeta_a$ *and* $\upsilon_5(\zeta_a)=0$; *every other raw slack is positive.*
(vii) $0<du/dG<1$.

Consequently $V(t)=\operatorname{arsinh} u(\cosh t)$ *for* $G_a\le\cosh t\le G_1$, *the measure* ν *is optimal, and* $|V'(t)|<1$ *there.*

Proof. Reduce polynomial moments modulo $c^2+\mathsf{a}c+\mathsf{b}$ and use Lemma B.12 for their rational signs. The derivatives satisfy

$$G_\zeta=\frac{r_+e_-e_+}{8\zeta^2r_-^2}<0,\qquad \frac{du}{dG}=\frac{32\zeta^2q_-(\zeta^4-2\zeta^3-4\zeta+4)}{r_+^3e_-e_+}\in(0,1).$$

The signs of the root polynomial at $-0.505,-0.48,0.25,0.285$ isolate x,y. The identity $\bar{c}^2+\mathsf{a}\bar{c}+\mathsf{b}=(\zeta^3-\zeta^2-2)R_p$, with $R_p<0$, and $\bar{c}+\mathsf{a}/2<0$ give $x\le\bar{c}<y$, with equality exactly at ζ_1. The bounds on G,u,W are the corresponding rational endpoint sign tests. For the minimizer margin verify $d>0$, $\bar{g}>0.28$ and $(\bar{g}-0.28)^2(G^2-1)>d^2$. Lemma B.10 proves the contacts and dual nonnegativity. For $m\le 9$, Table 5 and the signs of Table 4 give every remaining slack; υ_5 has the sign of N_{12}. For $m\ge 10$, (54) gives $\ell_m,\upsilon_m\ge 0.28T_{10}(1.0233)-1.05>0.17$. Complementary slackness proves optimality, and $|V'|=|(du/dG)\sinh t|/\sqrt{1+u^2}<1$. □

Theorem B.13 covers both copies of P_2, one for each sign of t. At G_1 the data of the arc reduce to the spiral,

$$\begin{aligned}u&=1-\frac{1}{2G_1},\quad x=-\frac{1}{2G_1},\quad y=\frac{2G_1-1}{4G_1},\quad p=0,\\ d&=\frac{(2G_1-1)(G_1^2-1)}{2G_1^3},\quad W=\frac{2(1+2G_1-2G_1^2)}{4G_1+1},\end{aligned}\tag{63}$$

and the six equations of (49) vanish exactly modulo $32G_1^3-36G_1^2+4G_1-1$. At G_a the slack v_5 vanishes, and the data of the arc, with $w_2 = W$, $w_3 = 1 - W$, $w_5 = 0$, solve the central system as well; after the rescaling (59), a rigorous enclosure of this point is contained in the continuation box of P_5, which identifies the two branches.

B.6. The central two-frequency family

The central family is solved exactly at the critical point.

Theorem B.14 (Closed form at κ_c)**.** *At $h = 0$ the system $F(0, Z) = 0$ decouples. The three dual equations involve only (y, w_2, w_3), are linear in (w_2, w_3), and consist of $-3w_2 + 22w_3 = 12$ and two equations whose 3×3 consistency determinant is $900(2y+1)^3 q_5(y)$, $q_5(y) = 32y^5 - 40y^4 + 16y^3 - 6y^2 - 12y + 3$. The primal block is linear in $(\mathcal{A}, \mathcal{B}, D, P)$: its four equations are the coefficients of h in the four primal equations of (52), after (59). Let y_0 be the root of q_5 in $(0.23188, 0.23189)$. The primal coordinates are the unique solution of the four coefficient equations just described at $y = y_0$; the dual coordinates are*

$$43243231\, w_5^{(0)} = 7851355 - 25581300y_0 - 5903400y_0^2 - 17737600y_0^3 - 41558400y_0^4,$$

and $w_2^{(0)} = 2/5 - 22w_5^{(0)}/25$, $w_3^{(0)} = 3/5 - 3w_5^{(0)}/25$. Numerically $y_0 \approx 0.2318891$, $\mathcal{A}_0 \approx 0.5146605$, $\mathcal{B}_0 \approx -0.2585419$, $D_0 \approx 2.1051110$, $P_0 \approx 2.5790145$, $w_2^{(0)} \approx 0.3743478$, $w_3^{(0)} \approx 0.5965020$ and $w_5^{(0)} \approx 0.0291502$. The Jacobian $F_Z(0, Z_0)$ is block triangular, with the dual block in (y, w_2, w_3) and the primal block in $(\mathcal{A}, \mathcal{B}, D, P)$, and both block determinants are nonzero elements of $\mathbb{Q}(y_0)$.

Proof. Setting $h = 0$ in F gives seven explicit polynomials; the dual ones do not contain $\mathcal{A}, \mathcal{B}, D, P$. The rest is linear algebra over $\mathbb{Q}[y]/(q_5)$. Signs of elements of $\mathbb{Q}(y_0)$ are decided from $q_5(0.23188) > 0 > q_5(0.23189)$, $q_5' < 0$ near y_0, and the Taylor-dominance criterion of Lemma B.12 on $[0.23188, 0.23189]$. □

Proposition B.15 (Dual nonnegativity at κ_c)**.** *At $h = 0$ the dual polynomial factors as $Q = (c + \frac{1}{2})^2(c - y_0)^2(q_2c^2 + q_1c + q_0)$ with $q_i \in \mathbb{Q}(y_0)$, $q_2 \approx 0.09328$, $q_1 \approx -0.05001$, $q_0 \approx 0.65052$; in particular $q_0 - |q_1| - |q_2| > 0$, so $Q \geq 0$ on $[-1, 1]$ with zeros exactly at $-\frac{1}{2}$ and y_0.*

Proof. Division of Q by $(c + \frac{1}{2})^2(c - y_0)^2$ in $\mathbb{Q}(y_0)[c]$ leaves no remainder, and the signs are decided as in the proof of Theorem B.14. □

At $h = 0$ the raw slacks of the three-cycle are 0 or $\frac{3}{2}$: for $\theta_m = 2\pi m/3$ the slack of sign σ ($\sigma = -1$ lower, $\sigma = +1$ upper) equals $\frac{1}{2}(1 - \cos\theta_m) + \sigma\frac{\sqrt{3}}{2}\sin\theta_m$. It vanishes for $m \equiv 0$, for $m \equiv 1$ with $\sigma = -1$, and for $m \equiv 2$ with $\sigma = +1$, the congruences being modulo 3; we call these pairs (m, σ) critical.

Proposition B.16 (First-order slacks)**.** *For a critical pair (m, σ) let s_m^σ be the limit, as $h \downarrow 0$, of the raw slack divided by h. With $Y_m^\sigma(y) = (\frac{3}{4} - y^2)(1 - T_m(y)) + \sigma(1 - y^2)U_{m-1}(y)$,*

$$\begin{aligned}
m \equiv 0: &\quad s_m^\sigma = \tfrac{m^2}{2} - \sigma m(\mathcal{B}_0 + D_0) + P_0 Y_m^\sigma(y_0),\\
m \equiv 1,\ \sigma = -1: &\quad s_m^- = \tfrac{m^2}{2} + (\mathcal{B}_0 - 3\mathcal{A}_0) - m\mathcal{B}_0 + mD_0 + P_0 Y_m^-(y_0),\\
m \equiv 2,\ \sigma = +1: &\quad s_m^+ = \tfrac{m^2}{2} + (\mathcal{B}_0 - 3\mathcal{A}_0) + m\mathcal{B}_0 - mD_0 + P_0 Y_m^+(y_0).
\end{aligned}$$

These vanish exactly for the active pairs $(1,-), (2,+), (3,+), (5,+)$ and are positive for every other critical pair; the four smallest positive values, in increasing order and rounded to two decimal places, are $s_8^+ \approx 11.22$, $s_6^+ \approx 11.49$, $s_3^- \approx 14.91$ and $s_4^- \approx 18.40$.

Proof. Differentiate the raw slack at $h=0$, using $T_m(1+h)=1+m^2h+O(h^2)$, $\omega_x=2\pi/3-2h\mathcal{B}_0/\sqrt{3}+O(h^2)$, $g(x)=1/2+h(\mathcal{B}_0-\mathcal{A}_0)+O(h^2)$ and $dU_{m-1}(G)=hD_0m+O(h^2)$. The addition formulas in the three residue classes give the displayed expressions. Reduction in $\mathbb{Q}[y_0]/(q_5)$ proves the four identities and all remaining signs for $m\le 7$. For $m\ge 8$, $|Y_m^\sigma|\le 5/2$, $|\mathcal{B}_0|+D_0<2.37$, $|\mathcal{B}_0-3\mathcal{A}_0|<1.81$ and $P_0<2.58$ give $s_m^\sigma\ge m^2/2-2.37m-8.26>0$. The same bound for $m\ge 10$, together with the exact comparisons for $m\le 9$, orders the four smallest positive slacks. □

Large lags along the central family are controlled by one estimate.

Lemma B.17 (Large lags near the three-cycle)**.** *Let $0<h\le 0.0236$, $t=\operatorname{arcosh}(1+h)$, and suppose that a two-atom spectrum with the parameters* (59) *satisfies*

$$\begin{gathered}0<\mathcal{A}<0.55,\quad -0.27<\mathcal{B}<0,\quad 0<P<3,\quad 0<D<2.2,\\ u<\tfrac{11}{20},\quad \bar g>0.49,\quad \bar g-|C|>\tfrac14,\quad |g(x)-\tfrac12|<h,\quad |\bar g-\tfrac12|<2h,\\ |2u\sin\omega_x-\tfrac{\sqrt3}{2}|<2h,\quad |\arccos x-\tfrac{2\pi}{3}|<\tfrac{h}{3}.\end{gathered}\tag{64}$$

Then $\ell_m,\upsilon_m>0$ for all $m\ge 24$.

Proof. For $mt\ge 3$, (43) gives $\ell_m,\upsilon_m\ge\cosh 3/4-(31/20)^2>0$. For $mt\le 3$, compare with the nonnegative three-cycle slacks. The gain $\bar g(\cosh mt-1)$ is at least $0.49m^2h$. The main trigonometric term changes by at most $3h+mh/2$, the second atom by at most $12h$, and the constant term by at most $2h$. Finally $|d|U_{m-1}(G)\le 2.2hm\sinh 3/3<7.7mh$. The resulting lower bound $h(0.49m^2-9m-17)$ is positive for $m\ge 24$. The numerical inequalities follow from $20<e^3<21$. □

Theorem B.18 (Central family near κ_c)**.** *There are $h_*>0$ and a real-analytic branch $Z(h)$ on a neighborhood of $[-h_*,h_*]$, with $Z(0)=Z_0$ and $F(h,Z(h))=0$, locally unique near $(0,Z_0)$. For $0\le h\le h_*$ the corresponding spectrum is feasible and optimal, so that $V(t)=\operatorname{arsinh}\big(\frac12+(\cosh t-1)\mathcal{A}(\cosh t-1)\big)$ for $\cosh t\le 1+h_*$. In particular V is even and real-analytic at $t=0$, and $V(0)=\log\varphi$.*

Proof. The nonsingular critical Jacobian gives the unique analytic branch by the implicit function theorem. Positivity of P_0,D_0 and the dual weights persists, as does the positive residual factor of Q. The defining contacts stay zero. For every other $m\le 23$, the slack either tends to $3/2$ or, after division by h, tends to the positive quantity in Proposition B.16. For the uniform tail, the divided limits in (64) are respectively

$$\mathcal{B}_0-\mathcal{A}_0,\quad \mathcal{B}_0-\mathcal{A}_0+P_0(1/4-y_0^2),\quad \sqrt3\mathcal{A}_0+\mathcal{B}_0/\sqrt3,\quad -2\mathcal{B}_0/\sqrt3.$$

Their absolute values are less than $1,2,2,1/3$, respectively, by exact comparison in $\mathbb{Q}(y_0,\sqrt3)$. Also $u\to 1/2$, $\bar g\to 1/2$ and $C=O(\sqrt h)$. Thus (64) holds on one common interval, and Lemma B.17 covers all $m\ge 24$. Duality proves optimality. Evenness in t and $G=\cosh t$ give the analytic formula for V, with $V(0)=\operatorname{arsinh}(1/2)=\log\varphi$. □

Proposition B.19 (The central family)**.** *The continued branch of P_5 on $0\le h\le G_a-1$ consists of feasible and optimal spectra, and $|V'|<1$ along it. At $h=G_a-1$ it coincides with the endpoint of P_2.*

Proof. The exact critical solution initializes the continuation, and Theorem B.18 identifies it near zero. Each slab verifies (64), $0.23<y<0.29$, $0.36<w_2<0.38$, $0.59<w_3<0.65$, $w_5<0.031$, $w_5'<0$, $|u_G|<1$ and $\sqrt{G^2-1}<\sqrt{1+u^2}$. The quotient $Q/((c-x)^2(c-y)^2)$ satisfies $q_0-|q_1|-|q_2|>0$. Inclusion of the exact endpoint of P_2 in the last continuation box gives $w_5(G_a)=0$, hence $w_5>0$ before it. All nonactive slacks with $m<24$ are positive. Near zero evaluate them after division by h, retaining the exact constant at $h=0$ and propagating the divided part through sums and products. For the main phase, the mean value theorem encloses $(\arccos x-2\pi/3)/h=-\mathcal{B}/\sqrt{1-\xi^2}$; addition formulas and the entire function sinc then enclose its divided trigonometric differences. Lemma B.17 covers $m\ge 24$. Duality gives optimality, and $V'=u_G\sinh t/\sqrt{1+u^2}$ gives the derivative bound. □

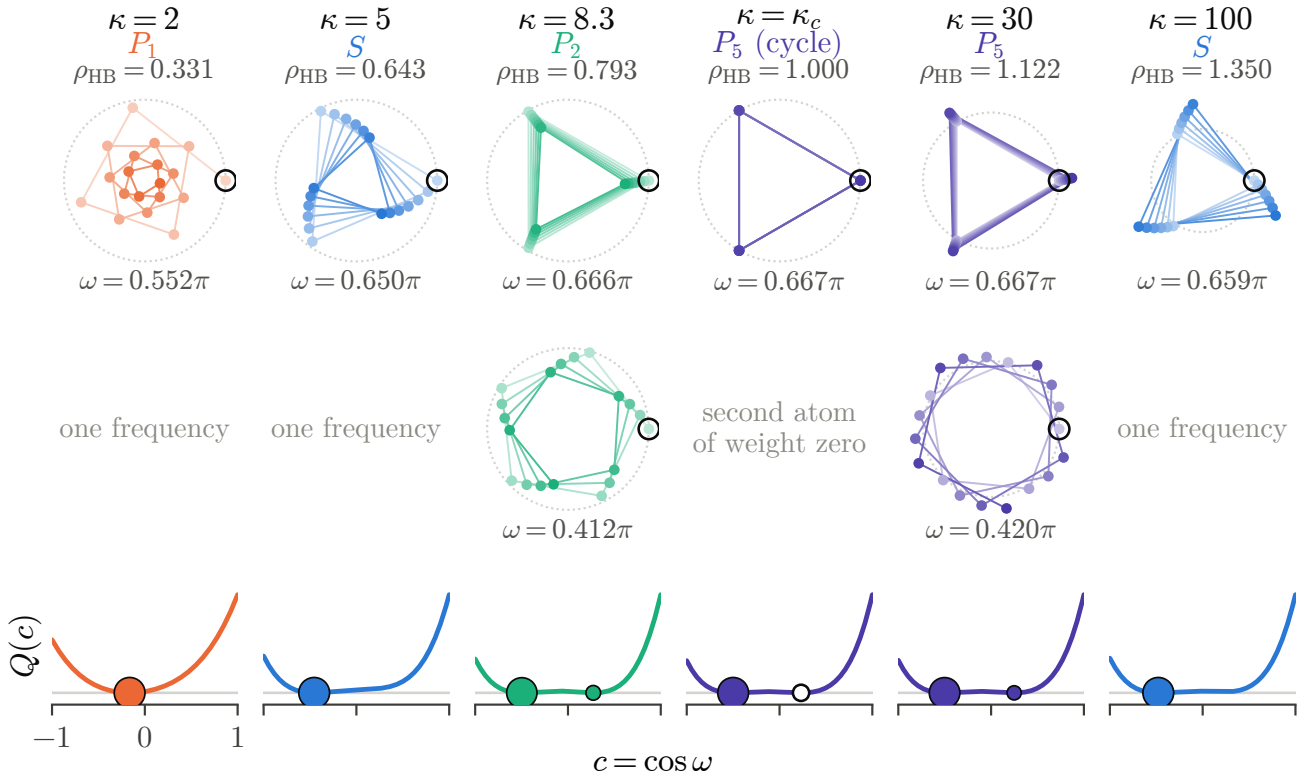


Figure 8: Extremal spectra by regime. Top: the first nineteen points in each frequency plane, with unit initial amplitude and displayed radius $\rho_{\mathrm{HB}}^{k/12}$. Bottom: the dual polynomial $Q(c) \geq 0$ and its selected double zeros; filled marker size indicates spectral mass. An open marker denotes the zero-mass site at κ_c.

B.7. Regularity of the optimal rate

Proof of Theorem 2.3, regularity. The inverse of the continuous strictly increasing map $t - V(t)$ is continuous and strictly increasing. On each open analytic arc its derivative $1 - V'$ is positive, so the analytic inverse function theorem applies. The critical point belongs to the interior analytic branch of Theorem B.18; the other junctions give precisely the five listed switches. Proposition B.5 gives strict improvement over the spiral everywhere else, and $\rho_{\mathrm{sp}} > q$. The expansions follow in Appendix C. □

C. Finite-horizon bounds and asymptotic expansions

This appendix completes the proofs of Theorems 2.1 and 2.3. Throughout, $q \in (0,1)$ is fixed, the parameters are those of Polyak's tuning, $T = T(q) = \log \rho_{\mathrm{HB}}$, and the recurrence is $e_{k+1} = 2q\sigma_k - q^2 e_{k-1}$.

C.1. A uniform lower bound from rest

We first prove the uniform lower bound (7). All variables below refer to the special spiral: $\rho = \rho_{\mathrm{sp}}$, $G = (\rho + \rho^{-1})/2$, $\cos\omega = -\rho/(1+\rho^2)$, $z = \rho e^{i\omega}$, $m = (z + q^2/z)/(2q)$, $g = 1 - |m|^2$, $C = g\tanh(\log\rho)$, and $\mathsf{h} = (C + 2\operatorname{Re} m)/4$. In $\mathbb{C} \oplus \mathbb{R}$ prescribe

$$\begin{gathered} x_0 = (1,2), \quad \sigma_0 = (m + \Delta, 0), \quad \psi_0 = \mathsf{h}, \\ x_k = (z^k, y_k), \quad \sigma_k = (mz^k, 0), \quad \psi_k = \mathsf{h}\rho^{2k} \quad (k \geq 1), \\ y_k = 2(-q^2)^{\lceil k/2 \rceil}, \quad \Delta = \tfrac{q}{2}(1 - z^{-1}). \end{gathered} \tag{65}$$

Since $m + \Delta = (z + q^2)/(2q)$, the first step from $x_{-1} = x_0$ gives x_1, and the later steps follow from $2qmz = z^2 + q^2$ and $y_{k+1} = -q^2 y_{k-1}$. Once interpolation is proved, $\|x_0\| = \sqrt{5}$ and $\|x_k\| \geq \rho^k$ give (7).

Pairs of positive indices and their minimizer pairs satisfy (34) by Proposition 2.7 and Lemma A.3, since the last coordinate carries zero gradient and only adds $\frac{1}{4}(y_i - y_j)^2$ to R_{ij}. Only the pairs with the index 0 remain. Write $X = \rho^k$, $Z = z^k$, and

$$\mathsf{p}_\pm = \tfrac{1}{2}\bar{\Delta}(m \pm 1), \quad \mathsf{a}_\mp = \frac{g\rho^{\mp 1}}{4G}, \quad \mathsf{b}_- = gGe^{-i\omega} + \mathsf{p}_+, \quad \mathsf{b}_+ = gGe^{i\omega} + \mathsf{p}_-,$$
$$\mathsf{c}_\mp = \frac{g\rho^{\pm 1}}{4G} - \operatorname{Re}\mathsf{p}_\pm - \frac{|\Delta|^2}{4}, \quad \mathsf{d} = (1-q^4)^2.$$

Lemma C.1 (Slacks of the initial pairs)**.** *For $k \geq 1$,*

$$R_{0k} \mp T_{0k} = \mathsf{a}_\mp X^2 + \operatorname{Re}(\mathsf{b}_\mp Z) + \mathsf{c}_\mp + (1 - (-q^2)^{\lceil k/2 \rceil})^2, \tag{66}$$

and $|T_{\star 0}| = |\frac{1}{2}\operatorname{Re}\Delta - \frac{C}{4}|$, $R_{0\star} = \frac{1}{4}(5 - |m+\Delta|^2)$. On $F(\rho, q) = 0$,

$$g = \frac{2\rho(\rho^2 - \rho + 1)}{(\rho^2+1)^2} = \frac{1}{2} - \frac{(\rho-1)^4}{2(\rho^2+1)^2},$$
$$\operatorname{Re}\Delta = \frac{q}{2}\frac{\rho^2+2}{\rho^2+1} \leq q, \quad |\Delta|^2 = \frac{q^2}{4}\frac{\rho^4 + 4\rho^2 + 1}{\rho^2(\rho^2+1)}. \tag{67}$$

In particular $0 < g \leq \frac{1}{2}$ and $|C| < \frac{1}{2}$.

Proof. Insert (65) into R_{0k} and T_{0k}, use $\langle u, v\rangle = \operatorname{Re}(u\bar{v})$ and collect the coefficients of X^2, Z and 1. The last coordinate contributes $(1-(-q^2)^{\lceil k/2\rceil})^2$. The remaining identities follow by clearing denominators and reducing modulo $F(\rho, q)$, which is quadratic in q. □

Table 6: Rational sign certificates for the initialized spiral. Every inequality holds for both signs on the entire closed interval in ρ.

Interval in ρ	Certified inequalities
$[33/100, 1]$	$\mathsf{c}_\mp + \mathsf{d} > 0$ and $(\mathsf{c}_\mp + \mathsf{d})^2 > \lvert\mathsf{b}_\mp\rvert^2$.
$[1, 6/5]$	$\lvert\mathsf{b}_\mp\rvert^2 < 1$, $\mathsf{a}_\mp - \epsilon > 0$, and $P_r^\mp(1) > 0$. For $(-,0),(-,1),(+,2)$, $(P_r^\mp)'(1) > 0$; for the other three pairs (σ, r), $4(\mathsf{a}_\mp - \epsilon)(\mathsf{c}_\mp + \mathsf{d}) > (\mathsf{b}_r^\mp + \epsilon)^2$.
$[6/5, 1713/1000]$	$4\mathsf{a}_\mp^2\rho^{20} > \lvert\mathsf{b}_\mp\rvert^2$, $\mathsf{z}_\mp := \mathsf{a}_\mp\rho^{20} + \mathsf{c}_\mp + \mathsf{d} > 0$, and $\mathsf{z}_\mp^2 > \lvert\mathsf{b}_\mp\rvert^2\rho^{20}$. The exact slacks (66) are positive for $1 \leq k \leq 9$.

In Table 6, $\mathsf{b}_r^\mp = \operatorname{Re}(\mathsf{b}_\mp e^{ir\omega})$ for $r = 0, 1, 2$,

$$\epsilon(\rho) = \frac{3}{40}\frac{(\rho^2-1)(\rho+1)^2}{\rho(\rho^2+1)}, \quad P_r^\mp(X) = (\mathsf{a}_\mp - \epsilon)X^2 + (\mathsf{b}_r^\mp + \epsilon)X + \mathsf{c}_\mp + \mathsf{d}.$$

Each quantity in the table is reduced modulo $F(\rho, q)$, which is quadratic in q, to $(\mathcal{P}(\rho) + \mathcal{R}(\rho)q)/\mathcal{D}(\rho)$. If $F = f_0 + f_1 q + f_2 q^2$, the zeros of the numerator on either algebraic branch are zeros of $f_0\mathcal{R}^2 - f_1\mathcal{P}\mathcal{R} + f_2\mathcal{P}^2$. Sturm sequences isolate these zeros and decide the signs of denominators. A simple resultant zero that is assigned to the other branch by a sign change on its isolating interval cannot also be a zero on the physical branch, since its multiplicity would then be at least two. Signs at rational points reduce to comparisons of squares in a quadratic number field. There are 44 such checks.

Proof of (7). For $q \leq 1/5$, $q < \rho < (1+\sqrt{2})q < 1/2$ gives $|\Delta|, |\mathsf{p}_\pm| < 3/5$. Altering the first gradient loses less than $99/100$ in any initial pair, while the scalar coordinate gains at least $(1-q^4)^2 > 99/100$. The minimizer pair has $R_{0\star} > 61/100 > |T_{\star 0}|$. For $\kappa \geq 2$, $q < \rho < 2q$ gives $R_{0\star} > 11/16 > |T_{\star 0}|$. For the other pairs use Table 6: if $\rho \leq 1$, then $X \leq 1$ and its first row gives positivity; if $1 \leq \rho \leq 6/5$, write $k = 3j + r$ and $\omega = 2\pi/3 + \vartheta$. The addition formulas and $\log\rho \geq 2(\rho-1)/(\rho+1)$ give $|\vartheta| \leq \epsilon\log\rho$, so $|e^{3ij\vartheta} - 1| \leq \epsilon\log X \leq \epsilon(X-1)$. The slack is therefore at least $P_r^\mp(X) > 0$ by the second row. For

$\rho \ge 6/5$, bound the oscillatory term by $-|\mathsf{b}_{\mp}|X$; the third row gives positivity for $k \ge 10$ and checks the first nine lags. The bounds $\rho_{\mathrm{sp}}(2) > 33/100$ and $\rho_* < 1713/1000$ show that these intervals cover $\kappa \ge 2$, overlapping the small-q case. Interpolation yields $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^3)$, and $\|x_k\| \ge \rho^k$, $\|x_0\| = \sqrt{5}$ give the assertion. □

C.2. Certificates and bounded-memory functionals

We turn to the upper bound in (6). Its first ingredient is a certificate with a gap linear in the distance to the optimal exponent. For a weight vector w and a point (G, u) write $\Phi_{G,u}$ for the symbol (46), and for $t \in \mathbb{R}$ let $u_q(t) = \sinh(t - \log q)$ and $u_V(t) = \sinh V(t)$.

Lemma C.2 (Linear gap)**.** *There are $a > 0$, $\delta_0 > 0$, and certificates* (14) *with lags at most five at the exponents $t = T + \delta$, $0 < \delta \le \delta_0$, whose coefficients are bounded uniformly in δ and whose symbols satisfy $\Phi_{T+\delta} \le -a\delta$ at every frequency.*

Proof. Use at $t = T + \delta$ the boundary weights of Theorem 2.8, supported on $\{2, 3, 5\}$, and replace $u_V = \sinh V(t)$ by $u_q = \sinh(t - \log q)$. On the adjacent arc choose $\theta < 1$ with $|V'| \le \theta$. Then $u_q^2 - u_V^2 \ge (1 - \theta)\delta$, since $u_V \ge 1/2$. Lemma B.1 gives

$$\Phi_{G,u_q}(c) \le -(G - c)(1 - \theta)\delta.$$

For $T \ne 0$, take $\delta \le |T|/2$ so that $G - c \ge \cosh(T/2) - 1 > 0$. For $T = 0$, add equal minimizer weights $b\delta$. On $c \ge 19/20$, $g \le -1/8$, giving the gap $b\delta/4$; on $c \le 19/20$, the gap is at least $\delta((1 - \theta)/20 - 2b)$. Any $0 < b \le (1 - \theta)/80$ works. All multipliers stay bounded. □

Fix $M_0 = 5$. We record the inner products and function values in a fixed number of consecutive iterates as a Gram window. This places the limit $\delta \downarrow 0$ in a finite-dimensional space.

Fix an exponent t. For positions e_l and values ψ_l put $f_l = e^{-lt} e_l$ and $h_l = e^{-2lt}\psi_l$. A banded functional of width $D \ge 0$ and offset $o \in \mathbb{Z}$ is

$$\mathcal{B}_j = \sum_{a,b=0}^{D} c_{ab} \langle f_{j+o+a}, f_{j+o+b} \rangle + \sum_{a=0}^{D} d_a h_{j+o+a}, \tag{68}$$

with real coefficients $c_{ab} = c_{ba}$ and d_a. Its symbol, charge and size are $\hat{\mathcal{B}}(\omega) = \sum_{a,b} c_{ab} \cos((a - b)\omega)$, $\chi(\mathcal{B}) = \sum_a d_a$ and $|\mathcal{B}| = \sum_{a,b} |c_{ab}| + \sum_a |d_a|$. The charge is the response to adding a constant to all scaled function values; zero charge allows these terms to telescope. The functional lies inside a segment $e_0, \dots, e_n$ if $1 \le j + o$ and $j + o + D \le n - 1$. After substituting the recurrence, the scaled certificate $e^{-2jt} Z_j$ has this form with offset -1, width $M' + 2$, zero charge and symbol Φ_t.

For a probability measure ν on $[-1, 1]$ and $C \in \mathbb{R}$, the realization of Lemma A.3 in $L^2([-1, 1], \nu; \mathbb{C})$, taken for all $l \in \mathbb{Z}$, is the stationary sequence of (ν, C) at the exponent t: after removing the factor e^{lt}, its pairwise inner products depend only on index differences. It satisfies the recurrence for every l, $f_l(c) = e^{il \arccos c}$, and $h_l = (C + 2\int \mathrm{Re}\, m \, d\nu)/4$; hence every banded functional takes on it the value

$$\mathcal{B}_j = \int \hat{\mathcal{B}}(\arccos c)\, d\nu(c) + \frac{\chi(\mathcal{B})}{4}\Big(C + 2\int \mathrm{Re}\, m \, d\nu\Big), \tag{69}$$

independent of j. The pair (ν, C) is H-feasible at (v, t) if $|C| \le \bar{g}$ and the lag conditions of (13) hold for $1 \le m \le H$. By Lemma A.3, the stationary sequence of an H-feasible pair satisfies every minimizer inequality and every interpolation inequality between indices at distance at most H.

Lemma C.3 (Telescoping with finite real filters)**.** *Let $D = M_0 + 2 = 7$ and let $\mathcal{B}$ be a banded functional of width D and offset o with $\chi(\mathcal{B}) = 0$. Suppose that, for some $\varepsilon \ge 0$, real coefficients ϖ_{ra} give the finite factorization*

$$-\hat{\mathcal{B}}(\omega) - \varepsilon = \sum_{r=1}^{R} \Big| \sum_{a=0}^{D} \varpi_{ra} e^{ia\omega} \Big|^2. \tag{70}$$

For each integer $0 \le a_0 \le D$ there is a banded functional $\mathcal{U}$ of width $D-1$ and offset o, with $|\mathcal{U}| \le c_D(|\mathcal{B}|+\varepsilon)$ for a constant c_D independent of R and of the chosen factors, such that for all sequences in every real Hilbert space and every j,

$$\mathcal{B}_j = -\varepsilon\|f_{j+o+a_0}\|^2 - \sum_{r=1}^{R}\Big\|\sum_{a=0}^{D}\varpi_{ra}f_{j+o+a}\Big\|^2 + \mathcal{U}_{j+1} - \mathcal{U}_j. \tag{71}$$

Proof. Fourier orthogonality gives $\sum_{r,a}\varpi_{ra}^2 = -\sum_a c_{aa} - \varepsilon \le |\mathcal{B}|$. Add the filter squares and $\varepsilon\|f_{j+o+a_0}\|^2$ to $\mathcal{B}_j$. Its quadratic coefficient matrix $\tilde{c}$ now has zero symbol. For each diagonal offset e, put $\mu_{0,a} = \tilde{c}_{aa}$ and $\mu_{e,a} = 2\tilde{c}_{a,a+e}$ for $e > 0$. Zero symbol means $\sum_a \mu_{e,a} = 0$. Each diagonal therefore telescopes by

$$\sum_a \mu_{e,a} g_{j+o+a} = R_{j+1} - R_j, \qquad R_j = \sum_a \mu_{e,a}\sum_{r=0}^{a-1} g_{j+o+r}, \quad g_l = \langle f_l, f_{l+e}\rangle.$$

The zero-charge value terms telescope by the same identity. Only indices up to $j+o+D-1$ occur. Finally $\sum_{a,b}|\sum_r \varpi_{ra}\varpi_{rb}| \le (D+1)\sum_{r,a}\varpi_{ra}^2$ gives a bound independent of R; for $D=7$, $c_D = 63$ suffices. □

For a nonnegative trigonometric polynomial of degree at most seven, the scalar Fejér–Riesz theorem gives a polynomial factor of degree at most seven [4, §1 and Theorem 2.1]. The symbols here are real and even. If the factor is $p = u + iv$, with real-coefficient polynomials u, v, averaging $|p(e^{i\omega})|^2$ at ω and $-\omega$ gives $|u(e^{i\omega})|^2 + |v(e^{i\omega})|^2$. Thus (70) holds with two real filters of width seven for every certificate of Lemma C.2, after subtracting its gap $\varepsilon = a\delta$. The algebra behind the minimizer correction is also explicit. With $\Delta = u'^2 - u^2$, $g_{u'} = 1 - c^2 - u'^2$ and equal minimizer weights $\gamma \ge 0$, subtraction of (47) gives

$$\begin{aligned} -\Phi_{G,u'} - 2\gamma g_{u'} - \varepsilon = {} & -\Phi_{G,u} + (G-c)\{\Delta(\alpha_J - 1) + 2(u'-u)(1-c^2)\beta_J\} \\ & + 2\gamma(1-c)^2 + (\Delta - 4\gamma)(1-c) \\ & + (G-1)\Delta + 2\gamma u'^2 - \varepsilon, \end{aligned} \tag{72}$$

The last terms are nonnegative for $4\gamma \le \Delta$ and $\varepsilon \le (G-1)\Delta + 2\gamma u'^2$, as in Lemma C.2. This retains the same width at the critical point.

C.3. The rate of bounded-memory trajectories

An H-segment retains every minimizer inequality and the interpolation inequalities of lags at most H. Its envelope Γ_H is finite and submultiplicative, with $\rho_H = \lim_p \Gamma_H(p)^{1/p} \ge e^T$. For $H \le k \le n-1$ record the Gram matrix of $e_{k-H}, \dots, e_{k+1}$ and the values $\psi_{k-H+1}, \dots, \psi_k$ as $Y_k \in \mathcal{E}_H = \mathrm{Sym}_{H+2}(\mathbb{R}) \times \mathbb{R}^H$. The admissible windows form a closed convex cone $\mathcal{K}_H$: positive semidefiniteness and every interpolation inequality are closed convex conditions on these entries. The minimizer inequalities imply $|\psi_l| \le \|e_l\|^2/2$, hence $|\mathcal{V}(Y)| \le |\mathcal{V}|\,\mathrm{tr}\,Y$ for a linear functional, with coefficient norm $|\mathcal{V}|$. The envelope and the old positions already in Y_k give

$$\mathrm{tr}\,Y_{k+p} \le (H+2)\max_{\max(0,p-H)\le r\le p+1}\Gamma_H(r)^2\,\mathrm{tr}\,Y_k \quad (p \ge 0,\ k+p \le n-1). \tag{73}$$

Lemma C.4 (Plateau for classes of segments)**.** *Let $\mathcal{G}$ be a class of H-segments in Euclidean spaces $\mathbb{R}^d$, over all finite d, closed under restriction to consecutive indices, and let $\Gamma_{\mathcal{G}}$ be defined as Γ, with the supremum over $\mathcal{G}$. Assume that $\Gamma_{\mathcal{G}}(p) > 0$ for every p and that $r = \lim_p \Gamma_{\mathcal{G}}(p)^{1/p} > 0$. Then there is no banded functional $\mathcal{B}$ at the exponent $\log r$ with $\chi(\mathcal{B}) = 0$, $\hat{\mathcal{B}} < 0$ on $[0,\pi]$, and $\mathcal{B}_j \ge 0$ whenever $\mathcal{B}_j$ lies inside a member of $\mathcal{G}$.*

Proof. If $\hat{\mathcal{B}} \le -\varepsilon < 0$, localization and Parseval give (39), exactly as in Lemma A.6. Zero charge cancels the interior values; fixed bandwidth and $|h_l| \le \|f_l\|^2/2$ control all endpoint terms. Closure

under restriction makes $\Gamma_{\mathcal{G}}$ submultiplicative; a zero state stays zero by the minimizer inequality. Thus $\Gamma_{\mathcal{G}}(p) \geq r^p$, and the block-energy argument of Lemma A.7 applies to this class without any change. It forces growth strictly below r, a contradiction. □

Lemma C.5 (Continuation)**.** *Every $Y \in \mathcal{K}_H$ is the window Y_H of an infinite segment, and hence of an infinite H-segment.*

Proof. Realize the Gram matrix by $e_0, \dots, e_{H+1}$ and set $\sigma_l = (e_{l+1} + q^2 e_{l-1})/(2q)$ for $1 \leq l \leq H$. These data and the minimizer datum satisfy every pairwise interpolation inequality. Theorem A.2 supplies a single $\psi \in \mathcal{S}$. Continue the recurrence with $\sigma_l = \nabla\psi(e_l)$ and $\psi_l = \psi(e_l)$; all subsequent inequalities hold by necessity. □

Lemma C.6 (Bounded memory preserves the rate)**.** *There is $H_0 \geq M_0 + 2$ such that $\rho_H = e^T$ for every $H \geq H_0$.*

Proof. Let t_H be the largest exponent admitting an H-feasible pair at $(t - \log q, t)$. The minimizer inequality bounds $|t - \log q| \leq \operatorname{arsinh} 1$. Compactness gives attainment, and a weak limit of maximizers shows $t_H \downarrow T$ as $H \to \infty$. Choose $H_0 \geq M_0 + 2$ with $t_{H_0} < T + \delta_0$. For $H \geq H_0$, the multipliers of Lemma C.2 exclude every $t \in (T, T + \delta_0]$ using only lags at most five; larger exponents exceed t_H. If $\rho_H > e^T$, finite-dimensional separation at $t = \log \rho_H$ supplies a strict certificate using lags at most H. Its local functional has zero charge and is nonnegative on every H-segment, contradicting Lemma C.4. □

C.4. A limiting functional at the exact rate

Lemma C.7 (Functionals above the rate)**.** *Let $H \geq H_0$. For $0 < \delta \leq \delta_0$ there are linear functionals $\mathcal{V}_\delta$ on $\mathcal{E}_H$ with $\sup_\delta |\mathcal{V}_\delta| < \infty$ such that, with $t = T + \delta$:*

(i) *for every H-segment and $H \leq k \leq n-2$, $\mathcal{V}_\delta(Y_k) - e^{-2t}\mathcal{V}_\delta(Y_{k+1}) \geq \delta \|e_{k-M_0}\|^2$;*

(ii) *for every infinite H-segment and $k \geq H$, $\mathcal{V}_\delta(Y_k) \geq \delta \sum_{i \geq 0} e^{-2it} \|e_{k+i-M_0}\|^2$.*

In particular, $\mathcal{V}_\delta \geq 0$ on $\mathcal{K}_H$.

Proof. The certificate gives a nonnegative banded functional $\mathcal{B}_j = e^{-2jt} Z_j$ of offset -1, width $M_0 + 2$, zero charge, and symbol at most $-a\delta$. Its coefficients are uniformly bounded. Apply Lemma C.3 with $\varepsilon = a\delta$, $a_0 = 1$: $\mathcal{U}^\delta_{j+1} - \mathcal{U}^\delta_j \geq a\delta \|f_j\|^2$. Define $\mathcal{V}_\delta(Y_k) = -a^{-1} e^{2(k-M_0)t} \mathcal{U}^\delta_{k-M_0}$. Since $H \geq M_0 + 2$, all its indices lie in Y_k and its coefficients are uniformly bounded. Multiplication of the preceding inequality gives (i). Iteration gives the partial sums in (ii). By Lemma C.6, $\Gamma_H(p) \leq K_\delta e^{p(T+\delta/2)}$; hence (73) makes the discounted terminal term tend to zero. Lemma C.5 then gives nonnegativity on the entire cone. □

Lemma C.8 (A functional at the exact rate)**.** *Let $H \geq H_0$. There is a linear functional $\mathcal{V}_*$ on $\mathcal{E}_H$ such that*

(a) *$\mathcal{V}_* \geq 0$ on $\mathcal{K}_H$;*

(b) *$\mathcal{V}_*(Y_{k+1}) \leq e^{2T} \mathcal{V}_*(Y_k)$ for every H-segment and $H \leq k \leq n-2$;*

(c) *if (ν, C) is H-feasible at $(T - \log q, T)$, the windows of its stationary sequence at the exponent T satisfy $\mathcal{V}_*(Y_k) \geq \frac{1}{2} e^{2(k-M_0)T} > 0$ for $k \geq H$.*

Proof. By the uniform bound in Lemma C.7, there are $\delta_n \downarrow 0$ such that $\mathcal{V}_{\delta_n}$ converges to a linear functional $\mathcal{V}_*$. Properties (a) and (b) follow from Lemma C.7 in the limit. For (c), the stationary sequence restricted to $l \geq 0$ is an infinite H-segment with $\|e_l\|^2 = e^{2lT}$, and Lemma C.7(ii) gives $\mathcal{V}_\delta(Y_k) \geq \delta \sum_{i \geq 0} e^{-2i(T+\delta)} e^{2(k+i-M_0)T} = \frac{\delta}{1 - e^{-2\delta}} e^{2(k-M_0)T} \geq \frac{1}{2} e^{2(k-M_0)T}$; the bound passes to the limit. □

Fix $H \ge H_0$ and let $\mathcal{N} = \{Y \in \mathcal{K}_H : \mathcal{V}_*(Y) = 0\}$. By (a) and (b), $Y_k \in \mathcal{N}$ implies $Y_{k+1} \in \mathcal{N}$. An $\mathcal{N}$-segment is an H-segment all of whose windows lie in $\mathcal{N}$. These segments form a class closed under restriction to consecutive indices; let $\Gamma_{\mathcal{N}} \le \Gamma_H$ be its growth function and $\rho_{\mathcal{N}} = \lim_p \Gamma_{\mathcal{N}}(p)^{1/p}$, with $\rho_{\mathcal{N}} = 0$ if $\Gamma_{\mathcal{N}}(p) = 0$ for some p.

Lemma C.9 (A smaller rate on the kernel)**.** $\rho_{\mathcal{N}} < e^T$.

Proof. Suppose $\rho_{\mathcal{N}} = e^T$. For stationary windows at T, linearity gives $\mathcal{V}_*(Y_H^{\nu,C}) = \int \vartheta \, d\nu + \vartheta_0 C$ with continuous ϑ. Lemma C.8(c) excludes the equation $\mathcal{V}_*(Y_H^{\nu,C}) = 0$ from the compact H-feasible set. Separate the finite system of lag and minimizer inequalities together with this one equality, as in Theorem 2.5(a). This gives a nonnegative combination Z_j of interpolation slacks and a real multiplier ζ for the equality, whose combined frequency symbol is strictly negative and whose value coefficient cancels. Thus $\mathcal{B}_j = e^{-2jT}(Z_j + \zeta \mathcal{V}_*(Y_{j+H}))$ has zero charge, negative symbol, and is nonnegative inside every $\mathcal{N}$-segment. Its symbol and charge can be read from (69) for $\nu = \delta_c$ and arbitrary C. Lemma C.4 contradicts the assumed rate. □

Lemma C.10 (Contraction near the kernel)**.** *There are an integer $p_* \ge H$ and a number $\eta > 0$ such that, if an H-segment and an index $k \ge H$ satisfy $k + p_* \le n - 1$ and $\mathcal{V}_*(Y_k) \le \eta \operatorname{tr} Y_k$, then $\operatorname{tr} Y_{k+p_*} \le \frac{1}{2} e^{2p_*T} \operatorname{tr} Y_k$.*

Proof. Choose $\rho_{\mathcal{N}} < r < e^T$ and $\Gamma_{\mathcal{N}}(p) \le K_r r^p$. For an initial window in $\mathcal{N}$, its forward invariance and (73) give

$$\operatorname{tr} Y_{k+p} \le (H+2) K_r^2 \max(1, r)^{2(H+1)} r^{2(p-H)} \operatorname{tr} Y_k \quad (p \ge H).$$

Choose p_* so this coefficient is at most $e^{2p_*T}/4$. If no $\eta > 0$ works, normalize failing segments to $\operatorname{tr} Y_H = 1$ and restrict them to length $H + p_* + 1$. The comparison bound and $|\psi_l| \le \|e_l\|^2/2$ bound all their Gram entries and values. A convergent subsequence therefore yields an admissible segment with $Y_H \in \mathcal{N}$ but $\operatorname{tr} Y_{H+p_*} \ge e^{2p_*T}/2$, contradicting the bound just proved. □

C.5. Uniform finite-horizon bounds

Proof of Theorem 2.1. Fix $H \ge H_0$ and let p_*, η be from Lemma C.10. Put $\mathsf{A} = |\mathcal{V}_*|/\eta$; choose $\mathsf{B} \ge 1$ bounding the discounted window growth over at most p_* steps by (73). For $\tilde{s}_l = e^{-2(l-k)T} \operatorname{tr} Y_l$, a window with $\mathcal{V}_*(Y_l) > \eta \operatorname{tr} Y_l$ satisfies $\tilde{s}_l \le \mathsf{A} \operatorname{tr} Y_k$. At every other window, Lemma C.10 gives $\tilde{s}_{l+p_*} \le \tilde{s}_l/2$. Advance by one step in the first case and by p_* steps in the second; bound the intervening windows by B. This proves

$$\operatorname{tr} Y_l \le \mathsf{B} \max(1, \mathsf{AB}) e^{2(l-k)T} \operatorname{tr} Y_k \quad (H \le k \le l \le n-1). \tag{74}$$

A trajectory from rest has initial window trace at most $c_H \|e_0\|^2$, by the comparison matrix. Thus (74) gives the upper bound uniformly in f, d, N; the finitely many smaller horizons are covered by the same comparison bound. For the lower bound choose the at-most-two-atom optimal spectrum, using the three-cycle if $T = 0$, and apply Lemma A.8. Embedding unused coordinates with $\mu \|w\|^2/2$ places the example in $\mathbb{R}^5$. The same function and start work at every horizon. Taking Nth roots proves both limits. The three-dimensional bound (7) follows from the explicit construction above. □

The constant C_κ is determined by the number η of Lemma C.10, which is produced by a compactness argument. The same argument gives $\Gamma(p) \le C \rho_{\mathrm{HB}}^p$ for segments with arbitrary initial states.

C.6. Asymptotic expansions

Proof of Theorem 2.3, expansions. At the threshold $V(t) = \log \varphi + O(t^2)$, so $T = \log(q/q_c) + O(\log^2(q/q_c))$, $q_c = \varphi^{-1}$. Since $d \log q / d\kappa = 1/(\sqrt{\kappa}(\kappa - 1))$ equals $1/(4\kappa_c)$ there, exponentiation gives (8). For $q \downarrow 0$, expand the regular system (48), (57) at $z = 0$: $u_1 = 1 - z/2 + z^2/8 - z^3/4 + O(z^4)$. Substitute $z = 2\rho/(1+\rho^2)$ into $\rho/q = u + \sqrt{1+u^2}$ and compare coefficients. The spiral uses $u = 1 - z/2$;

subtraction gives the stated cubic gap. Finally $q \uparrow 1$ lies on the spiral, and its unique positive root converges to the unique positive root of $F(\rho, 1)$. □

D. General tunings and rate certificates

Throughout, (3) holds and $A = 1 + \beta - (x + y)/2$, $B = (y - x)/2 > 0$. The roots of a, b lie in the unit disk by the quadratic Schur criterion; $a - b = (x - y)z$, $a(1) = x$ and $b(1) = y$.

Lemma D.1 (Dominant quadratic roots)**.** *Suppose $\rho_{\mathrm{q}} > \sqrt{\beta}$. If $\rho_a \geq \rho_b$, the roots of a are real and ρ_a is a positive root of a. If $\rho_b \geq \rho_a$, the roots of b are real and distinct, $-\rho_b$ is a root of b, and the other root is $-\beta/\rho_b \in (-\rho_b, 0]$.*

Proof. The root radius of $z^2 - pz + \beta$ is $\sqrt{\beta}$ for $|p| \leq 2\sqrt{\beta}$ and $(|p| + \sqrt{p^2 - 4\beta})/2$ otherwise. The latter is strictly increasing in $|p|$. Since $1 + \beta - x > 1 + \beta - y$, the dominant endpoint has the asserted sign; Vieta's formula gives its other root. □

Lemma D.2 (The angle at the threshold)**.** *For $\omega \in (0, \pi)$,*

$$\cot \Theta(\omega) = \frac{\Xi(\cos \omega)}{\sin \omega}, \quad \Xi(c) = \frac{B^2 - (1 - \beta)^2 - A^2 + 2A(1 + \beta)c - 4\beta c^2}{2B(1 - \beta)}. \tag{75}$$

The polynomial Ξ has degree at most two and is concave, $\Xi(1) < 0$ and $\Xi(-1) < 0$; consequently $\Theta(\pi) = \pi$ and $\Theta(\omega) \to \pi$ as $\omega \to 0$.

Proof of Lemma D.2. The identities $1 - m = -a(z)/(Bz)$ and $1 + m = b(z)/(Bz)$ prove (24). On the unit circle $Bm = (1 + \beta)\cos\omega - A + i(1 - \beta)\sin\omega$. Its imaginary part is positive, and $\cot\Theta = g/(2s)$ gives (75). The leading coefficient is nonpositive, while

$$\Xi(1) = -\frac{xy}{(y - x)(1 - \beta)}, \qquad \Xi(-1) = -\frac{(2 + 2\beta - x)(2 + 2\beta - y)}{(y - x)(1 - \beta)}.$$

Both are negative. Thus $\cot\Theta \to -\infty$ at both ends and the continuous endpoint angles equal π.

□

The comparison with $\omega/2$ is polynomial: with $c = \cos\omega$,

$$H_r(c) = \mathrm{Re}\big[(1 - z)a(z)\overline{b(z)}\big], \qquad z = re^{i\omega}, \tag{76}$$

has degree at most three and $H_1(c) = (1 - c)\mathcal{L}(c)$, where

$$\begin{aligned} \mathcal{L}(c) &= 4\beta c^2 + 2(y - 1 - 2\beta - \beta^2 + \beta x)c + 2 - 2x + xy + 2\beta^2 - 2\beta y \\ &= (y - x)(1 - \beta)\big(1 + c - \Xi(c)\big). \end{aligned} \tag{77}$$

Thus $\Theta(\omega) > \omega/2$ exactly when $\mathcal{L}(\cos\omega) > 0$.

Lemma D.3 (The lag-one cubic)**.** *Put $P = 1 + \beta - x$, $Q = 1 + \beta - y$ and $C = Qr^2 + P\beta$, $D = Pr^2 + Q\beta$, $E = r^4 + PQr^2 + \beta^2$. Then*

$$H_r(c) = E + r^2 D - r(C + D + E + \beta r^2)c + r^2(C + 2\beta)T_2(c) - \beta r^3 T_3(c).$$

At $r = 1$, $H_1(c) = (1 - c)\mathcal{L}(c)$ with $\mathcal{L}$ as in (77), $\mathcal{L}(-1) = (2 + 2\beta - x)(2 + 2\beta - y) > 0$ and $\mathcal{L}(1) = 2(1 - \beta)(y - x) + xy > 0$. For $\omega \in (0, \pi]$, $\Theta(\omega) > \omega/2$ exactly when $\mathcal{L}(\cos\omega) > 0$. The cycle-polynomial identity (78) holds with $\kappa' = x/y$.

Proof. Expand (76) using $z\bar{z} = r^2$ and $z + \bar{z} = 2rc$. Substitution $r = 1$ gives (77) and the endpoint values. Strict monotonicity of cotangent on $(0, \pi)$ gives the angle comparison. For the cycle polynomial of [10],

$$x^2 - 2\big(\beta - c + \kappa'(1 - \beta c)\big)x + 2\kappa'(1 - c)(1 + \beta^2 - 2\beta c) = \kappa'\mathcal{L}(c), \qquad \kappa' = x/y. \tag{78}$$

This follows by collecting the coefficients of c. □

D.1. Weighted integral quadratic constraints

Fix a strict certificate $(\alpha_n, \beta_n, \gamma_\pm, \varepsilon)$ at t, with finitely many lags, and put $\varrho = e^t$. Extend positions by zero to negative times and write $\tilde{e}_k = \varrho^{-k} e_k$, $\tilde{\sigma}_k = \varrho^{-k} \nabla\psi(e_k)$. The weighted sum of interpolation inequalities becomes a quadratic form in their Fourier transforms.

Lemma D.4 (The certificate as an integral quadratic constraint)**.** *There is a 2×2 Hermitian matrix $\Pi(\omega)$ of trigonometric polynomials with the following properties.*

(a) *For every $\psi \in \mathcal{S}$ and every sequence with $\tilde{e} \in \ell_2$,*

$$\mathcal{J}(\tilde{e}, \tilde{\sigma}) := \frac{1}{2\pi} \int_{-\pi}^{\pi} \begin{bmatrix} \hat{e}(\omega) \\ \hat{\sigma}(\omega) \end{bmatrix}^* \Pi(\omega) \begin{bmatrix} \hat{e}(\omega) \\ \hat{\sigma}(\omega) \end{bmatrix} d\omega \ \geq 0,$$

where $\hat{e}$ and $\hat{\sigma}$ are the Fourier transforms of $\tilde{e}$ and $\tilde{\sigma}$, and $\Pi(\omega)$ acts blockwise.

(b) $[1; m]^* \Pi(\omega) [1; m] = \Phi_t(\omega)$, *where* $m = m(t, \omega)$.

Proof. Sum $2\varrho^{-2k-n}\alpha_n (R+T)_{k,k+n}$ and $2\varrho^{-2k-n}\beta_n (R-T)_{k,k+n}$, together with the minimizer inequalities weighted by $4\varrho^{-2k}\gamma_\pm$. All sums converge absolutely because $|\psi(e)| \leq \|e\|^2/2$ and $\|\nabla\psi(e)\| \leq \|e\|$. The function-value coefficient is

$$4 \left[\sum_n (\alpha_n - \beta_n) \sinh(nt) + \gamma_+ - \gamma_- \right] \sum_k \varrho^{-2k} \psi(e_k) = 0.$$

The remaining form is a finite sum of shifts of $(\tilde{e}, \tilde{\sigma})$. Parseval gives its Hermitian trigonometric matrix Π. Evaluation on the wave $(e^{ik\omega}, me^{ik\omega})$ gives exactly Φ_t, since the same cancellation removes its function values. □

Lemma D.5 (Homotopy)**.** *Let $\mathcal{G}$ be the causal convolution operator on $\ell_2(\mathbb{Z}_{\geq 0}; \mathbb{R}^d)$ of a stable, strictly proper rational transfer function G, and let Δ be a causal map on sequences with $\|P_N \Delta\xi\| \leq \|P_N \xi\|$ and $\|P_N \Delta\xi - P_N \Delta\xi'\| \leq \|P_N(\xi - \xi')\|$ for every N, where P_N truncates to the indices $< N$, and with $\Delta 0 = 0$. Let Π be a bounded Hermitian matrix function such that $\mathcal{J}(\xi, \tau\Delta\xi) \geq 0$ for all $\xi \in \ell_2$ and $\tau \in [0, 1]$, with $\mathcal{J}$ as in Lemma D.4, and*

$$[G(e^{i\omega}); 1]^* \Pi(\omega) [G(e^{i\omega}); 1] \leq -\varepsilon < 0 \quad (\omega \in [-\pi, \pi]).$$

Then there is c_* such that for all $\tau \in [0, 1]$ and $f \in \ell_2$ the solution ξ of $\xi = \mathcal{G}\tau\Delta\xi + f$ satisfies $\|\xi\| \leq c_* \|f\|$.

Proof. This is the discrete-time IQC continuation argument [20]. Strict properness makes the loop well posed and causal. For a square-summable solution, put $\zeta = \tau\Delta\xi$. The strict frequency inequality, Parseval, and $\xi = \mathcal{G}\zeta + f$ give

$$\varepsilon \|\zeta\|^2 \leq C_1 \|f\| \|\zeta\| + C_2 \|f\|^2, \qquad \|\xi\| \leq C_* \|f\|,$$

with constants uniform in τ. To establish square summability, continue from $\tau = 0$ in steps $|\tau' - \tau| \leq (2C_* \|\mathcal{G}\|)^{-1}$. For a solution at τ', solve the τ-loop with forcing $P_N\{f + \mathcal{G}(\tau' - \tau)\Delta\xi'\}$. Causality identifies the first N entries, giving $\|P_N \xi'\| \leq C_* \|f\| + \frac{1}{2} \|P_N \xi'\|$. The resulting bound is uniform in N, so the solution lies in ℓ_2. Finitely many steps reach 1. □

Lemma D.6 (Weighted shift)**.** *Let $\varrho > 0$, and suppose that every trajectory $(e_k)_{k \geq -1}$ of (10), with arbitrary e_{-1}, e_0 and a fixed $\psi \in \mathcal{S}$, satisfies $\sum_{j \geq 0} \varrho^{-2j} \|e_j\|^2 \leq C_0(\|e_0\|^2 + \|e_{-1}\|^2)$ for some $C_0 > 0$. Put $C' = C_0 \max(1, \varrho^{-2})$ and $\theta = C'/(1 + C')$. Then $\|e_k\|^2 \leq C_0 \theta^{\lfloor k/2 \rfloor} \varrho^{2k} (\|e_0\|^2 + \|e_{-1}\|^2)$ for all $k \geq 0$.*

Proof. For $S_k = \sum_{j \geq k} \varrho^{-2j} \|e_j\|^2$, restarting at (e_{k-1}, e_k) gives $S_k \leq C'(S_{k-1} - S_{k+1})$. Since $S_{k+1} \leq S_k$, $S_{k+1} \leq \theta S_{k-1}$; induction yields $S_k \leq \theta^{\lfloor k/2 \rfloor} S_0$ and the claimed bound. □

D.2. One-sided multipliers and exact quadratic rates

For nonnegative $\lambda_1, \dots, \lambda_N$, not all zero, define

$$M(z) = \sum_{n=1}^{N} \lambda_n (1 - z^n).$$

This polynomial encodes the nonnegative weights of the lower-lag inequalities; in control terminology it is a finite causal Zames–Falb multiplier. The following criterion transfers its sign between circles by the harmonic minimum principle.

Theorem D.7 (Multiplier criterion)**.** *Assume* (3).

(a) *If* $\mathrm{Re}[M(z)a(z)/b(z)] > 0$ *on* $|z| = 1$, $z \neq 1$, *then* $\rho < 1$. *If* $\mathrm{Re}[M(z)a(z)/b(z)] \geq 0$ *on* $|z| = 1$, *then* $\rho \leq 1$.

(b) *If, in addition to the strict hypothesis of* (a), $\mathrm{Re}[M(z)a(z)/b(z)] \geq 0$ *on* $|z| = r$ *for some* $r \in (\rho_b, 1)$, *then* $\rho \leq r$.

The same holds with a *and* b *interchanged and* ρ_a *in place of* ρ_b, *for multipliers built from upper lags.*

The proof follows Lemma D.8.

Put $a^\sharp(u) = u^2 a(1/u)$ and $b^\sharp(u) = u^2 b(1/u)$. Both have constant coefficient 1 and no zero in the closed unit disk.

Lemma D.8 (Symbols of one-sided certificates)**.** *Let* $\varrho = e^t$ *and* $M(z) = \sum_{n=1}^{N} \lambda_n (1 - z^n)$ *with* $\lambda_n \geq 0$.

(a) *If* $t \geq 0$, *the weights* $\alpha_n = \lambda_n \varrho^{-n}$ *on* $\ell_n \geq 0$ *and* $\gamma_- = \sum_n \alpha_n \sinh nt$ *on* $\bar{g} - C \geq 0$ *cancel* C, *and, with* $u = e^{i\omega}/\varrho$, *the symbol is* $\Phi_t(\omega) = -\frac{\varrho^2}{B^2} |a^\sharp(u)|^2 \,\mathrm{Re}\, \frac{M(u) b^\sharp(u)}{a^\sharp(u)}$.

(b) *If* $t \leq 0$, *the weights* $\alpha_n = \lambda_n \varrho^{n}$ *on* $\ell_n \geq 0$ *and* $\gamma_+ = -\sum_n \alpha_n \sinh nt$ *on* $\bar{g} + C \geq 0$ *cancel* C, *and, with* $z = \varrho e^{i\omega}$, *the symbol is* $\Phi_t(\omega) = -\frac{1}{B^2 \varrho^2} |b(z)|^2 \,\mathrm{Re}\, \frac{M(z) a(z)}{b(z)}$.

At $t = 0$ *the two expressions agree, and on* $|z| = 1$ *the functions* $\mathrm{Re}[M b^\sharp / a^\sharp]$ *and* $\mathrm{Re}[Ma/b]$ *have the same sign. For upper lags the same holds with* a *and* b *interchanged.*

Proof. The stated minimizer weights cancel C. Use $\cosh(nt) \pm \sinh(nt) = \varrho^{\pm n}$ in the lag sum and then (24). For $t \geq 0$, substitute $z = 1/\bar{u}$; for $t \leq 0$, use $z = \varrho e^{i\omega}$ directly. On the unit circle $b^\sharp / a^\sharp = \overline{b/a} = |b/a|^2 a/b$. Conjugating (24) interchanges the lag orientations. □

Proof of Theorem D.7. The real part of $F = M b^\sharp / a^\sharp$ is harmonic on the disk, nonnegative on its boundary and positive at 0. The strong minimum principle makes it positive in the interior; Lemma D.8(a) gives strict certificates at every $t > 0$, hence $\rho \leq 1$. For the strict hypothesis, the zero at $z = 1$ is removed by a small positive minimizer weight: $g(0, 0) < 0$, and the remaining compact arc already has a strict margin. Theorem 3.2 gives $\rho < 1$. For part (b), Ma/b is analytic on $r < |z| < 1$ and has nonnegative real part on both boundary circles. The strong minimum principle gives strict positivity inside; Lemma D.8(b) supplies certificates for $\log r < t < 0$. Together with the positive exponents they imply $\rho \leq r$. Interchanging a, b proves the upper-lag case.

□

Proposition D.9 (Single-harmonic spectra)**.** *Let* $\omega_0 = 2\pi j/k$ *with* $\gcd(j, k) = 1$ *and* $1 \leq j < k/2$. *At* $t = 0$ *the pair* $(\delta_{\omega_0}, 0)$ *is feasible if and only if* $\Theta(\omega_0) \leq \pi/k$.

Proof. For one atom, the signed lag slacks are positive multiples of $\cos\Theta - \cos(2\pi r/k - \Theta)$, $r = 0, \dots, k-1$. All are nonnegative exactly when $\Theta \leq \pi/k$; then $\cos\Theta > 0$ also gives the minimizer constraint. □

Proof of Theorem 3.3. For $M = 1 - z$, the real part in Theorem D.7 is $H_1(c)/|b|^2$, giving (i). Proposition D.9 and start-up realization give the single $\mathbb{R}^3$ trajectory in (ii). If the phase inequality is strict, the nonzero lag residues have a common positive margin. Writing $g_0 = g(0, \omega_0) > 0$, at a small positive exponent multiples of k have slack $g(t)(\cosh(nt) - 1) \geq 0$. For other residues the lower slack satisfies

$$\ell_n(t) \geq \ell_n(0) + g(t)(\cosh(nt) - 1) - 2|g(t) - g_0| - 2|s(t) - s(0)|,$$

and the upper slack has the same estimate. The last two terms tend to zero independently of n, whereas the original nonzero residues have a common positive minimum. All lags therefore remain feasible for some $t > 0$, proving $\rho \geq e^t > 1$. On the relative closure of Ω_1, $\mathcal{L} \geq 0$, so the weak multiplier criterion gives $\rho \leq 1$. A phase-locked harmonic gives the reverse inequality and an actual nonconverging trajectory, while nearby points of Ω_1 converge.

□

For the lag-one multiplier write $\mathsf{G}(z) = (1 - z)a(z)/b(z)$, which is analytic for $|z| > \rho_b$. Where $b(re^{i\omega}) \neq 0$, $H_r(\cos\omega) = |b(re^{i\omega})|^2 \operatorname{Re} \mathsf{G}(re^{i\omega})$.

Lemma D.10 (Near the unit circle)**.** *Assume $\Theta > \omega/2$ on $(0, \pi]$. There is $\eta > 0$ with $\operatorname{Re} \mathsf{G} > 0$ on $\{1 - \eta \leq |z| < 1\}$. If $H_r \geq 0$ on $[-1, 1]$ for some $r \in (\rho_b, 1)$, then $H_{r'} \geq 0$ on $[-1, 1]$ for all $r' \in [r, 1)$.*

Proof. Write $\mathsf{G} = (1 - z)\gamma(z)$, $\gamma = a/b$. For $z = (1 - \delta)e^{i\theta}$,

$$\operatorname{Re} \mathsf{G} = \frac{x}{y}\delta + \frac{\mathcal{L}(1)}{2y^2}\theta^2 + O(\delta^2 + \delta\theta^2 + |\theta|^3).$$

Both displayed coefficients are positive. This proves positivity near 1 inside the disk; compactness supplies a thin positive annulus on the remaining arc. The second assertion is the minimum principle on $r \leq |z| \leq 1$. □

Proof of Theorem 3.4. Lemma D.10 makes the admissible radii a nonempty upper interval with supremum 1. The multiplier criterion gives $\rho \leq r$ at each such radius, proving (a). Continuity gives the necessity in (b). For sufficiency, when $\rho_a > \rho_b$, apply the annulus argument at $r = \rho_a$. When $\rho_b \geq \rho_a$, the only possible boundary pole is $z_b = -\rho_b$, with residue $(1 - z_b)a(z_b)/b'(z_b) < 0$. Let its residue be $v < 0$ and write $z = z_b + w$. The condition $|z| > \rho_b$ implies $\operatorname{Re} w < |w|^2/(2\rho_b)$, so $\operatorname{Re}(v/w) > v/(2\rho_b)$. The real part is thus bounded below near the pole. If d is the annulus diameter, the harmonic function $\operatorname{Re} \mathsf{G} - \epsilon \log(|z - z_b|/d)$ tends to $+\infty$ there and has nonnegative lower limits at all other boundary points. The minimum principle and $\epsilon \downarrow 0$ give nonnegativity in the annulus; the strong minimum principle gives strict positivity. Thus $r_1 = \rho_{\mathrm{q}}$. For (c), $\beta > 0$ and $|A| < 2\sqrt{\beta}$. Choose $2\sqrt{\beta}\cos\omega_0 = A$; at $t_0 = \log\sqrt{\beta}$, this mode has $m = 0$, $g = 1$. Its signed slacks are at least $\cosh(t_0) - 1 > 0$. The estimate $\ell_n, \upsilon_n \geq g(\cosh t - 1) - 2|s|$ retains positivity at a slightly larger negative exponent, uniformly in n, so realization gives $\rho > \sqrt{\beta}$.

□

Corollary D.11 (Explicit regions)**.** *Assume* (3).

(a) *$\mathcal{L}(-1) = (2 + 2\beta - x)(2 + 2\beta - y) > 0$ and $\mathcal{L}(1) = 2(1 - \beta)(y - x) + xy > 0$. Hence Ω_1 is the set where $\beta = 0$, or the vertex $c_* = (1+2\beta+\beta^2-\beta x-y)/(4\beta)$ lies outside $(-1, 1)$, or the discriminant $(y - 1 - 2\beta - \beta^2 + \beta x)^2 - 4\beta(2 - 2x + xy + 2\beta^2 - 2\beta y)$ is negative. On the boundary of Ω_1 the tangency frequency is $\omega_* = \arccos c_*$, and the phase-locked points of Theorem 3.3*(iii) *are those with $c_* = \cos(2\pi/K)$.*

(b) *With $c_{jk} = \cos(2\pi j/k)$ and $h_{jk} = \cot(\pi/k)\sin(2\pi j/k)$,*

$$\begin{aligned} \Theta(2\pi j/k) \leq \pi/k &\iff \Xi(c_{jk}) \geq h_{jk} \\ &\iff \mathcal{L}(c_{jk}) \leq (y - x)(1 - \beta)(1 + c_{jk} - h_{jk}). \end{aligned}$$

For $j = 1$ the right side vanishes, and the K-tongue is $\{\mathcal{L}(\cos(2\pi/K)) \leq 0\}$.

(c) *If $\rho_a \geq \rho_b$ and $\rho_q > \sqrt{\beta}$, then with $r = \rho_a$, $H_r(c) = (1-c)K_\mu(c)$ for a convex polynomial K_μ of degree at most two with $K_\mu(-1) \geq 0$; within Ω_1 the set $\mathcal{Q}$ is given there by $K_\mu(1) \geq 0$ together with the vertex-or-discriminant condition for K_μ. If $\rho_b > \rho_a$ and $\rho_q > \sqrt{\beta}$, then with $r = \rho_b$, $H_r(c) = (1+c)K_L(c)$ for a concave polynomial K_L of degree at most two with $K_L(1) \geq 0$, and within Ω_1 the set $\mathcal{Q}$ is given there by $K_L(-1) \geq 0$.*

The proof is given together with Lemma D.12 below.

Lemma D.12 (Quotients at the dominant real roots)**.** *Let $r > 0$. If $a(r) = 0$, equivalently $x = (1-r)(1-\beta/r)$, then $H_r(c) = (1-c)K_\mu(c)$, where K_μ has degree at most two, quadratic coefficient $4\beta r^3$, and $K_\mu(-1) = (\beta+r^2)(1+r)b(-r)$. If $b(-r) = 0$, equivalently $y = (1+r)(1+\beta/r)$, then $H_r(c) = (1+c)K_L(c)$, where K_L has degree at most two, quadratic coefficient $-4\beta r^3$, and $K_L(1) = (\beta + r^2)(1-r)a(r)$. Both quotients are determined by polynomial division of the cubic in Lemma D.3.*

Proof of Lemma D.12 and Corollary D.11. Substitute $x = (1-r)(1-\beta/r)$ or $y = (1+r)(1+\beta/r)$ in (76) and divide by $1-c$ or $1+c$. The displayed quotients and endpoint identities follow by coefficient comparison. Lemma D.1 gives the required endpoint signs. A convex polynomial of degree at most two is positive on $[-1,1]$ precisely when its endpoints are positive and either its vertex lies outside $(-1,1)$ or its discriminant is negative. For nonnegativity replace the strict signs by weak signs; a concave polynomial is nonnegative exactly when both endpoints are. These tests prove (a),(c), including the linear case $\beta = 0$. For (b), use the monotonicity of cotangent and (77). □

Proof of Corollary 3.5. For Polyak's tuning, put $u_0 = (q^{-1} - q)/2$. Then $2u_0(1+c-\Xi(c)) = (c+u_0)^2 + 2u_0 - 1$. Its minimum is positive exactly when $u_0 > 1/2$; at equality the double contact is $c = -1/2$, giving the three-tongue threshold. For $\beta = 0$, the linear $\mathcal{L}$ has positive endpoints. Normalization gives $\mathrm{Lip}(I - \alpha\nabla f) \leq |1 - (x+y)/2| + (y-x)/2 = \max(|1-x|, |1-y|)$. The endpoint scalar quadratics attain that rate, giving the stated formula and, by the quotient tests, membership in $\mathcal{Q}$.

□

E. Threshold geometry and periodic approximation

For signed $n \neq 0$, the basic polynomial identity is

$$\Lambda_n(\omega) = \frac{\sin\Theta}{\sin\omega}\left[\Xi(c)(1 - T_{|n|}(c)) - \mathrm{sgn}(n)(1-c^2)U_{|n|-1}(c)\right], \quad c = \cos\omega. \tag{79}$$

At the endpoints, $\Lambda_n(0) = 0$, $\Lambda_n(\pi) = (-1)^n - 1 \leq 0$ and $\cos\Theta = -1$. Removing endpoint mass therefore preserves feasibility and leaves a nonzero measure.

Lemma E.1 (Ends)**.** *Let q be a tuning and $s \in \mathbb{R}$. Near $\omega = 0$ and $\omega = \pi$,*

$$\pi - \Theta_{q,s}(\omega) = \arctan\frac{\sin\omega}{-\Xi_q(\cos\omega) - s\sin\omega}, \tag{80}$$

with $-\Xi_q(\pm 1) > 0$. Hence $\Theta_{q,s}(\omega)$ is continuous in (q,s,ω) on (tunings)$\times\mathbb{R}\times[0,\pi]$ and extends to a real analytic function of (q,s,ω) on a neighborhood of every compact subset. With $\tau_q = -1/\Xi_q(1)$, every lag function of $\Theta_{q,s}$ satisfies $\Lambda_n(0) = \partial_\omega\Lambda_n(0) = 0$ and $\partial_\omega^2\Lambda_n(0) = -n(n+2\tau_q)$.

Proof. The denominator $-\Xi_q(\cos\omega) - s\sin\omega$ is positive near the endpoints. Formula (80) gives the local analytic extensions, agreeing with the interior arccotangent. At 0, $\pi - \Theta = \tau_q\omega + O(\omega^2)$; substitute in $\Lambda_n = -\cos(\pi - \Theta) + \cos(n\omega + \pi - \Theta)$. □

Lemma E.2 (Exact alternative)**.** *Let Θ be a curve and $\mathcal{N} \subset \mathbb{Z}\setminus\{0\}$. Then $\mathrm{FP}[\mathcal{N}](\Theta)$ is infeasible if and only if a strict certificate symbol with lags in $\mathcal{N}$ exists.*

Proof. For finitely many lags, take the compact convex hull of the moment curve

$$\omega \longmapsto \big((\Lambda_n(\omega))_n, \cos\Theta(\omega)\big).$$

Feasibility asks whether this hull meets the nonnegative orthant. Strict separation [3, §2.5.1] gives the nonnegative certificate coefficients when they are disjoint: the separating functional has nonnegative coordinates because the orthant is a cone. For arbitrary lags, normalize feasible measures to probability. Compactness and the finite-intersection property imply that finite-subsystem feasibility gives full feasibility. Integrating a strict symbol proves the converse. □

E.1. General spectra and monotone thresholds

Realization of general spectra at exponent zero. For the lower bound in Theorem 3.7, let $(\nu, 0)$ be feasible at $t = 0$, with ν normalized to have mass one. Work in the real Hilbert space $L^2(\nu;\mathbb{C}) \oplus \mathbb{R}$, and put $z(\omega) = e^{i\omega}$, $m = m(0,\omega)$ and $r = (z - A + \beta)/B$. As in Lemma A.8, choose $\tau \in (-1, 1)$ so that the recurrence $p_{-1} = p_0 = 1$, $p_{k+1} = (A + B\tau)p_k - \beta p_{k-1}$ satisfies $p_k \to 0$ and $\delta_0 = \inf_{k\geq 1}(1 - p_k)^2 > 0$. For $Y > 0$ prescribe

$$\begin{aligned} &e_0 = (1, Y), && \sigma_0 = (r, \tau Y), && \psi_0 = \tfrac{1}{2}\int \operatorname{Re} r\, d\nu + \tfrac{\tau}{2}Y^2, \\ &e_k = (z^k, p_k Y), && \sigma_k = (mz^k, \tau p_k Y), && \psi_k = \tfrac{1}{2}\int \operatorname{Re} m\, d\nu + \tfrac{\tau}{2}p_k^2 Y^2 \quad (k \geq 1). \end{aligned}$$

Include the minimizer datum $(0, 0, 0)$ and set $e_{-1} = e_0$. The identities $Br = z - A + \beta$ and $Bmz = z^2 - Az + \beta$ give (10) at every step.

All interpolation inequalities excluding index 0 follow by integrating the feasible spectral inequalities. The scalar summand is the quadratic $\tau u^2/2$: it contributes $T_{ij} = 0$ and $(1-\tau^2)(u_i - u_j)^2/4$ to R_{ij}. The exceptional first-summand deficits $|T_{0k}| - R_{0k}$, $k \geq 1$, and $|T_{0\star}| - R_{0\star}$ are bounded above by a finite $M \geq 0$, because z^k, m and r are uniformly bounded in $L^2(\nu)$ and their prescribed values are bounded. Thus

$$Y^2 \geq \frac{4M}{(1-\tau^2)\min\{1, \delta_0\}}$$

makes every interpolation inequality valid, with one Y independent of the horizon.

For each N, restrict the data to the finite-dimensional span of $e_0, \dots, e_N, \sigma_0, \dots, \sigma_N$. Theorem A.2 and Lemma A.1 give an objective in $\mathcal{F}_{\mu,L}$ realizing these first N steps. Since $\|e_N\|^2 \geq \int |z|^{2N}\, d\nu = 1$ and $\|e_0\|^2 = 1 + Y^2$, we obtain $W_N \geq (1 + Y^2)^{-1/2}$ for all N. The finite-dimensional objective and its dimension may depend on N; the lower-bound constant is common to all horizons.

Proposition E.3 (Monotonicity and phase floors)**.** (a) *Let Θ, Θ' be curves with $\Theta' \leq \Theta$ on $(0, \pi)$. If* $\mathrm{FP}(\Theta)$ *is feasible, then* $\mathrm{FP}(\Theta')$ *is feasible.*

(b) *For a tuning, a strict certificate at $t = 0$ exists if and only if there are finitely many $\lambda_n \geq 0$, $n \neq 0$, and $\gamma > 0$ such that $M(z) = \gamma + \sum_n \lambda_n(1 - z^n)$ satisfies $\Theta(\omega) > \arg M(e^{i\omega}) + \pi/2$ for all $\omega \in (0, \pi]$.*

(c) *For every such M and every $\omega = 2\pi j/k$, $\arg M(e^{i\omega}) + \pi/2 \geq \pi/k$. For $M = 1 - z$, $\arg M(e^{i\omega}) + \pi/2 = \omega/2$.*

Proof. Delete the phase-π set from a feasible measure; its contribution to every constraint is nonpositive, and its minimizer contribution is strictly negative. On the remainder, reweight by $\sin\Theta/\sin\Theta'$. This ratio is bounded: it is at most 1 where $\Theta' > \pi/2$, and at most $1/\sin(\min\Theta')$ elsewhere. The normalized lag $\Lambda_n/\sin\Theta = \ell_n[\cot\Theta]$ and normalized minimizer are nondecreasing in cotangent, proving (a). For (b), $\operatorname{Re} M \geq \gamma > 0$ and $S = \operatorname{Re}(e^{-i\Theta}M)$. Its negativity is exactly the phase-floor inequality; compactness supplies a uniform margin. A strict symbol has $\gamma > 0$ by its value at 0. At $2\pi j/k$, all nonzero summands $1 - z^n$ lie in the sector $[-\pi/2 + \pi/k, \pi/2 - \pi/k]$, as does their positive sum with γ. This proves (c), the discrete-time phase limitation of [34]; $M = 1 - z$ gives equality with $\omega/2$. □

E.2. The dip and coordinates of a gap

Theorem E.4 (Convexity). *Let q be a tuning, $c = \cos\omega$, and write $\Xi(c) = \Xi_0 + \Xi_1 c + \Xi_2 c^2$. Then*

$$h_q''(\omega)\,\sin^3\omega = \tfrac{1}{2}(1+c)^2\,\Xi(1) + \tfrac{1}{2}(1-c)^2\,\Xi(-1) + \Xi_2\,(1-c^2)^2 < 0 \quad (0<\omega<\pi),$$

so h_q is strictly concave on $(0,\pi)$. Consequently, for every shift s, $\Theta_{q,s}''(\omega) > 0$ at every $\omega \in (0,\pi)$ with $\Theta_{q,s}(\omega) \le \pi/2$, and $\{\Theta_{q,s} \le \pi/2\}$ is a compact interval in $(0,\pi)$ or empty.

Proof. Twice differentiating $h = \Xi(c)/\sin\omega$ and expressing Ξ_0, Ξ_1 through $\Xi(1), \Xi(-1)$ gives the displayed identity. Its sign follows from $\Xi(\pm 1) < 0$ and $\Xi_2 \le 0$. For $u = h + s$, differentiation gives $\Theta'' = -u''/(1+u^2) + 2u(u')^2/(1+u^2)^2 > 0$ where $u \ge 0$. The set $\{u \ge 0\}$ is a compact interval because u is strictly concave and tends to $-\infty$ at both ends. □

Corollary E.5 (Connected phase sublevel sets). *Let $\Theta = \Theta_{q,s}$ be a shifted curve, $\ell(\omega) = n\omega/2 - \pi m$ a lattice line and J an interval on which $\ell \le \pi/2$. Then $\{\omega \in J : \Theta(\omega) < \ell(\omega)\}$ is an interval. In particular the dip $\{\omega : \Theta(\omega) < \omega/2\}$ and the closed dip*

$$\bar\Delta = \{\omega \in (0,\pi] :\ \Theta(\omega) \le \omega/2\}$$

are intervals; $\bar\Delta$ is compact and contained in $(0,\pi)$, and if the dip is nonempty, $\bar\Delta$ is its closure.

Proof. On the interval $\{\Theta \le \pi/2\}$, the function $\Theta - \ell$ is strictly convex. Its strict and weak sublevel sets are intervals. Taking $\ell = \omega/2$ gives compact containment and the closure statement. □

Lemma E.6 (Dip lemma). *Let Θ be a curve and $\bar\Delta = \{\omega \in (0,\pi] : \Theta \le \omega/2\}$. Every solution of $\mathrm{FP}(\Theta)$ gives positive mass to $\bar\Delta$.*

Proof. Outside $\bar\Delta \cup \{0\}$ the lag-one function is strictly negative. A measure giving no mass to $\bar\Delta$ violates that lag unless it is carried by 0, where it violates the minimizer constraint. □

Proof of Theorem 3.8. Strict convexity makes ω_* the unique nonzero zero of $\Lambda_1 \le 0$. At that frequency $\Lambda_{-K} = 2\sin(K\omega_*/2)\sin((K+1)\omega_*/2) < 0$. Choose a small positive weight ϵ on $-K$. Near ω_* it removes the zero; away from $0, \omega_*$ the lag-one strict margin is preserved. Near 0, the expansion in Lemma E.1 retains a negative quadratic coefficient for small ϵ. Adding a still smaller $\gamma\cos\Theta$, $\gamma > 0$, makes the symbol strictly negative also at 0. The exact criterion gives convergence. The remaining boundary contacts are precisely the locked ones of Theorem 3.3(iii).

□

Proof of Theorem 3.10. The compact interval $\bar\Delta$ either is empty, contains one of the frequencies $2\pi/K'$, or lies in the unique interval between consecutive such frequencies. The first two cases are Theorem 3.3. In the last case a singleton is a tangency handled by Theorem 3.8; otherwise its interior is the dip and the endpoint inequalities define I_K. For a singleton, decreasing κ strictly raises the angle curve by Lemma E.10, placing the point on $\partial\Omega_1$.

□

Proposition E.7 (Coordinates of a gap). (a) *The map $q \mapsto \Xi_q$ is a homeomorphism from the tunings with $\beta > 0$ onto the set of real quadratics $\Xi(c) = \Xi_0 + \Xi_1 c + \Xi_2 c^2$ with $\Xi_2 < 0$, $\Xi(1) < 0$ and $\Xi(-1) < 0$. Tunings with $\beta = 0$ have $\Xi_2 = 0$ and lie in Ω_1.*

(b) *Let $K \ge 2$. A tuning lies in I_K if and only if $\beta > 0$ and*

$$1 + c - \Xi_q(c) = a\,(c - r_1)(c - r_2) \quad \text{with} \quad c_K < r_1 < r_2 < c_{K+1}.$$

Then $a = -\Xi_2 > a_0(r_1, r_2) := 2/((1-r_1)(1-r_2))$, the closed dip is $[\arccos r_2, \arccos r_1]$, and $q \mapsto (a, r_1, r_2)$ is a homeomorphism of I_K onto

$$\mathcal{G}_K = \{(a, r_1, r_2) :\ c_K < r_1 < r_2 < c_{K+1},\ a > a_0(r_1, r_2)\}.$$

In particular I_K is connected.

(c) *Consider the closure of $\mathcal{G}_K$ within $\{a > a_0(r_1, r_2),\ r_1 > -1\}$, where the quadratic $1 + c - a(c - r_1)(c - r_2)$ still corresponds to a tuning by (a). Its face $\{r_1 = r_2 = r,\ c_K < r < c_{K+1}\}$ consists exactly of the points of $\partial\Omega_1$ that are not phase-locked and whose tangency frequency $\arccos r$ lies in the window. Its face $r_2 = c_{K+1}$ lies on the edge $\Theta(2\pi/(K+1)) = \pi/(K+1)$ of the $(K+1)$-tongue, and $r_1 = r_2 = c_{K+1}$ is a phase-locked point. For $K \geq 3$ the face $r_1 = c_K$ lies on the edge of the K-tongue, $r_1 = r_2 = c_K$ is a phase-locked point, and $(r_1, r_2) = (c_K, c_{K+1})$ is a junction of the two tongues. For $K = 2$, as $r_1 \downarrow -1$ the tunings approach the line $\alpha L = 2(1 + \beta)$.*

(d) *For every fraction j/k in lowest terms with $1/(K+1) < j/k < 1/K$, put $c_{jk} = \cos(2\pi j/k)$ and $h_{jk} = \cot(\pi/k)\sin(2\pi j/k)$. Then $h_{jk} > 1 + c_{jk}$, and the tuning with coordinates $(a, r_1, r_2) \in \mathcal{G}_K$ satisfies $\Theta(2\pi j/k) \leq \pi/k$ if and only if*

$$r_1 < c_{jk} < r_2 \qquad \text{and} \qquad a \geq A_{jk}(r_1, r_2) := \frac{h_{jk} - 1 - c_{jk}}{(c_{jk} - r_1)(r_2 - c_{jk})}.$$

(e) *The face $a = a_0(r_1, r_2)$ is $\Xi(1) = 0$, and $\Xi_q(1) = -\alpha L/((\kappa - 1)(1 - \beta))$. Along every sequence of tunings whose quadratics converge to a quadratic with $\Xi(1) = 0$ and $\Xi(-1) < 0$, the condition number tends to infinity.*

The proof follows Lemma E.8.

For a quadratic Ξ, put

$$E_\Xi(z) = \Xi\left(\frac{z + z^{-1}}{2}\right) - \frac{z - z^{-1}}{2}, \qquad p_\Xi(z) = z^2 E_\Xi(z). \tag{81}$$

Lemma E.8. *For every tuning and $z \neq 0$, $2B(1 - \beta)E_{\Xi_q}(z) = -a(z)\, b(z^{-1})$.*

Proof. Expand in the five Laurent monomials $z^{-2}, \ldots, z^2$; their coefficients agree by (75). □

Proof of Proposition E.7. For (a), p_Ξ has no unit-circle zero: on that circle $E_\Xi = \Xi(\cos\omega) - i\sin\omega$, whose real-axis intersections are the two negative values $\Xi(\pm 1)$. Its winding number is zero, so p_Ξ has two inner and two outer roots. Since $p_\Xi(0) = \Xi_2/4 < 0$ and $p_\Xi(\pm 1) < 0$, the inner roots are conjugate or have the same real sign. Their product is $\beta \in (0, 1)$. Take them as the roots of $a = z^2 - Pz + \beta$; reciprocal outer roots give $b = z^2 - Qz + \beta$, since the quartic's leading and constant coefficients agree. Coefficient comparison in Lemma E.8 gives $(y - x)(1 - \beta) > 0$, where $x = 1 + \beta - P$, $y = 1 + \beta - Q$. Schur stability gives $0 < x < y < 2(1 + \beta)$. This is the unique tuning; separation of the inner and outer root multisets also gives continuous dependence, including multiple roots. At $\beta = 0$, $\mathcal{L}$ is linear and positive.

For (b), the dip is exactly the negative set of $1 + c - \Xi(c) = a(c - r_1)(c - r_2)$. The window and strict tongue exclusions give the ordered roots; $\Xi(1) < 0$ is $a > a_0$, while $\Xi(-1) < 0$ is automatic. These identities give both continuous coordinate maps. The region above the continuous graph a_0 over its open triangle is path connected. Part (c) follows by allowing the ordered roots to coincide or reach window endpoints. For $K = 2$, the face $r_1 = -1$ is $\Xi(-1) = 0$, hence the excluded stability boundary.

For (d), $h_{jk} - 1 - c_{jk} = 2\cos(\pi j/k)\sin(\pi(j-1)/k)/\sin(\pi/k) > 0$. Substitute $\Xi(c) = 1 + c + a(c - r_1)(r_2 - c)$. For (e), the endpoint identity is in the proof of Lemma D.2. If $\Xi_n \to \bar{\Xi}$ with $\bar{\Xi}(1) = 0$ and $\bar{\Xi}(-1) < 0$, the negative-endpoint identity rules out $y_n \to 0$. Hence $\kappa_n - 1 = y_n/(|\Xi_n(1)|(1 - \beta_n)) \to \infty$. □

E.3. Monotonicity, periodicity and the threshold

Proposition E.9 (Monotone form)**.** *For a tuning q and $s \in \mathbb{R}$, $\mathrm{FP}(\Theta_{q,s})$ is feasible if and only if there is a nonzero finite measure $\nu \geq 0$ on $(0, \pi)$ with $\int |h_q + s|\, d\nu < \infty$ and*

$$\int \ell_n[h_q + s]\, d\nu \geq 0 \quad (n \neq 0), \qquad \int (h_q + s)\, d\nu \geq 0.$$

If ν satisfies these conditions for a function h, and h' is ν-measurable with $h' \geq h$ on $\operatorname{supp}\nu$ and bounded above there, then $\int \ell_n[h']\,d\nu = \int \ell_n[h]\,d\nu + \int (h'-h)(1-\cos n\omega)\,d\nu$, and ν satisfies them for h'.

Proof. Delete the endpoint atoms and put $\nu = \sin\Theta\, w$. Then $\int |h|\,d\nu = \int |\cos\Theta|\,dw < \infty$, and the constraints transform as stated. Conversely $w = \nu/\sin\Theta$ is finite because $1/\sin\Theta = \sqrt{1+h^2} \leq 1+|h|$. Measurability and $h \leq h' \leq M$ give $|h'| \leq |h| + M^+$, hence integrability. The displayed increment formula follows by linearity. □

Lemma E.10 (The condition number lowers the curve)**.** *With $y = \alpha L$,*

$$\frac{\partial \Xi_q(c)}{\partial \kappa} = \frac{N(c)}{y(1-\beta)(\kappa-1)^2},$$

where $N(c) = \big(y-(1+\beta)(1-c)\big)^2 + (1-\beta)^2(1-c^2) > 0$ for $-1 \leq c \leq 1$. Hence $h_{(\kappa',\alpha L,\beta)} \geq h_{(\kappa,\alpha L,\beta)} + \varepsilon$ on $(0,\pi)$ for $\kappa' > \kappa$ and some $\varepsilon > 0$, and the curve Θ_q decreases strictly as κ grows.

Proof. Differentiate (75) using $x = y/\kappa$. Completing the square gives $N(c)$. It is positive for $|c| < 1$ and has endpoint values $N(1) = y^2 > 0$, $N(-1) = (y-2(1+\beta))^2 > 0$. Compactness supplies a uniform positive increment of Ξ; division by $0 < \sin\omega \leq 1$ gives the cotangent increment. □

Proposition E.11 (Threshold)**.** *For every tuning q there is a number $\sigma(q)$, the threshold, such that $\mathrm{FP}(\Theta_{q,s})$ is feasible if and only if $s \geq \sigma(q)$. Moreover:*

(a) $\mathcal{D} = \{q :\ \sigma(q) \leq 0\}$;
(b) *if $h_{q'} \geq h_q + e$ on $(0,\pi)$ for a constant e, then $\sigma(q') \leq \sigma(q) - e$;*
(c) *the set $\{(q,s) :\ s \geq \sigma(q)\}$ is closed, and $\mathrm{FP}(\Theta_{q,\sigma(q)})$ is feasible.*

Proof of Proposition E.11. The monotone form makes feasibility increasing in s. For large s, the atom at $2\pi/3$ is feasible; for $s < -\sup h_q$, the minimizer constraint fails. Normalize feasible measures to probability. Joint continuity from Lemma E.1 and weak compactness show that the feasible epigraph is closed, so its finite lower endpoint is attained. The exact criterion proves (a), and monotonicity proves (b). The same reasoning applies to arbitrary lag subsets and finite periodic grids; for P_L the atom at $2\pi/L$ gives the upper witness.

□

Proposition E.12 (Periodic spectra)**.** *Let $k \geq 3$, $\omega_j = 2\pi j/k$, $\theta_j = \Theta(\omega_j)$, and let w be a nonzero measure on $\{\omega_j : 1 \leq j < k/2\}$ with weights w_j. Then w solves $\mathrm{FP}(\Theta_q)$ if and only if $\sum_j w_j \Lambda_n(\omega_j) \geq 0$ for $n = 1,\dots,k-1$ and $\sum_j w_j \cos\theta_j \geq 0$. In that case there are $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^{2r+1})$, with r the number of nonzero weights, and x_0 whose trajectory converges to a k-periodic orbit and stays at distance at least $c\|x_0 - x^\star\| > 0$ from $x^\star$. In particular $\mathcal{T}_k \subset \mathcal{D}$.*

Proof. The lag values depend only on $n \bmod k$, with value zero in residue zero. Reweight by $|g-2is|^{-1}$ and use start-up realization: the iterates are $((\sqrt{\nu_j}e^{iN\omega_j})_j, p_N Y)$, $p_N \to 0$. The planar part has fixed positive norm. An atom at π for even k can be deleted by its nonpositive lag and minimizer contributions. □

Lemma E.13 (Pinning identity)**.** *For a curve Θ, $k \geq 1$, $r, m \in \mathbb{Z}$ and $\omega \in [0,\pi]$, with $\Lambda_0 :\equiv 0$,*

$$\Lambda_{r+mk}(\omega) = \cos\Theta(\omega) - \cos\big(mk\omega - \psi_r(\omega)\big), \qquad \psi_r = \Theta - r\omega.$$

In particular, for $k \geq 3$, $\Lambda_n = \Lambda_r$ on Ω_k for $n \equiv r \pmod k$, and, if Θ is differentiable, for $n = r+mk$, $\partial_\omega \Lambda_n = \partial_\omega \Lambda_r + mk\sin(r\omega - \Theta)$ on Ω_k.

Proof. Substitute $n = r + mk$ in (25) and differentiate. On the grid, $mk\omega \in 2\pi\mathbb{Z}$. □

E.4. Smoothing and sampling

Extend $\Theta = \Theta_{q,s}$ oddly to $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and put $\zeta = e^{i\Theta}$. The phase is represented locally by real-valued arguments of ζ. Lemma E.1 shows that ζ is C^1 with Lipschitz derivative, and analytic between $0, \pi$. Put $M_1 = \sup|\Theta'|$, $M_2 = \sup_{\mathbb{T}\setminus\{0,\pi\}}|\Theta''|$. Symmetrizing a measure and folding it back preserve all constraints. Write

$$F_\mu(n) = \int_{\mathbb{T}} e^{in\omega}\bar{\zeta}(\omega)\,d\mu, \quad c_0(\mu) = \int \cos\Theta\,d\mu, \quad C_n(\mu) = \mathrm{Re}(F_\mu(0) - F_\mu(n)).$$

A feasible probability measure is nondegenerate of order n_0 when $c_0 > 0$ and $C_n > 0$ for $0 < |n| < n_0$. Choose an even nonnegative smooth bump ψ of integral 1 supported in $[-1/4, 1/4]$. Set $\varphi = \psi * \psi$, $K_\varepsilon = \varepsilon^{-1}(\varphi * \varphi)(\cdot/\varepsilon)$, periodized for $0 < \varepsilon < \pi$, and $m_\varphi = \int t^2\varphi(t)\,dt$, $m_K = \int t^2(\varphi * \varphi)(t)\,dt$. Then $\widehat{K_\varepsilon}(\xi) = \hat{\varphi}(\varepsilon\xi)^2 \in [0, 1]$.

Lemma E.14 (Smoothing)**.** *For every symmetric probability measure μ on $\mathbb{T}$ and every $n \in \mathbb{Z}$, the density $f = K_\varepsilon * \mu$ satisfies*

$$C_n(f) \geq \widehat{K_\varepsilon}(n)\,C_n(\mu) + (1 - \widehat{K_\varepsilon}(n))c_0(\mu) - D_0 - 3D_n - M_2 m_K \varepsilon^2,$$

where $D_n = \sup_{|\xi-n|\leq M_1}|\widehat{K_\varepsilon}(\xi) - \widehat{K_\varepsilon}(n)| \leq M_1 \min\{2m_\varphi\varepsilon^2(|n| + M_1), \varepsilon\}$. Moreover $\int \cos\Theta\, f \geq c_0(\mu) - D_0 - M_2 m_K\varepsilon^2/2$.

Proof. Use Fubini and the uniform local Taylor estimate $\Theta(v + t) = \Theta(v) + \Theta'(v)t + r$, $|r| \leq M_2 t^2/2$. The modulated convolution differs from $e^{inv}\bar{\zeta}(v)\widehat{K_\varepsilon}(n - \Theta'(v))$ by at most $M_2 m_K\varepsilon^2/2$. Replace this kernel value by $\widehat{K_\varepsilon}(n)$ and use $|\Lambda_n| \leq 2$: the lag error is $D_0 + 3D_n + M_2 m_K\varepsilon^2$. For the derivative bound, use $|\hat{\varphi}'(\eta)| \leq \min(m_\varphi|\eta|, 1/2)$ in $\widehat{K}'_\varepsilon = 2\varepsilon\hat{\varphi}\hat{\varphi}'$. The same estimate at $n = 0$ gives the minimizer inequality. □

Lemma E.15. *There is $c_\varphi > 0$ with $1 - \widehat{K_\varepsilon}(\xi) \geq c_\varphi \min\{\varepsilon^2\xi^2, 1\}$ for all real ξ.*

Proof. For $|\eta| \leq 2\pi$, $1 - \hat{\varphi}(\eta) \geq 2m_\varphi\eta^2/\pi^2$ by $1 - \cos u \geq 2u^2/\pi^2$ on $[-\pi, \pi]$. Away from zero, strict loss, continuity and the Riemann–Lebesgue lemma give a positive lower bound. Since $1 - \hat{\varphi}^2 \geq 1 - \hat{\varphi}$, combine the two ranges. □

Sampling on L equally spaced points identifies Fourier frequencies that differ by multiples of L, producing the aliasing estimate below.

Lemma E.16 (Aliasing)**.** *Let $f \geq 0$ be a smooth even function on $\mathbb{T}$ and $\mu_L = \frac{2\pi}{L}\sum_{j=0}^{L-1} f(2\pi j/L)\,\delta_{2\pi j/L}$. Then μ_L is symmetric, F_{μ_L} is L-periodic in n, and there is $C(f, \Theta)$ with $|F_{\mu_L}(n) - F_f(n)| \leq C(f, \Theta)L^{-2}$ for $|n| \leq L/2$.*

Proof. For $h = f\bar{\zeta}$, the derivative is Lipschitz and the periodic boundary terms cancel. Two integrations by parts give $|\hat{h}(m)| \leq C(1 + |m|)^{-2}$. Its Fourier series therefore converges absolutely, and finite-grid orthogonality gives the aliasing formula

$$F_{\mu_L}(n) = 2\pi\sum_{m\in\mathbb{Z}}\hat{h}(mL - n), \qquad F_f(n) = 2\pi\hat{h}(-n).$$

For $|n| \leq L/2$, the omitted terms are bounded by $CL^{-2}\sum_{m\neq 0}(|m| - 1/2)^{-2}$. This is the standard trapezoidal aliasing argument of [29, §3], here using the proved polynomial Fourier decay. Symmetry and periodicity follow directly from the weights. □

Lemma E.17 (Sampling)**.** *Let $\Theta = \Theta_{q,s}$ and let μ be a feasible symmetric probability measure for* $\mathrm{FP}(\Theta)$ *that is nondegenerate of order n_0, where n_0 is the least integer exceeding the larger root of $P(x) = c_\varphi c_0 x^2 - 6M_1 m_\varphi x - (8M_1^2 m_\varphi + M_2 m_K)$, $c_0 = c_0(\mu)$. Then there is L_0 such that* $\mathrm{P}_L(\Theta)$ *is feasible for every $L \geq L_0$.*

Proof. Choose one small $\varepsilon \le 1/n_0$ and apply the smoothing estimate in three ranges. For $|n| \ge 1/\varepsilon$, the kernel loss gives a fixed positive fraction of c_0. For $n_0 \le |n| < 1/\varepsilon$, it gives the lower bound $\varepsilon^2 P(|n|) \ge \varepsilon^2 P(n_0) > 0$. For the finitely many $0 < |n| < n_0$, nondegeneracy survives the $O(\varepsilon^2)$ error. The smoothed density thus has one positive lag margin δ for all signed $n \ne 0$, and a positive minimizer margin. Lemma E.16 preserves these margins on every sufficiently fine grid: choose balanced representatives $|n| \le L/2$ for nonzero residues; residue zero is identically zero. Fold the sampled measure and delete its endpoints. □

Proof of Theorem 3.11. Use the monotone-form measure ν at a feasible profile. If $\int (1-\cos n\omega)\, d\nu = 0$ for some $n \ne 0$, its support lies on a periodic grid, which remains feasible under every weak increase of κ. Otherwise every such integral is positive. A uniform strict profile increase gives strictly positive lag and minimizer increments; the reciprocal-sine weighting is finite by Proposition E.9. The resulting measure is nondegenerate of every finite order, so Lemma E.17 gives every sufficiently large period. Apply the same argument at the attained shift $\sigma(q)$ and any additional $\varepsilon > 0$ to obtain $\inf_L \sigma_L = \sigma$. Finally $\mathcal{T}_L \subset \mathcal{D}$, $\mathcal{D}$ is closed, and increasing κ by arbitrarily small amounts gives periodic feasible tunings. This proves the closure identity.

□

Proof of Theorem 3.12. On a fixed finite grid, compactness gives lower semicontinuity of σ_L. A positive shift makes all constraints strict except those identically zero on the same support; these remain zero for every tuning. This gives upper semicontinuity. Thus $\sigma = \inf_L \sigma_L$ is upper semicontinuous; closedness of its epigraph gives lower semicontinuity. Lemma E.10 gives strict decrease in κ. As $\kappa \downarrow 1$, $B \downarrow 0$ and the numerator of Ξ tends uniformly to $-N(c) < 0$, hence $h_q \to -\infty$ uniformly and $\sigma \to +\infty$. The zero-crossing graph of this continuous strictly decreasing function is continuous into $(1, \infty]$: at a finite test κ, its positive or negative sign persists locally.

□

Corollary E.18 (Periodic thresholds of the condition number)**.** *For $L \ge 3$ put $\kappa_L(\alpha L, \beta) = \inf\{\kappa > 1 :\ \sigma_L(\kappa, \alpha L, \beta) \le 0\} \in (1, \infty]$. Then $\{(\kappa, \alpha L, \beta) \in \mathcal{T}_L\} = \{\kappa \ge \kappa_L(\alpha L, \beta)\}$ and*

$$\kappa_\partial = \inf_{L \ge 3} \kappa_L.$$

Proof of Corollary E.18. Each σ_L is continuous and strictly decreasing in κ, so its sublevel set is the stated closed upper ray. The inequality $\sigma \le \sigma_L$ gives $\kappa_\partial \le \kappa_L$. For $\kappa > \kappa_\partial$, strict decrease gives $\sigma(\kappa) < 0$; the infimum formula produces an L with $\sigma_L(\kappa) < 0$. Let $\kappa \downarrow \kappa_\partial$. The same inequalities include the infinite case.

□

E.5. Limiting symbols

Theorem E.19 (Limit symbol)**.** *For every tuning q there are $\lambda_n \ge 0$, $n \ne 0$, and $\gamma \ge 0$ with $\sum_n \lambda_n + \gamma = 1$ such that $S_q = \sum_n \lambda_n \Lambda_n^{q,\sigma(q)} + \gamma \cos \Theta_{q,\sigma(q)}$ is continuous, $S_q \le 0$ on $[0, \pi]$, $\max S_q = 0$, and every threshold extremal w is carried by $\{S_q = 0\}$ and satisfies $\int \Lambda_n^{q,\sigma(q)}\, dw = 0$ whenever $\lambda_n > 0$, and $\int \cos \Theta_{q,\sigma(q)}\, dw = 0$ if $\gamma > 0$. Here $\Lambda_n^{q,s}$ are the lag functions of $\Theta_{q,s}$.*

The proof follows Lemma E.20.

Lemma E.20 (Limits of symbols)**.** *Let $\mathcal{N} \subset \mathbb{Z} \setminus \{0\}$, let $\Theta_j \to \Theta$ uniformly on $[0, \pi]$, all curves, and let $S_j = \sum_{n \in \mathcal{N}} \lambda_n^{(j)} \Lambda_n^{(j)} + \gamma_j \cos \Theta_j \le 0$ on $[0, \pi]$, with finitely many $\lambda_n^{(j)} \ge 0$, $\gamma_j \ge 0$ and $\sum_n \lambda_n^{(j)} + \gamma_j = 1$. Then, along a subsequence, $\lambda_n^{(j)} \to \lambda_n$ for every n, $\sum_n \lambda_n \le 1$, and $S = \sum_{n \in \mathcal{N}} \lambda_n \Lambda_n + \gamma_* \cos \Theta$, $\gamma_* = 1 - \sum_n \lambda_n$, is continuous with $S \le 0$ on $[0, \pi]$. If moreover $\mathrm{FP}[\mathcal{N}](\Theta)$ is feasible, then $\max S = 0$, and every solution w of $\mathrm{FP}[\mathcal{N}](\Theta)$ is carried by $\{S = 0\}$ and satisfies $\int \Lambda_n\, dw = 0$ whenever $\lambda_n > 0$, and $\int \cos \Theta\, dw = 0$ if $\gamma_* > 0$.*

Proof. Take a coordinatewise convergent subsequence of the coefficients. Fatou gives $\sum \lambda_n \le 1$, and $|\Lambda_n| \le 2$ gives uniform convergence of the limiting series. For any nonnegative $\phi \in L^1$, the functions $e^{-i\Theta_j}\phi$ form a compact subset of L^1. The Riemann–Lebesgue estimate is uniform on this compact set (approximate it by a finite L^1 net). Split the lag sum into finite and high-frequency parts to obtain $\int S\phi \le 0$, hence $S \le 0$ by continuity. The missing coefficient mass contributes exactly $(1-\sum \lambda_n)\cos\Theta$. If $\max S < 0$, a finite truncation is a strict symbol, contradicting feasibility. Finally integrate S against any feasible measure: every weighted slack is nonnegative, so all positive-coefficient slacks vanish and the measure is carried by $\{S = 0\}$. □

Proof of Theorem E.19. Take normalized strict symbols at shifts $s_j \uparrow \sigma(q)$, available by Lemma E.2. The curves converge uniformly by Lemma E.1. Apply Lemma E.20 and attainment of the threshold. □

F. Farey programs and polynomial certificates

Throughout, the exponent is zero and $K \ge 2$.

Lemma F.1 (Positions)**.** *Let Θ be a curve, $k \ge 3$, $\omega_i = 2\pi i/k$, and let $\mathsf{a}_i = k\Theta(\omega_i)/\pi \in (0,k]$ be the phase index of ω_i. For $\mathsf{a} \in (0,k]$ and $\varrho \in \{0,\dots,k-1\}$ put $F_{\mathsf{a}}(\varrho) = 2\sin(\pi\varrho/k)\sin(\pi(\varrho-\mathsf{a})/k)$.*

(a) *$\Lambda_n(\omega_i) = F_{\mathsf{a}_i}(in \bmod k)$ for all $n \in \mathbb{Z}$.*

(b) *For $1 \le \varrho \le k-1$, $F_{\mathsf{a}}(\varrho) < 0$ if and only if $\varrho < \mathsf{a}$, and $F_{\mathsf{a}}(\varrho) = 0$ if and only if $\varrho = \mathsf{a}$; $F_{\mathsf{a}}(0) = 0$.*

(c) *If $\gcd(j,k) = 1$, then $n \mapsto jn \bmod k$ is a bijection of $\mathbb{Z}/k\mathbb{Z}$; position 1 is the class $b + k\mathbb{Z}$ of the parent lags of Theorem F.3, position 0 is the class $k\mathbb{Z}$, and position j is the class of the lag 1. If $p \equiv jn$ is the position of the lag n, then $in \equiv \tau_i p \pmod k$ with $\tau_i \equiv ib$.*

Proof. Reduce in modulo k in (25). The first sine in $F_{\mathsf{a}}(\varrho)$ is positive for $0 < \varrho < k$, so its second factor determines the sign. Multiplication by the unit $j \bmod k$ permutes residues; its inverse gives the position formula. □

Theorem F.2 (Lattice lines)**.** *Let Θ be a curve.*

(a) *$\Lambda_n < 0$ at a point of $\mathcal{C}$ if and only if $0 < n\xi - \ell\eta < 1$ for some integer ℓ. The sign of Λ_n changes only across the lattice lines $n\xi - \ell\eta = 0$ and $n\xi - \ell\eta = 1$.*

(b) *The points $(2\pi j/k, \pi/k)$ become the lattice points (j,k). The condition $\Theta(2\pi j/k) \le \pi/k$ says that $\mathcal{C}$ meets the ray through (j,k) at or beyond (j,k), and Ω_1 is the set of tunings whose curve lies in the half plane $\xi < 1$.*

Proof of Theorem F.2. Substitution in (25) gives $\Lambda_n = 2\sin(\pi n\xi/\eta)\sin(\pi(n\xi-1)/\eta)$. Since $0 < 1/\eta \le 1$, the two sines have opposite signs exactly when $0 < n\xi - \ell\eta < 1$ for an integer ℓ. At a rational frequency, $(\xi,\eta) = \pi(j,k)/(k\Theta)$, proving (b).

□

Theorem F.3 (Farey parents and first violations)**.** (a) *The lattice lines $n\xi - \ell\eta = 1$ through (j,k) have $n \equiv j^{-1} \pmod k$. Among them, those with $|n| < k$ are the lower lag $n = b$, whose line also passes through (c,d), and the upper lag $n = -d$, whose line also passes through (a,b).*

(b) *For a single atom at $\omega = 2\pi j/k$ with phase $\Theta \in (0,\pi)$, the lag n is violated if and only if $jn \bmod k \in \{1,\dots,\lceil k\Theta/\pi\rceil - 1\}$. No lag is violated if and only if $\Theta \le \pi/k$; when Θ exceeds π/k, the first violated lags are $n \equiv b \pmod k$, the class of the parent lags b and $-d$. In particular, for every curve, $\delta_{2\pi j/k}$ solves $\mathrm{FP}(\Theta)$ if and only if $\Theta(2\pi j/k) \le \pi/k$.*

Proof of Theorem F.3. The incidence equation is $nj - \ell k = 1$. The Farey determinant identities give the solutions $(n,\ell) = (b,a), (-d,-c)$; all others differ by (k,j). The mediant relations and neighbor identities are the standard unimodular characterization of Farey parents [12, Chapter III]. For (b), write $r = jn \bmod k$. The inequality $\cos(2\pi r/k - \Theta) > \cos\Theta$ is equivalent to $0 < r < k\Theta/\pi$. The first

residue entering this interval is $r = 1$, whose lag class is $b \bmod k$. When no lag is violated, $\Theta \le \pi/k$ also ensures the minimizer constraint. At $\Theta = \pi$ that constraint fails.

□

Lemma F.4 (The tongue of a fraction)**.** *For every tuning $q = (\kappa, \alpha L, \beta)$ and $c \in [-1, 1]$,*

$$\Xi_q(c) = \Xi_\infty(c) - \frac{N(c)}{\alpha L\,(1-\beta)(\kappa - 1)}, \tag{82}$$

where N is the polynomial of Lemma E.10 and

$$\Xi_\infty(c) = \frac{(1-c)\big(\alpha L\,(1+\beta) - 2(1 - 2\beta c + \beta^2)\big)}{\alpha L\,(1-\beta)}.$$

Consequently $\Theta_q(\omega_0) \le c'$ if and only if $\kappa \ge \kappa_{j/k}(\alpha L, \beta)$, where

$$\kappa_{j/k}(\alpha L, \beta) = 1 + \frac{\big(\alpha L - (1+\beta)(1-c_0)\big)^2 + (1-\beta)^2 s_0^2}{(1-c_0)\big(\alpha L\,(1+\beta) - 2(1 - 2\beta c_0 + \beta^2)\big) - h_0\,\alpha L\,(1-\beta)} \tag{83}$$

when the denominator is positive, and $\kappa_{j/k} = \infty$ otherwise. In particular $\kappa_\partial \le \kappa_{j/k}$ for every fraction j/k.

Proof. At fixed y, β, the numerator of (75) is affine in $x = y/\kappa$. Separate its constant part and its $(\kappa - 1)^{-1}$ part to obtain (82). Since $N > 0$, solving $\Xi(c_0) \ge h_0$ gives (83), including the empty case when its denominator is nonpositive. The exact threshold criterion gives $\kappa_\partial \le \kappa_{j/k}$. □

Proposition F.5 (Farey localization)**.** *Let $q \in I_K$. Then $\bar\Delta \subset (2\pi/(K+1), 2\pi/K)$. If w solves $\mathrm{P}_L(\Theta_q)$, then w has an atom $2\pi j/k$ with $\gcd(j,k) = 1$, $k \mid L$, $1/(K+1) < j/k < 1/K$, $k \ge 2K+1$ and $\Theta(2\pi j/k) \le \pi j/k$; moreover $\Theta(2\pi j/k) > \pi/k$ unless $\delta_{2\pi j/k}$ alone solves $\mathrm{FP}(\Theta_q)$.*

The proof follows Definition F.6.

Definition F.6 (Anchors)**.** The anchors of a solution w of the full program are its atoms in the closed dip $\bar\Delta$.

Proof of Proposition F.5. The closed dip lies strictly in the window by the gap definition and Corollary E.5. The dip lemma supplies a grid atom in it; reduce its fraction. Any fraction strictly between the Farey neighbors $1/(K+1), 1/K$ has denominator at least $2K+1$. The last assertion is the single-atom criterion. □

In the positions of an anchor, the complete periodic program is

$$\sum_i w_i F_{\mathsf{a}_i}(\tau_i p) \ge 0 \quad (p \in \mathbb{Z}/k\mathbb{Z}), \qquad \sum_i w_i \cos\Theta(\omega_i) \ge 0, \qquad \tau_i \equiv ib \pmod k. \tag{84}$$

The anchor has $\tau_j = 1$ and violates precisely the lag constraints at positions $1, \dots, \lceil \mathsf{a}_j \rceil - 1$, where a position is the residue $jn \bmod k$. At the lag-one position j, its value is nonnegative exactly when the anchor lies in the closed dip.

F.1. Fans and two-atom compensation

The fans of j/k are the integer points $(a,b) + m(j,k)$ and $(c,d) + m(j,k)$, $m \ge 1$, on $k\xi - j\eta = -1$ and $k\xi - j\eta = 1$. Their consecutive determinants are ∓ 1, so the members are reduced Farey neighbors and approach j/k from opposite sides. For the endpoints of I_K their lines are $U : \eta = K\xi + 1$ and $V : (K+1)\xi - \eta = 1$. In angle coordinates these are $\Theta = \pi - K\omega/2$ and $\Theta = (K+1)\omega/2 - \pi$. At each fan fraction their height is π/k'; crossing at or beyond that point is precisely the single-atom criterion. On the window both heights are at most $\pi/2$. Corollary E.5 thus makes their below-line sets Δ_U, Δ_V intervals. A level-two gap has a nonempty such interval whose closure lies strictly between consecutive points of the corresponding fan.

Lemma F.7 (Fan geometry)**.** *Let $K \geq 2$ and $k_p = pK + 1$. The fractions $(p-1)/k_{p-1} < p/k_p < 1/K$ are Farey neighbors, and the Farey parents of p/k_p are $(p-1)/k_{p-1}$ and $1/K$. The points (p, k_p) lie on the line $U : \eta = K\xi + 1$, which is the lattice line $-K\xi + \eta = 1$ of the lag $-K$. The lattice line of the lag $mK + 1$ through $(1, K)$ is $(mK + 1)\xi - m\eta = 1$, and it passes through $(m + 1, k_{m+1})$. Modulo k_p, $-pK \equiv 1$ and $(p-1)K + 1 \equiv -K$.*

Proof. The parent determinant is $pk_{p-1} - (p-1)k_p = 1$; the mediant relations identify the parents. Substitution proves the line equations and congruences. □

For integers τ, write $r_\tau(p) \in \{0, \dots, k-1\}$ for the residue of $p\tau$ modulo k.

Lemma F.8 (Record lows and Farey parents)**.** *Let $2 \leq j < k$, $\gcd(j, k) = 1$, $\tau \in \mathbb{Z}$, and let $i = r_\tau(j)$ and $m = (j\tau - i)/k$, so that $j\tau = mk + i$. Assume $i \geq 1$. Then $r_\tau(p) > i$ for all $1 \leq p \leq j-1$ if and only if $\gcd(m, j) = 1$ and $i\, r' < k$, where r/r' is the right Farey parent of m/j, that is, the fraction with $rj - mr' = 1$ and $1 \leq r' \leq j-1$. In that case*

$$\min_{1 \leq p \leq j-1} r_\tau(p) = \frac{k + l'i}{j}, \qquad l' = j - r',$$

and the minimum is attained at $p = l'$ only.

Proof. Use the determinant-k lattice $\Gamma = \{(p, y) : y \equiv p\tau \pmod{k}\}$. If $g = \gcd(m, j) > 1$, the point $(j/g, i/g)$ violates the record condition. Otherwise the Farey identities give a lattice basis

$$u = \left(l', \frac{k + l'i}{j}\right), \qquad v = \left(r', \frac{r'i - k}{j}\right), \quad u + v = (j, i), \quad \det(u, v) = -k.$$

If $ir' \geq k$, v violates the condition. If $ir' < k$, then $u_2 > i > v_2$ with $v_2 < 0$. For $au + bv$ with first coordinate in $[1, j-1]$, either $a \geq 1, b \leq 0$ and its second coordinate is at least u_2, uniquely at u, or $a \leq 0, b \geq 1$ and it is negative. This proves both the equivalence and the unique minimum. □

The next result identifies the admissible companion frequencies and mass ratios for two-atom measures on the periodic grid.

Theorem F.9 (Two-atom combs)**.** *Let $2 \leq j < k/2$, $\gcd(j, k) = 1$, $b \equiv j^{-1} \pmod{k}$, $\omega_a = 2\pi j/k$ with phase index $\mathsf{a} = k\Theta(\omega_a)/\pi$, and $\omega_c = 2\pi i/k$, $1 \leq i < k/2$, $i \neq j$, with phase index $\mathsf{a}_c = k\Theta(\omega_c)/\pi$. Assume that for some integer t with $2 \leq t \leq j$*

$$t - 1 < \mathsf{a} \leq t \qquad \text{and} \qquad i < \mathsf{a}_c,$$

that is, the anchor lies in the closed dip with bad positions $1, \dots, t-1$, and the companion lies outside the closed dip. Put $\tau \equiv ib \pmod{k}$ and $m = (j\tau - i)/k$ with $0 \leq \tau < k$, so that $r_\tau(j) = i$.

(a) *There are $w_0, w_c > 0$ such that $w = w_0\delta_{\omega_a} + w_c\delta_{\omega_c}$ satisfies all lag constraints of $\mathrm{FP}(\Theta)$ if and only if*

$$r_\tau(p) > \mathsf{a}_c \quad (1 \leq p \leq t-1) \tag{85}$$

and $\underline{\mathsf{r}} \leq \bar{\mathsf{r}}$, where

$$\underline{\mathsf{r}} = \max_{1 \leq p \leq t-1} \frac{-F_{\mathsf{a}}(p)}{F_{\mathsf{a}_c}(r_\tau(p))},$$

$$\bar{\mathsf{r}} = \min\Big\{ \frac{F_{\mathsf{a}}(p)}{-F_{\mathsf{a}_c}(r_\tau(p))} : \ 1 \leq p \leq k-1, \ 1 \leq r_\tau(p) < \mathsf{a}_c \Big\};$$

the admissible ratios are then $w_c/w_0 \in [\underline{\mathsf{r}}, \bar{\mathsf{r}}]$. Together with $w_0 \cos\Theta(\omega_a) + w_c \cos\Theta(\omega_c) \geq 0$ this characterizes the two-atom solutions of P_k on $\{\omega_a, \omega_c\}$.

(b) (Farey selection rule.) *For $t = j$, condition* (85) *holds if and only if* $\gcd(m, j) = 1$, $ir' < k$ *and* $\mathsf{a}_c < (k + l'i)/j$, *where* $l/l' < m/j < r/r'$ *are the Farey parents of* m/j. *For* $t < j$, (85) *is the condition* $\min_{1\le p\le t-1} r_\tau(p) > \mathsf{a}_c$ *and is implied by the case* $t = j$.

(c) (Two fan sequences.) *For the fan point* $p/(pK + 1)$ *with the companion* $i = p - 1$, $\tau \equiv K + 1$, *and* (85) *holds for every* $t \in [2, p]$ *if and only if* $\mathsf{a}_c < K + 1$; *this forces* $p \le K + 1$, *and then every lag is decided by (a). For* $(\ell + 1)/(K + \ell(K + 1))$ *with the companion* $i = \ell$, $\tau \equiv -K$, *and* (85) *holds if and only if* $\mathsf{a}_c < k - (t - 1)K$; *for* $t = \ell + 1$ *this is* $\mathsf{a}_c < K + \ell$.

Proof. At anchor position p, the two lag values are $F_{\mathsf{a}}(p)$ and $F_{\mathsf{a}_c}(r_\tau(p))$. The first is negative exactly at $1 \le p < t$. Positive compensation forces (85); under that condition the negative anchor values give the lower ratio bounds and the negative companion values give the upper bounds. All other constraints have two nonnegative terms. The upper set is nonempty because it contains $p = j$; the lower bound is strictly positive. Adding the minimizer constraint proves (a), including equality cases. For $t = j$, Lemma F.8 is exactly (b). For the first fan sequence, $\tau = K + 1 \bmod k$ and $r_\tau(p') = (K + 1)p'$ for $1 \le p' < p$ once $\mathsf{a}_c < K + 1$; this also forces $p \le K + 1$. For the second fan sequence, $\tau = -K \bmod k$ and $r_\tau(p') = k - Kp'$ for $1 \le p' \le \ell$. Taking the minimum gives (c). □

Theorem F.10 (Fan tongues)**.** *Let* $p \ge 2$ *and* $k = k_p = pK + 1$.

(a) *If there are* $w_j \ge 0$, $1 \le j < k/2$, *not all zero, with* $\sum_j w_j \Lambda_n(\omega_j) \ge 0$ *for* $n = 1, \dots, k - 1$ *and* $\sum_j w_j \cos\theta_j \ge 0$, *then there are* $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^{2r+1})$, r *the number of nonzero weights, and* x_0 *whose trajectory converges to a* k*-periodic orbit; in particular* $\rho \ge 1$.

(b) *Let* $p \le K + 1$, *and suppose* $\pi/k < \theta_p < \pi p/k$ *and* $\pi(p - 1)/k < \theta_{p-1} < \pi(K + 1)/k$. *Then* $\Lambda_1(\omega_p) > 0 > \Lambda_{-K}(\omega_p)$ *and* $\Lambda_{-K}(\omega_{p-1}) > 0 > \Lambda_1(\omega_{p-1})$, *and the lags* 1 *and* $-K$ *are satisfied by some measure on* $\{\omega_{p-1}, \omega_p\}$ *if and only if*

$$\Delta_p := \sin\tfrac{\pi p}{k}\,\sin\!\big(\tfrac{\pi p}{k} - \theta_p\big)\,\sin\tfrac{\pi(K+1)}{k}\,\sin\!\big(\tfrac{\pi(K+1)}{k} - \theta_{p-1}\big) \\ - \sin\tfrac{\pi(p-1)}{k}\,\sin\!\big(\theta_{p-1} - \tfrac{\pi(p-1)}{k}\big)\,\sin\tfrac{\pi}{k}\,\sin\!\big(\theta_p - \tfrac{\pi}{k}\big) \;\ge 0.$$

If moreover $\theta_{p-1} \le \pi/2$, *which is automatic for* $p \ge 3$, *and the lags* $n \in \{2, \dots, k-1\}\backslash\{(p-1)K+1\}$ *are satisfied for one ratio* w_{p-1}/w_p *of the admissible interval, then the conclusion of* (a) *holds.*

(c) *For* $p = 2$, $k = 2K + 1$, *and* $\pi/k < \theta_1, \theta_2 \le 2\pi/k$, *the remaining lags are satisfied automatically, and the program of* (a) *with the two harmonics* ω_1, ω_2 *is feasible if and only if*

$$2\cos\tfrac{\pi}{k}\cos\tfrac{\pi}{2k}\,\sin\!\big(\tfrac{\pi}{k} - \vartheta_2\big)\cos\!\big(\vartheta_1 + \tfrac{\pi}{2k}\big) \ge \sin\tfrac{\pi}{k}\,\sin\vartheta_1\sin\vartheta_2, \quad \vartheta_j = \theta_j - \tfrac{\pi}{k}.$$

Proof. Part (a) is Proposition E.12. For (b), the four relevant lag values are

$$\begin{aligned} \Lambda_1(\omega_p) &= F_{k\theta_p/\pi}(p), & \Lambda_{-K}(\omega_p) &= F_{k\theta_p/\pi}(1), \\ \Lambda_1(\omega_{p-1}) &= F_{k\theta_{p-1}/\pi}(p-1), & \Lambda_{-K}(\omega_{p-1}) &= F_{k\theta_{p-1}/\pi}(K+1). \end{aligned}$$

The phase bounds give their opposite signs. The two inequalities in the positive weight ratio are compatible exactly when $\Lambda_1(\omega_p)\Lambda_{-K}(\omega_{p-1}) - \Lambda_1(\omega_{p-1})\Lambda_{-K}(\omega_p) \ge 0$; division by 4 is the displayed Δ_p. The remaining residue conditions and phases at most $\pi/2$ give all other constraints. For $p = 2$, each atom violates only the residue with position 1, so all other residues are automatically nonnegative. Use $\sin(2u) = 2\sin u\cos u$ and $\pi(K + 1)/k = \pi/2 + \pi/(2k)$ in Δ_2 to obtain (c). □

F.2. Dual certificates and pinning

Theorem F.11 (Fan-lag certificates)**.** *Assume* (3) *and let* $K \geq 2$*. If there are finitely many weights* $\lambda_n \geq 0$*,* $n \in \mathcal{N}_{\mathrm{fan}}(K)$*, and* $\gamma > 0$ *such that*

$$\Xi(c)\Big[\sum_{m\geq 0} \lambda_{mK+1}(1 - T_{mK+1}(c)) + \sum_{m\geq 1} \lambda_{-mK}(1 - T_{mK}(c)) + \gamma\Big] \\ < (1-c^2)\Big[\sum_{m\geq 0} \lambda_{mK+1} U_{mK}(c) - \sum_{m\geq 1} \lambda_{-mK} U_{mK-1}(c)\Big] \quad (-1 \leq c \leq 1), \quad (86)$$

then $\rho < 1$*. The choice* $\lambda_1 = 1$*,* $\lambda_{-K} = r \geq 0$ *gives* (30)*.*

Proof. Formula (79) identifies the symbol with $\sin\Theta/\sin\omega$ times the left-minus-right difference. It is negative on the open interval. At both endpoints the lag terms are nonpositive and the minimizer term equals $-\gamma < 0$. Compactness gives a strict uniform margin, and Theorem 3.2 gives convergence. □

Lemma F.12 (Pinning lags)**.** *Let* $k = k_p$ *and* $\omega_j = 2\pi j/k$*. Then* $-pK \equiv 1$ *and* $(p-1)K + 1 \equiv -K \pmod{k}$*, so on every* k*-periodic spectrum the lags* 1 *and* $-pK$ *give the same constraint, and so do the lags* $-K$ *and* $(p-1)K + 1$*. The differences*

$$\mathsf{D} = \Lambda_{-pK} - \Lambda_1, \quad \mathsf{E} = \Lambda_{(p-1)K+1} - \Lambda_{-K}$$

vanish at every frequency ω_j*, and there*

$$\mathsf{D}'(\omega_j) = -k\sin(\omega_j - \theta_j), \quad \mathsf{E}'(\omega_j) = -k\sin(K\omega_j + \theta_j).$$

Proof. Use the congruences of Lemma F.7 and the derivative formula of Lemma E.13. Transferring a weight between congruent lags preserves grid values and adds the corresponding multiple of D or E to the off-grid symbol. Nonnegative transfers are exactly those that keep the two affected weights nonnegative. □

If both contacts of an optimal two-lag symbol lie on Ω_L and conditions (E1)–(E4) of Theorem G.12 hold, that symbol is optimal for the full program; the transfers using $1 \pm L, -K \pm L$ yield L-regularity.

F.3. Exact arithmetic for the two rational examples

For an interval $[u, v]$, write $P(u + (v-u)t) = \sum_{i=0}^{d} b_i \binom{d}{i} t^i (1-t)^{d-i}$. Nonnegative Bernstein basis functions sum to 1, so $\min b_i \leq P \leq \max b_i$ [5]. The following sign checks use only this identity and rational Taylor bounds for sine and cosine.

Proof of Proposition 3.13. For the stated tuning,

$$\mathcal{L}(c) = \frac{274486940640000c^2 - 89129341428596c + 2155051415249}{110124036000000}.$$

Its vertex lies in $(0, (\sqrt{5}-1)/4)$, has negative value, and both endpoint values on that interval are positive. Hence the tuning lies in I_4. Its angle quadratic is

$$\Xi(c) = \frac{88897198775653}{91052250190902} + \frac{242832333719}{122711927481}c - \frac{26156560000}{8676600933}c^2.$$

On $[-1, -1/2]$ and $[127/128, 1]$, $-\Xi$ has positive Bernstein coefficients. On each interval with consecutive endpoints

$$-\tfrac{1}{2},\ 0,\ \tfrac{1}{2},\ \tfrac{3}{4},\ \tfrac{7}{8},\ \tfrac{15}{16},\ \tfrac{31}{32},\ \tfrac{63}{64},\ \tfrac{127}{128},$$

the polynomial $3(1-c^2) - \Xi(c)^2$ has positive Bernstein coefficients. Thus $\Theta > \pi/6$ throughout. This excludes all denominators $k \geq 6$; the four primitive frequencies with $3 \leq k \leq 5$ are excluded by the same quadratic signs and the I_4 window. An irrational rotation also fails the single-atom constraints, since its lag phases are dense modulo 2π.

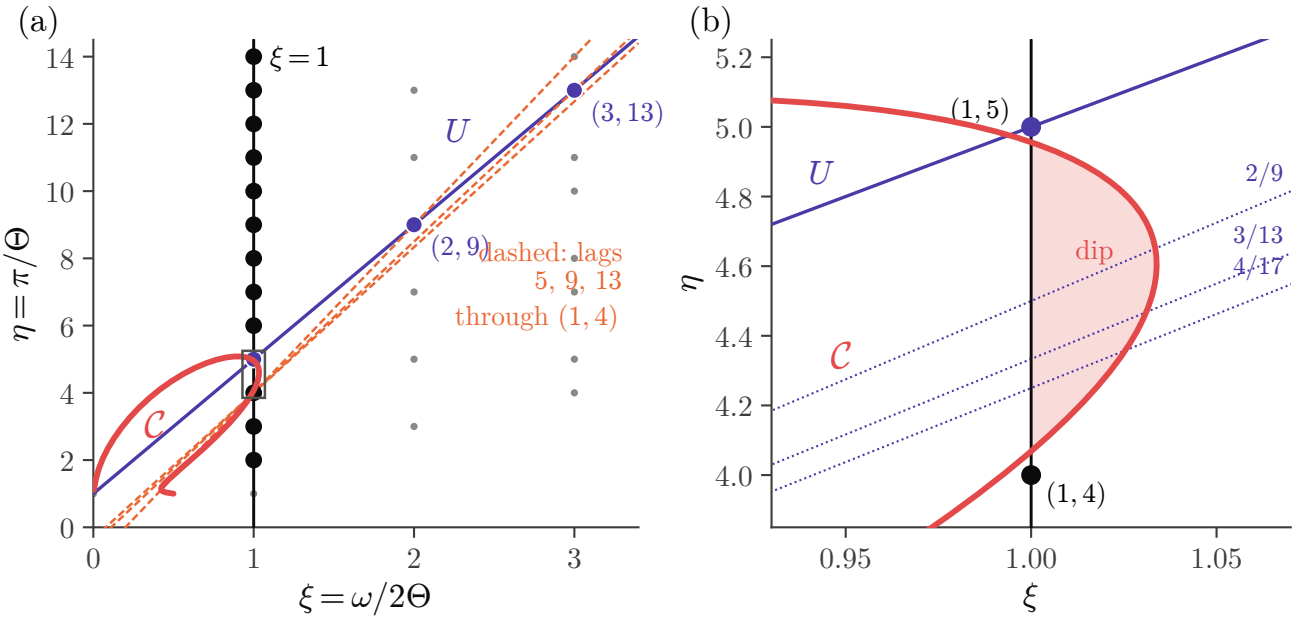


Figure 9: The example of Proposition 3.13 in lattice coordinates. The line $U : \eta = 4\xi + 1$ carries the fan $p/(4p+1)$; dashed lines correspond to its lower parent lags $5, 9, 13$. The right panel resolves the dip and the fan directions $2/9, 3/13, 4/17$.

Put $c_j = \cos(2\pi j/13)$ and $s_j = \sin(2\pi j/13)$. After division by the common positive factor $2(1-\beta)/B$, the raw lag and minimizer slacks are

$$\sum_{j=2,3} v_j\big[\Xi(c_j)(1 - c_{nj}) - s_j s_{nj}\big], \qquad \sum_{j=2,3} v_j \Xi(c_j), \quad (v_2, v_3) = (149, 648)/797,$$

with indices reduced modulo 13. All thirteen displayed quantities (the minimizer and $n = 1, \dots, 12$) are strictly positive. For a reproducible rational check, enclose π between $31415926535897932384\mathbf{6}/10^{20}$ and $31415926535897932384\mathbf{7}/10^{20}$, reduce angles to $[-\pi, \pi]$, and use the Taylor polynomials through degree 32 with their Lagrange remainder bounds. Substitution into these finite formulas gives strict positive interval lower endpoints. Periodicity gives every signed lag; multiples of 13 vanish. Proposition E.12 realizes the stated two harmonics in $\mathbb{R}^5$.

□

Proof of Proposition 3.14. Here

$$\Xi(c) = -\frac{620000}{208791}c^2 + \frac{14967}{7733}c + \frac{203570}{208791}, \qquad \mathcal{L}(c) = \frac{620000c^2 - 195318c + 5221}{250000}.$$

The same vertex and endpoint test gives I_4. Take $r = 13/100$, $\gamma = 1/1000$ in (30). The right-minus-left difference is

$$\begin{aligned} P(c) = &-\tfrac{644800}{208791}c^6 + \tfrac{23608}{7733}c^5 + \tfrac{389324}{94905}c^4 - \tfrac{34149683}{5219775}c^3 \\ &+ \tfrac{1007042}{347985}c^2 - \tfrac{92371789}{208791000}c + \tfrac{45613}{1898100}. \end{aligned}$$

Its degree-six Bernstein coefficients are strictly positive on each interval $[-1 + j/8, -1 + (j+1)/8]$, $0 \le j < 16$. Therefore $P > 0$ on $[-1, 1]$, and Theorem F.11 gives $\rho < 1$.

□

Figure 9 displays the lattice curve of Proposition 3.13 and its endpoint fan.

G. Boundary structure and finite certificates

For shifted curves, write $\Lambda_n^{q,s} = \Lambda_n[\Theta_{q,s}]$, $\theta_\omega(q,s) = \Theta_{q,s}(\omega)$ and $\tau_q = -1/\Xi_q(1) > 0$. The lag functions are jointly real analytic through the endpoints, with $\Lambda_n(0) = \Lambda_n'(0) = 0$ and $\Lambda_n''(0) = -n(n+2\tau_q)$. The calculation used throughout is

$$\frac{\sum_n \lambda_n \Lambda_n^{q,s} + \gamma\cos\Theta_{q,s}}{\sin\Theta_{q,s}} = \frac{\sum_n \lambda_n \Lambda_n^{q,s'} + \gamma\cos\Theta_{q,s'}}{\sin\Theta_{q,s'}} + (s-s')\Big[\sum_n \lambda_n(1-\cos n\omega) + \gamma\Big] \tag{87}$$

on $(0,\pi)$. It follows directly from $\cot\Theta_{q,s} = h_q + s$. In particular, if $S = \sum_n \lambda_n \Lambda_n^{q,s}$ and $P = \sum_n \lambda_n(1-\cos n\omega)$, then

$$\partial_s S = (\partial_s \log\sin\Theta_{q,s})S + \sin\Theta_{q,s} P. \tag{88}$$

G.1. Regularity and continuation

Definition G.1 (Regular periodic point)**.** Let $L \geq 3$. A pair (q_*, s_*) is L-regular if there are a finite set $S \subset \Omega_L \cap (0,\pi)$, a finite set $N \subset \mathbb{Z}\setminus\{0\}$ and weights $w^* \in \mathbb{R}^S_{>0}$, $\lambda^* \in \mathbb{R}^N_{>0}$ with $\sum_\omega w^*_\omega = \sum_n \lambda^*_n = 1$, such that, with $N_0 = N \cap L\mathbb{Z}$, $\mathcal{A} = (N \setminus N_0) \bmod L$ and $S^* = \sum_{n\in N} \lambda^*_n \Lambda_n^{q_*,s_*}$:

(R1) primal: w^* is feasible for $\mathrm{P}_L(\Theta_{q_*,s_*})$; the constraints of the residues in $\mathcal{A}$ are tight, the constraints of all other residues $\varrho \not\equiv 0$ and the minimizer constraint are strict; $|\mathcal{A}| = |S|$, and $M_* = \big(\Lambda_\varrho^{q_*,s_*}(\omega)\big)_{\varrho\in\mathcal{A},\ \omega\in S}$ has rank $|S|-1$;

(R2) dual: $S^* < 0$ on $(0,\pi]\setminus S$, $\partial_\omega^2 S^* < 0$ on S, $\sum_n \lambda^*_n n(n+2\tau_{q_*}) > 0$, and $P^*(\omega) = \sum_n \lambda^*_n(1-\cos n\omega) > 0$ on S;

(R3) pinning rank: every class $N_\varrho = N \cap (\varrho + L\mathbb{Z})$, $\varrho \in \mathcal{A}$, has at least two elements, and the vectors $v_\varrho = \big(\sin(\varrho\omega - \theta_\omega(q_*,s_*))\big)_{\omega\in S}$, $\varrho\in\mathcal{A}$, form a basis of $\mathbb{R}^S$ (for one fixed representative ϱ of each class).

A tuning q_* is L-regular if $(q_*, \sigma(q_*))$ is.

The values and first derivatives of S^* vanish on S: positivity of w^* and complementarity give the values, and the global sign gives the derivatives. The moment inequality is precisely $S^{*\prime\prime}(0) < 0$, controlling curvature at the endpoint. If $|S| \geq 2$, the condition $P^* > 0$ on S already follows from (R1).

Proof of Theorem 3.16. Let $M(q,s) = (\Lambda_\varrho^{q,s}(\omega))_{\varrho\in\mathcal{A},\omega\in S}$. Its entries are real analytic by Lemma E.1. At the base point, $M_* w^* = 0$ and $\operatorname{rank} M_* = |S| - 1$. Put $\mu^*_\varrho = \sum_{n\equiv\varrho} \lambda^*_n > 0$. Since the zero-class lag functions vanish on Ω_L, ${\mu^*}^\top M_* = (S^*(\omega))_{\omega\in S}$. Every entry is nonpositive by (R2), while its product with the positive vector w^* is zero. Hence $S^* = 0$ on S and ${\mu^*}^\top M_* = 0$. The adjugate of M_* is therefore $c\, w^* {\mu^*}^\top$ for some $c \neq 0$. Jacobi's formula and (88) give

$$\partial_s \det M(q_*,s_*) = c\sum_{\omega\in S} w^*_\omega \sin\theta_\omega(q_*,s_*) P^*(\omega) \neq 0.$$

The analytic implicit function theorem [17, Section 2.3] gives an analytic $s(q)$ with $\det M(q,s(q)) = 0$. A fixed nonzero minor keeps the rank equal to $|S|-1$ after the neighborhood is reduced. Nonzero adjugate columns and rows give analytic positive kernel vectors $w(q), \mu(q)$, normalized to agree with w^*, μ^* at the base point. The inactive residue constraints and minimizer constraint remain strict by continuity. Thus $w(q)$ solves FP at shift $s(q)$ and $\sigma(q) \leq \sigma_L(q) \leq s(q)$.

It remains to continue a dual symbol with fixed contacts. For each nonzero class write $\mu_\varrho = \sum_{n\equiv\varrho}\lambda_n$ and $m_\varrho = \sum_{n\equiv\varrho}\lambda_n(n-\varrho)/L$. Lemma E.13 gives

$$S'(\omega) = \sum_{\varrho\in\mathcal{A}} \mu_\varrho \Lambda_\varrho'(\omega) + L\sum_{\varrho\in\mathcal{A}} m_\varrho \sin(\varrho\omega - \Theta(\omega)) + \sum_{n\in N_0} \lambda_n \Lambda_n'(\omega), \qquad \omega\in S. \tag{89}$$

Keep the zero-class weights fixed at their original values and use the continued $\mu(q)$. By (R3), the equations $S'|_S = 0$ uniquely determine analytic moments $m(q)$ near the base point. Each class has two distinct lags, so its linear map from lag weights to its mass and moment is onto. A fixed affine right

inverse through the original weights lifts $(\mu(q), m(q))$ to analytic positive weights $\lambda(q)$. Normalize their sum. The resulting S_q has zero values and slopes on S.

Choose disjoint fixed neighborhoods of S and 0. On them the negative second derivative of S^* has a uniform margin, by (R2) and the endpoint moment condition. On their compact complement, S^* has a strictly negative maximum. Uniform C^2 continuity preserves both estimates. Since the continued values and slopes at $S \cup \{0\}$ are zero, Taylor's theorem gives $S_q \le 0$ with precisely that zero set; also $P_q > 0$ near S. For $s < s(q)$ sufficiently close, (87) gives

$$\frac{\sum_n \lambda_n(q)\Lambda_n^{q,s} + \epsilon \cos\Theta_{q,s}}{\sin\Theta_{q,s}} = \frac{S_q}{\sin\Theta_{q,s(q)}} - (s(q) - s)P_q + \epsilon(h_q + s).$$

Near 0, $h_q + s < 0$. Near S the second term has a strict negative margin; on the remaining compact set the first term does. The positive part of $h_q + s$ is bounded there. Choosing $\epsilon > 0$ smaller than these margins makes this expression negative throughout $(0, \pi)$; the unscaled symbol equals $-\epsilon$ at 0 and is at most $-\epsilon$ at π. It is therefore strict. Monotonicity excludes every smaller shift, proving $\sigma(q) = \sigma_L(q) = s(q)$.

For any threshold measure v, feasibility and the dual sign give $0 \le \int S_q \, dv \le 0$. Positivity of the dual weights forces every lag in N to be tight and v to be carried by $S \cup \{0\}$. The rank condition gives $v|_S = cw(q)$; the minimizer constraint excludes $c = 0$ and is exactly $0 \le v(\{0\}) \le c \int \cos\Theta_{q,s(q)} \, dw(q)$. Conversely every measure satisfying this description is feasible. This proves (c).

Differentiate the determinant equation. Because the parameter (q, s) enters each column only through $h_q(\omega) + s$,

$$\nabla s(q) = -\frac{\sum_{\omega \in S} w_\omega \sin\theta_\omega P_q(\omega) \nabla_q h_q(\omega)}{\sum_{\omega \in S} w_\omega \sin\theta_\omega P_q(\omega)}.$$

All weights in this quotient are positive. Lemma E.10 therefore gives $\partial_\kappa s < 0$. The strict conditions (R1)–(R3) persist, proving (e). A final application of the implicit function theorem to $s(\kappa, \alpha L, \beta) = 0$ gives the analytic boundary graph and completes the proof. □

Corollary G.2 (Open plateaus)**.** *The set of regular tunings is open. Its intersection with $\partial\mathcal{D}$ is relatively open in $\partial\mathcal{D}$; on it κ_∂ is real analytic, and on each connected component the support S of the threshold extremal, its least common denominator L and the tight classes $\mathcal{A}$ are constant, and κ_∂ solves $\det(\Lambda_\varrho(\omega))_{\varrho \in \mathcal{A},\, \omega \in S} = 0$ at the shift 0. A plateau is such a component. If the support has exactly one atom in the window $(2\pi/(K+1), 2\pi/K)$, then on the part of the plateau in I_K this atom is the unique anchor.*

Proof. Persistence gives openness and local constancy of the unique support and tight classes, hence constancy on each component. The least period is intrinsic by Theorem G.7(a). The determinant equation and analyticity follow from Theorem 3.16. Every extremal charges the dip, so a single support point in the gap window is its unique anchor. □

Corollary G.3 (Periods on a plateau)**.** *Let $q \in \partial\mathcal{D}$ be regular with threshold extremal carried by S, and let L be the least common multiple of the denominators of the points of $S/2\pi$.*

(a) *$q \in \mathcal{T}_{L'}$ if and only if $L \mid L'$.*

(b) *If $q \in I_K$, then the fan program of Theorem F.10*(a) *of some period $pK + 1$ is feasible at q if and only if $\gcd(L, K) = 1$.*

Proof of Corollary G.3. A multiple of L contains the support grid. Conversely every feasible periodic measure at threshold has the same positive support by Theorem 3.16(c), so its period is divisible by L. The congruence $pK \equiv -1 \pmod{L}$ is solvable exactly when $\gcd(K, L) = 1$; a solution can be increased to have $p \ge 2$. □

G.2. Boundary points and heavy atoms

Theorem G.4 (Structure of a boundary point)**.** *Let $q \in \mathcal{B}_K$.*

(a) (Limit symbol.) *There are $\lambda_n \geq 0$, $n \neq 0$, and $\gamma \geq 0$ with $\sum_n \lambda_n + \gamma = 1$ such that $S_q = \sum_n \lambda_n \Lambda_n + \gamma \cos \Theta_q$ is continuous, $S_q \leq 0$ on $[0, \pi]$, every extremal is carried by $Z_q = \{S_q = 0\}$, $\int \Lambda_n \, dw = 0$ for every extremal w whenever $\lambda_n > 0$, and $\int \cos \Theta_q \, dw = 0$ for every extremal if $\gamma > 0$.*

(b) (Dip.) *Every extremal charges the closed dip $\bar{\Delta} \subset (2\pi/(K+1), 2\pi/K)$; hence $Z_q \cap \bar{\Delta} \neq \emptyset$.*

(c) (Periodic points.) *$q \in \mathcal{P}$ if and only if some extremal is carried by some Ω_L; then each such extremal has an anchor $2\pi j/k$ with $1/(K+1) < j/k < 1/K$, $k \mid L$ and $k \geq 2K+1$.*

(d) (Non-periodic points.) *If $q \notin \mathcal{P}$, then every extremal has an atom at an irrational multiple of 2π, or infinitely many atoms, or a non-atomic part; and $\sigma_L(q) \to 0$ as $L \to \infty$.*

(e) (Closures of plateaus.) *Let Π be a plateau in I_K with support $S_\Pi \subset \Omega_L$. Then $\sigma_L = 0$ on $\operatorname{cl} \Pi \cap \mathcal{B}_K$, and at every $q \in \operatorname{cl} \Pi \cap \mathcal{B}_K$ some extremal is carried by S_Π. Hence the relative boundary of every plateau lies in $\mathcal{P} \setminus \mathcal{R}$.*

Proof. The limit symbol and its complementary slackness are Theorem E.19; dip carriage is Lemma E.6. A grid-carried extremal is equivalent to $\sigma_L = 0$, and Proposition F.5 gives its anchor. Finitely many frequencies at rational multiples of 2π lie on one grid; the absence of such a support, after removing a possible zero atom, gives the stated measure alternatives. The sampling argument of Theorem 3.11, applied to a measure having no such carrier, then works for every sufficiently large period.

On a plateau $\sigma = \sigma_L = 0$. Continuity preserves both equalities at its closure, and compactness of probability vectors on its fixed finite support gives an extremal there. A regular limit point has a neighborhood with the same support inside its own open plateau; if it is approached by the given plateau, the components coincide. Hence a relative endpoint is periodic and nonregular. □

Proof of Theorem 3.15. Strict decrease of σ in κ gives the three signs. The finite alternative and exact threshold theorem give (i); sampling a negative threshold gives (ii), and grid inclusion gives all multiples of its period. At equality use Theorem G.4. Every feasible grid measure has the Farey anchor of Proposition F.5. □

Proposition G.5 (Heavy atoms move continuously)**.** *Let $C \subset \mathcal{B}_K$ be connected and $m_0 > 1/2$, and suppose that at every $q \in C$ every extremal w with $w(\{0\}) = 0$ and $w((0, \pi]) = 1$ has an atom of mass at least m_0. Then:*

(a) *at each $q \in C$ all such extremals have their heavy atom at the same point $r(q) \in (0, \pi]$;*

(b) *$r : C \to (0, \pi]$ is continuous;*

(c) *if $q \in \mathcal{P}$, then $r(q) \in 2\pi\mathbb{Q}$;*

(d) *if r is not constant on C, then $C \setminus \mathcal{P}$ is uncountable.*

Proof. Normalize by removing the zero atom and setting the remaining mass to one. If two extremals had their atoms of mass at least $m_0 > 1/2$ at different points, their convex interpolation would either have no such atom or would have two, of total mass greater than one. Thus the location is unique. The minimizer constraint bounds $\cos \Theta_q(r(q)) \geq -(1 - m_0)/m_0 > -1$, keeping it locally away from 0. Weak compactness and the closed program constraints pass a heavy atom to every subsequential limit; removing a possible zero atom preserves its normalized mass lower bound. Uniqueness identifies the limit, proving continuity. At a periodic point the common location is rational. A nonconstant continuous image of a connected set is an interval, whose irrational points give uncountably many nonperiodic preimages. □

G.3. Finite decisions and localization

Proposition G.6 (Polynomial certificates). *Let q be a tuning, $N \subset \mathbb{Z}\setminus\{0\}$ finite, $\lambda_n \ge 0$ $(n \in N)$ and $\gamma \ge 0$. The symbol $\sum_{n\in N} \lambda_n \Lambda_n + \gamma \cos\Theta_q$ is strict if and only if*

$$\Xi_q(c)\Big[\sum_{n\in N} \lambda_n(1 - T_{|n|}(c)) + \gamma\Big] < (1-c^2)\sum_{n\in N} \operatorname{sgn}(n)\,\lambda_n\, U_{|n|-1}(c) \qquad (-1 \le c \le 1).$$

Proof. Put $\mathcal{P} = (1-c^2)\sum_n \operatorname{sgn}(n)\lambda_n U_{|n|-1} - \Xi_q[\sum_n \lambda_n(1-T_{|n|})+\gamma]$. The Chebyshev identities give $S = -(\sin\Theta_q/\sin\omega)\mathcal{P}(\cos\omega)$ in the interior. At the endpoints $S(0) = -\gamma$, $S(\pi) = -\sum_n \lambda_n(1-(-1)^n) - \gamma$, and $\Xi_q(\pm 1) < 0$. These identities give the claimed equivalence including strictness at both ends; compactness makes the negative margin uniform. □

Theorem G.7 (Exact relaxations near a regular point). *Let (q_*, s_*) be L-regular with data S, N, w^*, λ^*, and let U and s be as in Theorem 3.16.*

(a) *L is the least common denominator of the points of $S/2\pi$.*

(b) *For every $q \in U$ and every set of lags $N' \supset N$, $\sigma_{N'}(q) = \sigma_L(q) = \sigma(q) = s(q)$, and the solutions of $\mathrm{FP}[N'](\Theta_{q,\sigma(q)})$ are those of $\mathrm{FP}(\Theta_{q,\sigma(q)})$.*

(c) *For $q \in U$ the following are equivalent: $\rho(q) < 1$; $\mathrm{FP}[N](\Theta_q)$ is infeasible; $\mathrm{P}_L(\Theta_q)$ is infeasible; $s(q) > 0$.*

Proof. Let $L_0 \mid L$ be the least support period. If $L = mL_0$ with $m > 1$, translation by L_0 preserves all tight classes and their pinning vectors. Independence forces $m = 2$ and a single tight class L_0; then every certificate lag is a multiple of L_0 and $P^* = 0$ on the support, a contradiction. Thus $L = L_0$. The continued symbol and the strict lower-shift symbols in the proof of Theorem 3.16 use only N. They give the same threshold for every $N' \supset N$. Complementarity again gives exactly the same extremals, and shift monotonicity together with Theorem 3.7 gives (c). □

Theorem G.8 (Quantitative locking). *Let (q, s_*) be L-regular with data S, N, w^*, λ^*, let $\Theta = \Theta_{q,s_*}$, $S^* = \sum_n \lambda_n^* \Lambda_n^{q,s_*}$, $P^* = \sum_n \lambda_n^*(1-\cos n\omega)$, and $R^* = -S^*/(\sin\Theta\, P^*)$ on $(0,\pi)$.*

(a) *R^* is continuous and positive on $(0,\pi)\setminus S$, vanishes on S, tends to $+\infty$ at 0 and π, and $R^*(\omega+h) = \frac{|\partial_\omega^2 S^*(\omega)|}{2\sin\Theta(\omega)P^*(\omega)}h^2(1+o(1))$ at $\omega \in S$.*

(b) *If w solves $\mathrm{FP}(\Theta_{q,s})$, then w charges $(0,\pi)$ and*

$$s - s_* \;\ge\; \frac{\int_{(0,\pi)} \sin\Theta_{q,s}\, P^* R^*\, dw}{\int_{(0,\pi)} \sin\Theta_{q,s}\, P^*\, dw} \;\ge\; \inf_{\operatorname{supp} w \cap (0,\pi)} R^*.$$

(c) *For every $L' \ge 3$ with $\Omega_{L'} \cap S = \emptyset$, with $d(\Omega_{L'}, S) = \min\{|\omega - \omega'| :\ \omega \in \Omega_{L'},\ \omega' \in S\}$,*

$$\sigma_{L'}(q) - s_* \;\ge\; c_*\, d(\Omega_{L'}, S)^2(1-o(1)) \quad \text{as } d(\Omega_{L'}, S) \to 0,$$

where $c_* = \min_{\omega\in S} |\partial_\omega^2 S^*(\omega)|/(2\sin\Theta(\omega)P^*(\omega))$; and $d(\Omega_{L'}, S) \ge 2\pi/\operatorname{lcm}(L, L')$.

Proof. If P^* vanished in $(0,\pi]$, all lag phases there would be multiples of 2π, hence $S^* = 0$, contradicting either strictness off S or positivity of P^* on S. Taylor's formula at S, and the endpoint expansions of the lag functions, prove (a). Integrate (87) against a feasible finite measure. Its endpoint contributions are nonpositive; it must charge $(0,\pi)$, since a measure carried by the endpoints violates either this symbol or the minimizer constraint. Dividing gives (b). Apply it to a periodic threshold optimizer and use the quadratic minima in (a). Distinct fractions of denominators L, L' have distance at least $1/\operatorname{lcm}(L, L')$, proving (c). □

Theorem G.9 (Localization). *Let q be a tuning, $N \subset \mathbb{Z}\setminus\{0\}$ finite, and $\sigma_N = \sigma_N(q)$.*

(a) *There are $\lambda_n \ge 0$ $(n \in N)$ and $\gamma \ge 0$ with $\sum_n \lambda_n + \gamma = 1$ such that $T = \sum_{n\in N} \lambda_n \Lambda_n^{q,\sigma_N} + \gamma\cos\Theta_{q,\sigma_N} \le 0$ on $[0,\pi]$.*

(b) *Let $s \geq \sigma_N$ and let $w \geq 0$ satisfy the constraints of* $\mathrm{FP}[N](\Theta_{q,s})$*. Then, with $P = \sum_n \lambda_n(1-\cos n\omega)$ and $T_s(\pi) = \sum_n \lambda_n((-1)^n - 1) - \gamma \leq 0$,*

$$\int_{(0,\pi)} \frac{\sin \Theta_{q,s}}{\sin \Theta_{q,\sigma_N}} (-T)\, dw + \gamma\, w(\{0\}) + |T_s(\pi)|\, w(\{\pi\}) \\ \leq\ (s - \sigma_N) \int_{(0,\pi)} \sin \Theta_{q,s}\, (P + \gamma)\, dw. \quad (90)$$

(c) *If, in addition, w has an atom of mass w_a at $\omega_a \in (0,\pi)$, then*

$$-T(\omega_a)\ \leq\ \frac{2(s - \sigma_N)\, w([0,\pi])}{w_a} \cdot \frac{\sin \Theta_{q,\sigma_N}(\omega_a)}{\sin \Theta_{q,s}(\omega_a)}.$$

Proof. Normalized finite-dimensional separating vectors at shifts increasing to σ_N have a convergent subsequence in the simplex. Their limit gives $T \leq 0$. Integrating (87) gives (90), including the two displayed endpoint masses. Keeping one atomic contribution on its left and using $\sin \Theta(P + \gamma) \leq 2$ on the right proves (c). □

Corollary G.10 (Quadratic localization)**.** *In the setting of Theorem G.9, assume that $T(\pi) < 0$, that T has finitely many zeros $z_1, \ldots, z_r$ in $(0,\pi)$, all with $T''(z_i) < 0$, and that $T''(0) < 0$ if $\gamma = 0$. Then there is $c > 0$ such that every w as in Theorem G.9(b) satisfies*

$$\int_{[0,\pi]} \mathrm{dist}\big(\omega, \{0, z_1, \ldots, z_r\}\big)^2\, dw(\omega)\ \leq\ \frac{2(s - \sigma_N)}{c}\, w([0,\pi]).$$

Proof. Nondegenerate zeros and compactness give $-T \geq c' \,\mathrm{dist}(\omega, \{0, z_1, \ldots, z_r\})^2$. On the shift interval $[\sigma_N, \sigma_N + 1]$, the sine ratio in (90) has a positive lower bound, including its endpoint limits of one. The negative value at π controls its endpoint mass. This proves the estimate there; decreasing c so that $c \leq 2/\pi^2$ covers all larger shifts by the trivial distance bound. □

At a fixed shift, separate the frequencies according to which of the two lag constraints they satisfy: write $A_+ = \{\Lambda_1 > 0 > \Lambda_{-K}\}$, $A_- = \{\Lambda_{-K} > 0 > \Lambda_1\}$, both in $(0,\pi]$, and $R_+ = \Lambda_1/(-\Lambda_{-K})$, $R_- = \Lambda_{-K}/(-\Lambda_1)$.

Proposition G.11 (Decoupling)**.** *Assume that no point of $(0,\pi]$ has $\Lambda_1 \geq 0$ and $\Lambda_{-K} \geq 0$ at once, and that $\hat{R}_- := \sup_{A_-} R_-$ is attained at some $u \in A_-$ with $\cos \Theta(u) > 0$. Then $\hat{R}_+ := \sup_{A_+} R_+$ is attained if $A_+ \neq \emptyset$, and $\hat{R}_+ := 0$ otherwise, and:*

(a) $\mathrm{FP}[\{1, -K\}]$ *is feasible if and only if $\hat{R}_+ \hat{R}_- \geq 1$; for $\omega_1 \in A_+$, $\omega_2 \in A_-$ the measure $\delta_{\omega_1} + r\delta_{\omega_2}$ satisfies the two lag constraints exactly for $-\Lambda_{-K}(\omega_1)/\Lambda_{-K}(\omega_2) \leq r \leq \Lambda_1(\omega_1)/(-\Lambda_1(\omega_2))$, an interval that is nonempty if and only if $R_+(\omega_1) R_-(\omega_2) \geq 1$;*

(b) *if* $\mathrm{FP}[\{1, -K\}]$ *is feasible and $T = \lambda_1\Lambda_1 + \lambda_{-K}\Lambda_{-K} + \gamma \cos\Theta \leq 0$ with $\lambda_1, \lambda_{-K}, \gamma \geq 0$ not all zero, then $\gamma = 0$, $\lambda_1, \lambda_{-K} > 0$, $\hat{R}_+ = \lambda_{-K}/\lambda_1$, $\hat{R}_- = \lambda_1/\lambda_{-K}$, and the zeros of T in $(0,\pi]$ are exactly the maximizers of R_+ on A_+ and of R_- on A_-.*

Proof. On the complement of $A_+ \cup A_-$ both lags are nonpositive. Every feasible measure charges both sign sets. Integrating the two ratio bounds gives $\hat{R}_+ \hat{R}_- \geq 1$. Conversely the stated interval for r enforces both lag inequalities; choosing a maximizer of R_- with positive minimizer value also enforces that constraint, since $\cos\Theta > 0$ on A_+. The maximum of R_+ exists because it extends by zero to the boundary of its compact interior sign set. For (b), equality against this two-atom feasible measure forces $\gamma = 0$, both lag weights positive, and the reciprocal ratio bounds. Their equality sets are exactly the zero set away from 0. □

Theorem G.12 (Exact two-lag points)**.** *Let q be a tuning, $K \geq 2$, $s_2 \in \mathbb{R}$, and write $\Lambda_n = \Lambda_n^{q,s_2}$, $\Theta = \Theta_{q,s_2}$. Assume:*

(E1) *there are* $\lambda_1, \lambda_{-K} > 0$, $\lambda_1 + \lambda_{-K} = 1$, *such that* $T = \lambda_1\Lambda_1 + \lambda_{-K}\Lambda_{-K} \le 0$ *on* $[0,\pi]$, $T''(0) < 0$, *and* T *vanishes in* $(0,\pi)$ *exactly at two points* u_1, u_2 *with* $T''(u_1), T''(u_2) < 0$, *where* $\Lambda_1(u_1) > 0 > \Lambda_{-K}(u_1)$ *and* $\Lambda_{-K}(u_2) > 0 > \Lambda_1(u_2)$;

(E2) $u_1, u_2 \in \Omega_L$ *for some* $L \ge K+2$;

(E3) *with* $w_1, w_2 > 0$ *defined up to scaling by* $w_1\Lambda_1(u_1) + w_2\Lambda_1(u_2) = 0$, *one has* $w_1\Lambda_\varrho(u_1) + w_2\Lambda_\varrho(u_2) > 0$ *for every residue* $\varrho \not\equiv 0, 1, -K \pmod L$, *and* $w_1\cos\Theta(u_1) + w_2\cos\Theta(u_2) > 0$;

(E4) $\det\bigl(\sin(\varrho u_i - \Theta(u_i))\bigr)_{i=1,2;\ \varrho=1,-K} \neq 0$.

Then (q, s_2) *is* L*-regular with* $S = \{u_1, u_2\}$ *and* $N = \{1, 1\pm L, -K, -K\pm L\}$. *Consequently* $\sigma(q) = \sigma_L(q) = \sigma_{\{1,-K\}}(q) = s_2$, *the threshold extremal is carried by* $\{u_1, u_2\}$ *and unique up to scaling and an atom at* 0, *and the conclusions of Theorem 3.16 hold near* q.

Proof. The contact matrix has rank one: the nonzero dual vector annihilates it, and the signs in (E1) give a positive right kernel. Condition (E3) gives the remaining strict primal slacks. Replace each original lag $\varrho \in \{1, -K\}$ by the three lags $\varrho, \varrho \pm L$, placing weights $\lambda_\varrho - 2\varepsilon, \varepsilon, \varepsilon$ on them. Their added symbol and first derivative vanish on Ω_L. For sufficiently small $\varepsilon > 0$, C^2 persistence preserves the strict sign and curvatures of (E1); the lag 1 keeps $P > 0$. Condition (E4) supplies pinning rank. Theorem 3.16 applies. The unperturbed two-lag symbol and (87), with a small minimizer term, exclude all lower shifts using only $1, -K$; hence the relaxation threshold is equal as well. □

Corollary G.13 (Exact points on the boundary)**.** *In Theorem G.12 assume moreover* $s_2 = 0$ *and* $u_1/2\pi \in (1/(K+1), 1/K)$. *Then* $q \in \mathcal{B}_K$, q *lies on a plateau, and on this plateau* κ_∂ *is the analytic solution of*

$$\Lambda_1(u_1)\Lambda_{-K}(u_2) - \Lambda_1(u_2)\Lambda_{-K}(u_1) = 0$$

at the shift 0, *an equation in the values of* Θ *at the two fixed frequencies* u_1 *and* u_2. *If also* $u_2/2\pi < 1/(K+1)$, *then* $u_1/2\pi$ *is the unique anchor at every point of this plateau that lies in* I_K.

Proof. The symbol is strictly negative at $2\pi/K$ and $2\pi/(K+1)$, so both points lie outside the dip, while u_1 lies inside it. The dip is an interval, giving $q \in I_K$ and $\sigma(q) = 0$. The determinant equation holds near q; analyticity and connectedness of the plateau extend it throughout its projection. If u_2 lies below the window, only u_1 can be an anchor. □

For $K \ge 2$ and $r > 0$ put

$$W_K(c) = \frac{1 - T_K(c)}{1 - c}, \qquad \Psi_r^{(K)}(c) = \frac{(1+c)(1 - rU_{K-1}(c))}{1 + rW_K(c)}.$$

The polynomial W_K is nonnegative on $[-1, 1]$ and $W_K(1) = K^2$.

Proposition G.14 (Two-lag symbols as double tangencies)**.** *Let* $K \ge 2$, $0 < u_2 < u_1 < \pi$, $c_i = \cos u_i$, *let* q *be a tuning, and let* $T = \lambda_1\Lambda_1 + \lambda_{-K}\Lambda_{-K}$ *be a symbol of* Θ_q *with* $\lambda_1, \lambda_{-K} > 0$ *and* $r = \lambda_{-K}/\lambda_1$.

(a) *On* $(0,\pi)$, $T = \lambda_1 \dfrac{\sin\Theta_q\,(1-c)(1+rW_K(c))}{\sin\omega}\,(\Xi_q(c) - \Psi_r^{(K)}(c))$ *with* $c = \cos\omega$, *and* $T(\pi) < 0$.

(b) $T \le 0$ *on* $[0,\pi]$ *with zero set* $\{0, u_1, u_2\}$, $T''(u_1), T''(u_2) < 0$ *and* $T''(0) < 0$ *hold if and only if* $\Xi_q < \Psi_r^{(K)}$ *on* $[-1,1] \setminus \{c_1, c_2\}$, $\Xi_q(c_i) = \Psi_r^{(K)}(c_i)$ *and* $(\Psi_r^{(K)} - \Xi_q)''(c_i) > 0$ $(i = 1, 2)$, *where the value at* $c = 1$ *is* $\Psi_r^{(K)}(1) = 2(1 - rK)/(1 + rK^2)$.

(c) *In that case* Ξ_q *is the quadratic with values* $\Xi_q(c_i) = \Psi_r^{(K)}(c_i)$ *and slopes* $\Xi_q'(c_i) = \Psi_r^{(K)\prime}(c_i)$, *and* r *is a root of*

$$G_K(r) = \Psi_r^{(K)\prime}(c_1) + \Psi_r^{(K)\prime}(c_2) - 2\,\frac{\Psi_r^{(K)}(c_1) - \Psi_r^{(K)}(c_2)}{c_1 - c_2}.$$

(d) *Let* $\Xi_q(c_i) = \Psi_r^{(K)}(c_i)$ *and* $p_i = \sin\frac{Ku_i}{2}\sin\frac{(K+1)u_i}{2}$. *Then* $\Lambda_1(u_i) > 0 > \Lambda_{-K}(u_i)$ *if and only if* $p_i < 0$, *and* $\Lambda_{-K}(u_i) > 0 > \Lambda_1(u_i)$ *if and only if* $p_i > 0$. *In particular both sign conditions of* (E1) *hold when* $1/(K+1) < u_1/2\pi < 1/K$ *and* $u_2/2\pi < 1/(K+1)$.

Proof. Factor the two-lag symbol using $1 - T_K = (1-c)W_K$ and $\sin K\omega = \sin\omega U_{K-1}(c)$. The factor is positive in the interior. Taylor's formula at the contacts gives the curvature equivalence; at 0, its endpoint condition is $\Xi_q(1) < 2(1-rK)/(1+rK^2)$. A quadratic with the prescribed values and first derivatives exists exactly when the average of its two prescribed slopes equals the secant slope, namely $G_K(r) = 0$. Finally

$$U_{K-1}(\cos u) + W_K(\cos u) = \frac{\sin(Ku/2)\sin((K+1)u/2)}{\sin^2(u/2)\cos(u/2)},$$

which gives the sign assertions and both stated frequency windows. □

G.4. Verification of the mixed-spectrum examples

Proof of Theorem 3.17. We give finite polynomial certificates for the two rows. Set

$$G_n(c) = \Xi_q(c)(1 - T_{|n|}(c)) - \operatorname{sgn}(n)(1-c^2)U_{|n|-1}(c).$$

For the row of period 18 take the lag pairs $(1,19),(-4,14)$; for the row of period 21 take $(1,-20),(-12,9),(-4,-25)$. In each pair denote the first and second entries by r_i, t_i, and put $c_j = \cos(2\pi j/L)$ for the support numerators in Table 2. Form $M(\kappa) = (G_{r_i}(c_j))_{i,j}$. The determinant has opposite strict signs at the two endpoints of $[\hat\kappa - 10^{-7}, \hat\kappa + 10^{-7}]$, where

$$\hat\kappa_{18} = 100.55082118856557, \qquad \hat\kappa_{21} = 106.80551194382778.$$

Both intervals are strictly contained in the displayed open intervals. At any determinant root, the $(0,0)$ cofactor is nonzero. Normalize its adjugate column and row to obtain right and left kernels ρ, μ with sums one. To construct the dual weights, solve

$$\sum_i [G'_{t_i}(c_j) - G'_{r_i}(c_j)]v_i = -\sum_i \mu_i G'_{r_i}(c_j), \qquad \lambda_{t_i} = v_i, \quad \lambda_{r_i} = \mu_i - v_i.$$

Then $\sum_n \lambda_n G_n$ and its first derivative vanish at every contact. The determinants of this system lie in $(-580,-579)$ and $(-46714,-46713)$, respectively. Its columns are nonzero scalar multiples of the pinning columns in (R3), so (R3) follows as well.

Here are rational lower bounds valid throughout the respective brackets; ρ, μ are the polynomial kernels just defined.

L	$\min\rho_j$	$\min\lambda_n$	$\min_{r\neq 0, r_i}\sum_j \rho_j G_r(c_j)$	$\sum_j \rho_j \Xi_q(c_j)$	$\lvert\operatorname{cof}_{00} M\rvert$
18	6/25	$7/10^6$	3/50	6/5	17/50
21	9/100	$5/10^5$	1/25	11/10	1/2

The left kernels are positive too. All these inequalities are strict. Since $\Lambda_n = (\sin\Theta_q/\sin\omega)G_n$, the masses $w_j = \rho_j \sin\omega_j/\sin\Theta_q(\omega_j)$ give the required positive primal measure; its minimizer slack is $\sum_j \rho_j \Xi_q(c_j)$. Thus the displayed bounds verify all inactive primal constraints.

For the global dual sign, factor

$$\sum_n \lambda_n G_n(c) = (1-c)H(c), \qquad H(c) = \sum_n \lambda_n \left[\Xi_q(c)\frac{1 - T_{|n|}(c)}{1-c} - \operatorname{sgn}(n)(1+c)U_{|n|-1}(c)\right].$$

Enclose each c_j by the rational interval $[c_j^-, c_j^+]$ obtained by rounding outward to 14 decimal places and extending one further unit in the last place. On each contact interval $[c_j^- - 1/200, c_j^+ + 1/200]$, the upper Bernstein coefficient bound for H'' is less than $-7/10$ for $L = 18$ and $-7/25$ for $L = 21$. On each intervening closed interval, the upper Bernstein coefficient bound for H is less than $-9\cdot 10^{-6}$

and $-3\cdot10^{-6}$, respectively. The contact values and slopes are exactly zero, so Taylor's theorem proves the required strict sign and negative contact curvatures. The bound at $c=1$ is negative and, as $H(1)=\Xi_q(1)\sum_n\lambda_n n(n+2\tau_q)$, it also gives the endpoint moment condition.

For clarity, the coefficient test used here is the following standard one [5, Section 5.1]. If a polynomial on $[a,b]$ is written as $P(a+(b-a)x)=\sum_{j=0}^{d}p_jx^j$, its Bernstein coefficients are $b_i=\sum_{j\le i}p_j\binom{i}{j}/\binom{d}{j}$. Their minimum and maximum bound P on that interval. Applying this formula to the displayed polynomials, with rational interval operations, gives the stated strict bounds. The trigonometric constants are enclosed by argument reduction and alternating Taylor bounds. The same substitution in Ξ_q gives strict gap endpoint inequalities and a point with $\Theta_q<\omega/2$, hence membership in I_4.

Consequently (R1)–(R3) hold at every bracketed determinant root. Theorem 3.16 gives an analytic plateau, and strict decrease of σ in κ makes the boundary value unique. The fixed supports give the unique anchors. Their least periods are 18 and 21 by Theorem G.7; Corollary G.3 excludes periods $4p+1$ in the first row, and strict positivity at all three support points proves the final assertion for the second row. □

Theorem G.15 (An exact two-lag point of 3/13)**.** *There is a tuning $q^\star=(\kappa^\star,\alpha L^\star,\beta^\star)\in I_4$ with*

$$|\kappa^\star-83.5343|<5\cdot10^{-5},\quad |\alpha L^\star-2.218909|<5\cdot10^{-7},\quad |\beta^\star-0.622768|<5\cdot10^{-7},$$

for which, at shift 0*, $u_1=6\pi/13$ and $u_2=4\pi/13$ satisfy*

$$T(u_2)=T'(u_1)=T'(u_2)=0,\qquad T=-\Lambda_{-4}(u_1)\Lambda_1+\Lambda_1(u_1)\Lambda_{-4}.$$

The Jacobian of these three equations with respect to $(\kappa,\alpha L,\beta)$ is nonsingular at $q^\star$, which is their unique solution in an open neighborhood. Conditions (E1)–(E4) *of Theorem G.12 hold. Consequently*

$$\sigma(q^\star)=\sigma_{13}(q^\star)=\sigma_{\{1,-4\}}(q^\star)=0,$$

and $q^\star$ lies on an analytic plateau with unique anchor 3/13 *and unique threshold spectrum $\{2,3\}\cdot2\pi/13$, up to scaling and an admissible atom at* 0*.*

Proof of Theorem G.15. For $K=4$ write $u=6\pi/13$, $v=4\pi/13$, $c_1=\cos u$, $c_2=\cos v$, and let $Q_r(c)=d+bc+ac^2$ be the quadratic matching the contact values and slopes in Proposition G.14. The scalar compatibility equation has a root

$$0.136418311438750<r<0.136418311438754.$$

Rational interval bounds give $a<0$, $Q_r(\pm1)<0$, and $b+1+2a(c_1+c_2)<0$, so Q_r is the quadratic of a tuning. The global inequality requires only the factorization

$$(1+rW_4(c))Q_r(c)-(1+c)(1-rU_3(c))=(c-c_1)^2(c-c_2)^2\,8r[ac+b+a+1+2a(c_1+c_2)].$$

For $c\ge-1$ the last factor is negative. Thus the only interior zeros are the two contacts, and both curvatures are strictly negative. The endpoint condition, ten inactive residue slacks, minimizer slack, and pinning determinant are strict interval inequalities on the root enclosure. This proves (E1)–(E4) and membership in I_4.

For isolation, let $X_C(c)=C_0+C_1c+C_2c^2$ and put $A_C=1+c_1-X_C(c_1)$, $B_C=X_C(c_1)W_4(c_1)+(1+c_1)U_3(c_1)$. Eliminating $r=A_C/B_C$ from the first contact gives the polynomial profile

$$\mathcal{H}_C(c)=B_C[X_C(c)-1-c]+A_C[X_C(c)W_4(c)+(1+c)U_3(c)].$$

The three-dimensional system is $(\mathcal{H}_C(c_2),\mathcal{H}_C'(c_1),\mathcal{H}_C'(c_2))=0$. A preconditioned interval-Jacobian inclusion gives a unique root and a nonsingular derivative by the Krawczyk criterion [24, Theorem 13.3]. Replacing the polynomial profile by the trigonometric symbol multiplies the contact equations by a triangular matrix with nonzero diagonal; the change from tuning parameters to the coefficients of Ξ_q has nonsingular Jacobian for $\beta>0$. Consequently the original three equations have nonsingular Jacobian and the same open local uniqueness. Inverting the chart gives the three displayed rational parameter enclosures. Theorem G.12 and Corollary G.13 finish the proof. □

H. Single-harmonic certificates and fan asymptotics

Use the notation of Subsections 3.4 and 3.6.

H.1. The edge criterion

Lemma H.1 (Edge identities). *Let Θ be a curve and $\omega = \omega_0 + \delta \in [0,\pi]$. For $n \in b + k\mathbb{Z}$ and $m \in \mathbb{Z} \setminus \{0\}$,*

$$\Lambda_n(\omega) = 2\sin\big(c' + \tfrac{n\delta}{2}\big)\sin\big(c' + \tfrac{n\delta}{2} - \Theta(\omega)\big), \qquad \Lambda_{mk}(\omega) = 2\sin\tfrac{mk\delta}{2}\,\sin\big(\tfrac{mk\delta}{2} - \Theta(\omega)\big),$$

and $n\omega \equiv 2c' + n\delta \pmod{2\pi}$ for $n \in b + k\mathbb{Z}$. If $\Theta(\omega_0) = c'$ and Θ is twice differentiable at ω_0 with $\bar n = 2\Theta'(\omega_0)$, then for $n \in b + k\mathbb{Z}$

$$\Lambda_n'(\omega_0) = \sin c'\,(n - \bar n), \qquad \Lambda_n''(\omega_0) = \cos c'\,n(n - \bar n) - 2\sin c'\,\Theta''(\omega_0),$$

and for $n = mk \neq 0$

$$\Lambda_n'(\omega_0) = -\sin c'\,n, \qquad \Lambda_n''(\omega_0) = \cos c'\,n(n - \bar n).$$

Proof. Reduce $n\omega_0$ modulo 2π in $\Lambda_n = 2\sin(n\omega/2)\sin(n\omega/2 - \Theta)$, then differentiate twice. Only the stated two derivatives of Θ are required. □

Theorem H.2 (Edge criterion at a Farey fraction). *Let q be a tuning with $\bar n(q) \notin b + k\mathbb{Z}$.*

(a) *$F_q(0) = F_q'(0) = F_q(\omega_0) = F_q'(\omega_0) = 0$, $F_q''(0) = \frac{1}{2}\big[(1-\vartheta)n_1(n_1 + 2\tau_q) + \vartheta n_2(n_2 + 2\tau_q)\big]$, and*

$$F_q''(\omega_0) = \sin c'\,\Theta_*''(\omega_0) - \tfrac{1}{2}\cos c'\,k^2\vartheta(1-\vartheta).$$

(b) *If $F_q \geq 0$ on $[0,\pi]$, then $\sigma(q) = \sigma_k(q) = s_*(q)$. If $F_q > 0$ on $(0,\pi] \setminus \{\omega_0\}$, every solution of $\mathrm{FP}(\Theta_*)$ is $b'\delta_{\omega_0} + a'\delta_0$ with $b' > 0$ and $0 \leq a' \leq b'\cos c'$. If moreover $F_q''(0) > 0$ and $F_q''(\omega_0) > 0$, then $(q, s_*(q))$ is k-regular with $S = \{\omega_0\}$ and $N = \{n_1, n_2\}$, and $\sigma = \sigma_k = s_*$ on a neighborhood of q.*

(c) *If $F_q''(\omega_0) \leq 0$, then every symbol $T = \sum_n \lambda_n\Lambda_n \neq 0$ of Θ_* with $\lambda_n \geq 0$, lags in $\mathcal{N}_{j/k}$ and $T'(\omega_0) = 0$ has $T''(\omega_0) \geq 0$. Consequently $(q, s_*(q))$ is not L-regular with support $\{\omega_0\}$, for any L.*

The proof is given together with Lemma H.3 below.

Lemma H.3 (Single-harmonic certificates). *Let q be a tuning, let a finite set $N \subset \mathcal{N}_{j/k}$ contain at least two lags of $b + k\mathbb{Z}$, and let $\lambda_n > 0$, $n \in N$, $\sum_n \lambda_n = 1$. Let $T = \sum_{n\in N}\lambda_n\Lambda_n$ be formed with the lag functions of Θ_*, and assume $T < 0$ on $(0,\pi] \setminus \{\omega_0\}$, $T''(\omega_0) < 0$ and $\sum_n \lambda_n n(n + 2\tau_q) > 0$. Then $(q, s_*(q))$ is k-regular with $S = \{\omega_0\}$ and the lag set N. There is a neighborhood U of q on which $\sigma = \sigma_k = s_*$ and every solution of $\mathrm{FP}(\Theta_{q',\sigma(q')})$, $q' \in U$, is $b'\delta_{\omega_0} + a'\delta_0$ with $b' > 0$ and $0 \leq a' \leq b'\cos c'$.*

Proofs of Lemma H.3 and Theorem H.2. At the edge the only tight nonzero residue is $b + k\mathbb{Z}$ (position 1); its one-by-one primal matrix is zero, and its pinning entry is $\sin c' > 0$. The assumed sign and curvatures give all remaining regularity conditions. Theorem 3.16 then continues the zero $\Lambda_b(\omega_0) = 0$, whose unique shift is s_*. This proves the lemma, including its precise classification of all finite extremals.

The edge identities and $(1-\vartheta)n_1 + \vartheta n_2 = \bar n$ give part (a), with variance $k^2\vartheta(1-\vartheta)$. For (b), $P = (1-\vartheta)(1-\cos n_1\omega) + \vartheta(1-\cos n_2\omega) > 0$ on $(0,\pi]$, since $\gcd(n_1, n_2) = 1$. Equation (87) makes the symbol negative there at every lower shift; a measure carried only by 0 violates the minimizer constraint. At the threshold, complementarity restricts the support to $\{0, \omega_0\}$ and gives the asserted mass bound. The additional curvature conditions allow application of the lemma.

For (c), pair positive and negative slope contributions in any finite nonnegative combination with zero slope. For a position-1 lag the ratio of second derivative to first derivative is

$$\rho_n = n\cot c' - \frac{2\Theta_*''(\omega_0)}{n - \bar n};$$

for a nonzero position-0 lag it is $(\bar n - n)\cot c'$. Here $\Theta_*''(\omega_0) > 0$ by Theorem E.4. Thus the first ratio is strictly increasing on either side of $\bar n$: among positive-slope position-1 lags its minimum is

ρ_{n_2}, and among negative-slope ones its maximum is ρ_{n_1}. Position-0 lags with positive slope have $n \le -k$ and ratio at least $(\bar{n}+k)\cot c' > \rho_{n_2}$; those with negative slope have $n \ge k$ and ratio at most $(\bar{n}-k)\cot c' < \rho_{n_1}$. Therefore the difference between the least positive-slope ratio and the greatest negative-slope ratio is

$$\rho_{n_2} - \rho_{n_1} = k\cot c' - \frac{2\Theta''_*(\omega_0)}{k\vartheta(1-\vartheta)},$$

which is nonnegative exactly when $F''_q(\omega_0) \le 0$. A zero-slope nonnegative combination is a sum of two-lag combinations with opposite slopes. Each has second derivative equal to the product of its positive slope magnitudes and this nonnegative ratio difference. Its curvature is therefore nonnegative. A regular singleton certificate would have all its lags tight at ω_0, hence in $\mathcal{N}_{j/k}$, and would contradict this conclusion. □

Corollary H.4 (Single-harmonic plateaus)**.** *Let Π be a plateau whose threshold extremal is carried by $\{\omega_0\}$. Then $\Theta_q(\omega_0) = c'$ at every $q \in \Pi$, and $\kappa_\partial = \kappa_{j/k}$ on the projection of Π to the $(\alpha L, \beta)$ plane. If $j/k \in (1/(K+1), 1/K)$, then j/k is the unique anchor at every point of $\Pi \cap I_K$.*

Proof. A singleton regular primal has exactly one tight nonzero class. Its position cannot be 0, since that would give $P^*(\omega_0) = 0$. The one-atom feasibility condition and its tight position force phase index 1. Hence $\Theta_q(\omega_0) = c'$, and the tongue formula gives $\kappa_\partial = \kappa_{j/k}$. The dip condition gives the anchor. □

H.2. Asymptotics of two-frequency periodic measures

Lemma H.5. *Let $k \ge 3$ and $c' = \pi/k$. The trigonometric polynomial*

$$\Upsilon_k(\theta) = k\cos c' - (k-1)\cos(\theta - c') - \cos\big((k-1)\theta + c'\big)$$

is nonnegative on $\mathbb{R}$, vanishes exactly at $\theta \equiv 0$ and $\theta \equiv 2c' \pmod{2\pi}$, and $\Upsilon''_k(0) = \Upsilon''_k(2c') = k(k-1)\cos c' > 0$.

Proof. Put $m = k-1$ and $\varphi = \theta - c'$. Maximizing $m\cos\varphi - \cos m\varphi$ gives the stationary factors $2m\cos((m+1)\varphi/2)\sin((m-1)\varphi/2)$. The first yields maximum $(m+1)\cos c'$ exactly at $\varphi \equiv \pm c'$; the second yields at most $m-1$, which is smaller because $(m+1)\cos(\pi/(m+1)) > m-1$. Twice differentiating gives the stated curvatures. □

Proof of Theorem 3.23. Write $s_c = \sin c'$, $c_c = \cos c'$, $\mathsf{u} = k(k-1)c_c/4$, $\mathsf{v} = s_c/2$. At $s = s_* - e\delta^2$ set $\Theta_a = c' + A$, $\Theta_c = c' + C$. Joint analyticity gives

$$\begin{aligned} A &= \tfrac{1}{2}\bar{n}\delta + (\tfrac{1}{2}\Theta''_* + es_c^2)\delta^2 + O(|\delta|^3), \\ C &= -\tfrac{1}{2}(k-1)\bar{n}\delta + (\tfrac{1}{2}(k-1)^2\Theta''_* + es_c^2)\delta^2 + O(|\delta|^3). \end{aligned} \tag{91}$$

For masses $w_a = k - 1 + z\delta$, $w_c = 1$, every integer lag constraint is $\Phi(\theta) \ge 0$ on $\theta \in (2\pi/L)\mathbb{Z}$, where

$$\Phi(\theta) = w_a[\cos\Theta_a - \cos(\theta - \Theta_a)] + \cos\Theta_c - \cos((k-1)\theta + \Theta_c).$$

Indeed $\gcd(J, L) = 1$ and $(k-1)\omega_a + \omega_c = 2\pi j$. The two exact local identities are

$$\tfrac{1}{2}\Phi(u) = w_a \sin\tfrac{u}{2}\sin(\tfrac{u}{2} - \Theta_a) + \sin\tfrac{(k-1)u}{2}\sin(\tfrac{(k-1)u}{2} + \Theta_c), \tag{92}$$

$$\tfrac{1}{2}\Phi(2c' + v) = w_a \sin(c' + \tfrac{v}{2})\sin(\tfrac{v}{2} - A) - \sin(c' - \tfrac{(k-1)v}{2})\sin(\tfrac{(k-1)v}{2} + C). \tag{93}$$

Near 0 the grid is $u \in k|\delta|\mathbb{Z}$, and near $2c'$ it is $v = y\delta$, $y \in b + k\mathbb{Z}$. Taylor expansion gives $\Phi(u)/2 = f_1 u + f_2 u^2 + O(|u|^3)$ with $f_1 = -\delta(\mathsf{u}\bar{n} + \mathsf{v}z) + O(\delta^2)$, $f_2 = \mathsf{u} + O(|\delta|)$, and

$$\frac{\Phi(2c' + y\delta)}{2\delta^2} = Q(y) + O\big(|\delta|(1+|y|^3)\big), \qquad Q(y) = \mathsf{u}(y^2 - \bar{n}y) + \mathsf{v}z(y - \bar{n}) - \mathsf{v}E, \quad E = k(k-1)\Theta''_* + 2ks_c^2 e. \tag{94}$$

The two nearest lattice values balance at $\mathsf{v}z_* = -\mathsf{u}[\bar{n} + k(1-2\vartheta)]$. Convexity gives

$$\min_{y \in b+k\mathbb{Z}} Q(y)\big|_{z=z_*} = \max_z \min\{Q(n_1), Q(n_2)\} = \mathsf{u}k^2\vartheta(1-\vartheta) - \mathsf{v}E = k\sin^3 c'(e_* - e). \tag{95}$$

For $e < e_*$ this is positive. Away from fixed neighborhoods of $0, 2c'$, Φ converges uniformly to $\Upsilon_k > 0$. For the grid points $u \in k|\delta|\mathbb{Z}$ near 0, put $\nu = 1 - |1-2\vartheta| > 0$. Since $|\mathsf{u}\bar{n} + \mathsf{v}z_*| = (1-\nu)\mathsf{u}k$ and $|u| \geq k|\delta|$ for $u \neq 0$, the Taylor expansion gives

$$\tfrac{1}{2}\Phi(u) \geq \nu\mathsf{u}u^2 - O(|\delta|u^2) - O(|u|^3) > 0$$

after one fixed neighborhood and then $|\delta|$ are made sufficiently small. The point $u = 0$ is an exact zero. Near $2c'$, first choose Y_0 so that $Q(y) \geq \mathsf{u}y^2/4$ for $|y| > Y_0$. For $|y| \leq Y_0$, the fixed positive minimum in (95) absorbs the remainder. For $|y| > Y_0$ and $v = y\delta$ in a sufficiently small fixed neighborhood, the unscaled remainder is $O(|v|^3 + |\delta|^3) = O(|v|^3)$ and is smaller than $\mathsf{u}v^2/8$. Thus every grid point is controlled, including lags increasing with L. The minimizer slack tends to $k\cos c' > 0$.

Conversely a nonzero feasible measure on the pair has both masses positive for large L: each singleton has reduced denominator at least $L/(k-1)$, incompatible with its limiting angle $c' > 0$. Normalize $w_c = 1$. Let $u = 2\pi/L$. The inequalities $\Phi(u), \Phi(-u) \geq 0$ imply

$$|w_a \sin u \sin\Theta_a - \sin((k-1)u)\sin\Theta_c| \leq 2w_a \sin^2(u/2)\cos\Theta_a + 2\sin^2((k-1)u/2)\cos\Theta_c.$$

Since both angles tend to $c' \in (0, \pi/2)$, division by u first bounds w_a uniformly and then gives $|w_a \sin\Theta_a - (k-1)\sin\Theta_c| = O(u)$. Consequently $w_a = k - 1 + z\delta$ with z bounded independently of L. For $e > e_*$, (95) then forces one of the two fixed lattice constraints n_1, n_2 to be negative for all sufficiently large L. Shift monotonicity now yields the attained pair threshold and its δ^2 expansion. If $F_q'' < 0$, choose $0 < e < e_*$ to obtain $\sigma < s_*$. Under the strict regular conditions, the potential $2F_q/(\sin\Theta_* P)$ has coefficient $F_q''/(2\sin^3 c')$ at ω_0; every missing grid point is at distance at least $2\pi/(kL)$, giving the lower periodic bound. The fan pair itself gives the upper bound.

On any compact tuning set where $n_1 < \bar{n} < n_1 + k$ for fixed n_1, ϑ stays away from $0, 1$, all Taylor bounds above are uniform, and the margin for $e = e_* \mp \eta_0$ is the fixed number $k\sin^3 c'\eta_0$. The same neighborhood sizes, mass bounds and L_0 therefore work for both orientations and all tunings in the compact set. In particular the $o(\delta^2)$ in the pair threshold is uniform on such compact sets. □

H.3. The exact plateau domain

Put $C_0 = h_0 - 1 - c_0 > 0$.

Lemma H.6 (The edge in the gap)**.** *The tuning $q(a, D)$ is defined and lies in I_K if and only if $a > a_E(D)$, where $a_E(D)$ is the largest of*

$$\frac{h_0 + (1+D)(1-c_0)}{(1-c_0)^2}, \quad \frac{h_0 - (1+D)(1+c_0)}{(1+c_0)^2},$$
$$\frac{C_0 + D(c_K - c_0)}{(c_K - c_0)^2}, \quad \frac{C_0 + D(c_{K+1} - c_0)}{(c_{K+1} - c_0)^2}.$$

The function a_E is convex, piecewise affine and positive. The second term never exceeds the larger of the last two.

Proof. The first two affine inequalities are $\Xi^{a,D}(\pm 1) < 0$. The last two say that the convex polynomial $1 + c - \Xi^{a,D}(c) = a(c - c_0)^2 - D(c - c_0) - C_0$ is positive at both gap endpoints, hence has its roots inside them. Their maximum is positive, convex and piecewise affine. The negative-endpoint condition follows already from the lower gap root lying above -1, proving redundancy of the second term. □

Definition H.7 (Admissible symbols)**.** Let $D \in \mathbb{R}$. An admissible symbol for D is a finitely supported $\lambda = (\lambda_n)_{n \in \mathcal{N}_{j/k}} \geq 0$ with $\sum_n \lambda_n = 1$, with positive weight on at least two lags of $b + k\mathbb{Z}$, whose support has greatest common divisor 1, and with

$$\sum_{n \in b+k\mathbb{Z}} \lambda_n (n - \bar{n}(D)) = \sum_{n \in k\mathbb{Z}} \lambda_n n.$$

For such λ and $c = \cos\omega$ put

$$\Psi_\lambda(c) = \frac{\sin\omega \sum_n \lambda_n \sin n\omega}{\sum_n \lambda_n (1 - \cos n\omega)} \quad (0 < \omega \leq \pi), \qquad \Psi_\lambda(1) = \frac{2\sum_n \lambda_n n}{\sum_n \lambda_n n^2},$$

$R_\lambda(c) = \big(h_0 + (1+D)(c - c_0) - \Psi_\lambda(c)\big)/(c - c_0)^2$ for $c \neq c_0$, and $R_\lambda(c_0) = -\frac{1}{2}\Psi''_\lambda(c_0)$. Finally

$$a_F(D) = \inf_\lambda \max_{c \in [-1,1]} R_\lambda(c).$$

When $\bar{n}(D) \notin b + k\mathbb{Z}$, the weights $1 - \vartheta$, ϑ on the lags n_1, n_2 of Definition 3.18 form the edge symbol λ^{e}, which is admissible because $\gcd(n_1, n_1 + k) = \gcd(b, k) = 1$, and we put $a_2(D) = R_{\lambda^{\mathrm{e}}}(c_0)$.

Lemma H.8 (The certificate criterion along the edge)**.** *Let λ be admissible for D. Then R_λ is continuous on $[-1,1]$, and $q(a,D)$ is defined for all large a. Let $q(a,D)$ be defined and let $T_\lambda = \sum_n \lambda_n \Lambda_n$ be formed with the lag functions of $\Theta_{q(a,D)}$. Then $T_\lambda < 0$ on $(0,\pi] \setminus \{\omega_0\}$, $T''_\lambda(\omega_0) < 0$ and $\sum_n \lambda_n n(n + 2\tau_q) > 0$ if and only if $a > \max_{[-1,1]} R_\lambda$; in that case $q(a,D)$ is k-regular with $S = \{\omega_0\}$ and the lag set* $\operatorname{supp}\lambda$*. If $\bar{n}(D) \notin b + k\mathbb{Z}$, then $R_\lambda(c_0) \geq a_2(D)$, with equality only for $\lambda = \lambda^{\mathrm{e}}$, and*

$$F''_{q(a,D)}(\omega_0) = 2 s_0 \sin^3 c' \,(a - a_2(D)),$$

$$a_2(D) = \frac{\cot c' \,(1 + c_0^2)}{2 s_0^3} + \frac{(1+D)\, c_0}{2 s_0^2} - \frac{\sin c' \cos c' \,\big(1 + D + \cot c' \cot\omega_0\big)^2}{s_0} + \frac{\cos c' \, k^2 \vartheta(1-\vartheta)}{4 s_0 \sin^3 c'}.$$

Proof. Write $P_\lambda = \sum \lambda_n (1 - \cos n\omega)$, $Q_\lambda = \sum \lambda_n \sin n\omega$ and $\mu = \sum_{n \in b+k\mathbb{Z}} \lambda_n > 0$. The gcd condition gives $P_\lambda > 0$ on $(0,\pi]$. The Chebyshev form and the endpoint limits make Ψ_λ analytic through $[-1,1]$. At c_0 its value is h_0; the slope constraint gives derivative $1 + D$. Thus R_λ extends continuously. The master identity is

$$T_\lambda = \frac{\sin\Theta_q P_\lambda}{\sin\omega}(c - c_0)^2 (R_\lambda(c) - a), \qquad c \neq c_0, \tag{96}$$

and at the contact it gives

$$T''_\lambda(\omega_0) = 4 \sin^3 c' \, s_0 \mu (R_\lambda(c_0) - a). \tag{97}$$

At 0, the moment condition is exactly $\Xi^{a,D}(1) < \Psi_\lambda(1)$; at π, both sides reduce to the strict tuning endpoint condition. This proves the criterion and regularity.

Differentiating $\cot\Theta = \Xi/\sin\omega$ gives (32); inserting its second derivative in the edge curvature gives the displayed a_2. To prove sharpness, admissibility and the edge identities give

$$4 \sin^3 c' \, s_0 \mu \,[R_\lambda(c_0) - a_2(D)] = \cos c' \Big[\sum_{n \in b+k\mathbb{Z}} \lambda_n (n - n_1)(n - n_2) + \sum_{m \neq 0} \lambda_{mk} k^2 m (m + 1 - 2\vartheta) \Big].$$

Every summand is nonnegative and the second group is strictly positive on its support. Equality requires precisely the adjacent-lag weights. □

Theorem H.9 (Single-harmonic plateaus at large curvature)**.** *Let* $D_{\max} \geq 0$. *Put*

$$\bar{N} = 2\sin^2 c'\big(1 + D_{\max} + \cot c' \,|\cot \omega_0|\big), \quad N_1 = \bar{N} + \tfrac{3}{2}k,$$
$$s_h = \sin\frac{\pi}{2k}, \quad H_A = 1 + \frac{(h_0+1)^2}{s_h^2},$$

and let $a_1 = a_1(D_{\max})$ *be the largest of*

$$\frac{(h_0 + 2 + 2D_{\max})\,k^2 N_1^2}{2s_h^2}, \quad \frac{12\,k^2 N_1^4}{s_h^2}, \quad \frac{(1+D_{\max})^2}{4}, \quad \frac{\pi H_A\,(9k^2 + 3\pi k)}{16\,s_h^2}.$$

If $|D| \leq D_{\max}$, $a > a_1$ *and* $a > a_E(D)$, *then* $q(a, D) \in \mathcal{B}_K$, *and the symbol with weights* $(\frac{1}{3} - t, \frac{1}{3}, \frac{1}{3} + t)$ *on the lags* $n_0 - k, n_0, n_0 + k$ *satisfies the conditions of Lemma H.8; here* $n_0 \in b + k\mathbb{Z}$ *is a member nearest to* $\bar{n}(D)$ *and* $t = (\bar{n}(D) - n_0)/(2k)$. *In particular, applied with* $D_{\max} = |D|$, *this gives* $a_F(D) \leq \max\{a_1(|D|), a_E(D)\} < \infty$ *for every* D.

Proof. For the three stated lags, $|n_i| \leq N_1$, each weight lies in $[1/12, 7/12]$, $\sum w_i n_i = \bar{n}$, and their gcd is one. Let $u_i = (n_i - \bar{n})/2$, $V = \sum w_i u_i^2$, $U_3 = \sum w_i |u_i|^3$. Then $k^2/24 \leq V \leq 9k^2/16$, $U_3 \leq 27k^3/64$, and $\sum w_i n_i^2 \geq k^2/6$. Put $\delta_B = \pi/(kN_1)$, $J_B = [\omega_0 - \delta_B, \omega_0 + \delta_B]$, and let ε_C be the smaller cosine distance from its endpoints to c_0. Since $\sin\omega \geq s_h$ on J_B, $\varepsilon_C \geq 2s_h/(kN_1)$, and the definition of a_1 gives

$$a\varepsilon_C^2 > 2(h_0 + 2 + 2D_{\max}), \quad a\varepsilon_C^2 > 48N_1^2, \quad a\varepsilon_C^2 > 6\pi^2 N_1/k^2, \quad a\varepsilon_C^2 > 24N_1/k^2. \tag{98}$$

Outside J_B, $\Xi \leq -a\varepsilon_C^2/2$. Write $\Theta = \pi - \varphi$; then $\tan\varphi \leq 2\sin\omega/(a\varepsilon_C^2)$ and $T \leq -\cos\varphi(P - \tan\varphi|Q|)$. For $\omega \leq \pi/N_1$, use $P \geq k^2\omega^2/(3\pi^2)$, $|Q| \leq N_1\omega$. For $\omega \geq \pi/N_1$, the consecutive-lag gcd gives $P \geq 1/(24N_1^2)$. Indeed, either the distance of $k\omega$ from $2\pi\mathbb{Z}$ is at least π/N_1, forcing an adjacent phase distance of at least $\pi/(2N_1)$, or ω is within $\pi/(kN_1)$ of a nonzero kth root; coprimality then gives a phase distance of at least π/k. In both cases use $1 - \cos x \geq 2x^2/\pi^2$ for $|x| \leq \pi$ and the weight lower bound. Also $|Q| \leq 1$. Equation (98) makes both bounds strict and also gives the positive endpoint moment at 0.

On $J_A = \{\Theta \leq \pi/2\} \subset J_B$, strict convexity gives $\Theta'' \geq as_h/H_A$. Set $\delta = \omega - \omega_0$, $\ell = c' + \bar{n}\delta/2$, $g = \Theta - \ell$. Then $g \geq as_h\delta^2/(2H_A)$, and Taylor's sine bound with $\sum w_i u_i = 0$ yields

$$T \leq \delta^2\left[-\frac{2as_h^2}{\pi H_A} + \frac{9}{8}k^2 + \frac{3\pi}{8}k\right] < 0 \quad (\delta \neq 0).$$

It also gives $T''(\omega_0) < 0$. On $J_B \setminus J_A$, every $e_i = c' + n_i\delta/2$ lies in $(0, \pi/2)$ while $\Theta > \pi/2$, so $\Lambda_{n_i} = 2\sin e_i \sin(e_i - \Theta) < 0$. The three zones prove the conditions of Lemma H.8, including resonant slope indices. Taking a down to the stated sufficient bound gives the majorant for a_F. □

Proof of Theorem 3.19. The certificate criterion shows every point of $\Pi_{j/k}$ is regular. Conversely a singleton regular extremal has phase index 1; its tight lag set lies in $\mathcal{N}_{j/k}$ and satisfies the slope condition. A common divisor larger than one would force a second symbol zero on $(0, \pi]$, contradicting its strict negativity away from the prescribed contact. Thus its symbol is admissible and $a > a_F(D)$. Its dip cannot contain an endpoint frequency $2\pi/K'$: that would give a different single-harmonic extremal, contradicting uniqueness. Hence it lies in I_K and $a > a_E(D)$. The continuous upper bound of Theorem H.9 joins any two vertical rays of the parameter domain, proving connectedness without requiring continuity of a_F. Local constancy of support makes it exactly one component of the regular boundary. The lower curvature bound and its strict subcritical alternative follow from Lemma H.8 and Theorem 3.23(d). □

Proof of Corollary 3.20. The large-curvature certificate makes every $\Pi_{j/k}$ nonempty. Every gap contains $2/(2K+1)$. □

Proof of Proposition 3.21. Put $\epsilon = 1-\beta$ and $t = a\epsilon^2$. Matching the quadratic coefficients in the chart gives $B = 2\beta/(a\epsilon)$, $A = 2\beta(1+D+2ac_0)/(a(1+\beta))$, and

$$\frac{4\beta^2}{t} - t\left(1 - \frac{4\beta c_0^2}{(1+\beta)^2}\right) - 4\beta h_0 + \frac{4\beta(1+D)c_0\epsilon^2}{(1+\beta)^2} - \frac{4\beta^2(1+D)^2}{a(1+\beta)^2} = 0.$$

The coefficient of t lies in $[s_0^2, 1]$, so this identity bounds both t and $1/t$ along the ray. Hence $\beta \to 1$, and every limit point solves $4/t - s_0^2 t - 4h_0 = 0$. Its unique positive root is $2\tan(c'/2)/s_0$. Now $B \to 0$, $A \to 2c_0$, giving the other two parameter limits and the stated limiting characteristic roots. □

I. Tips, gap crossings, and junctions

I.1. Tongue tips and the lag-one boundary

For the fraction $1/K'$, put $\chi = \Theta_*'(\omega_*) - 1/2 = (\bar{n}-1)/2$. Strict convexity of $\Theta_* - \omega/2$ on its nonpositive sublevel shows that its dip opens to the right of ω_* if $\chi < 0$ and to the left if $\chi > 0$; at $\chi = 0$ it is empty. For $0 < |\chi| < K'/2$ the adjacent lags are 1 and $1 + \operatorname{sgn}(\chi)K'$, with respective weights $1 - 2|\chi|/K'$ and $2|\chi|/K'$. Thus

$$F_q''(\omega_*) = \sin c'\, \Theta_*''(\omega_*) - (K' - 2|\chi|)|\chi| \cos c'. \tag{99}$$

The fan pairs in rotation numbers are $\{(m+1)/(K'm+K'-1), m/(K'm+K'-1)\}$ for $\varsigma = -1$, and $\{m/(K'm+1), (m+1)/(K'm+1)\}$ for $\varsigma = 1$, listing the nearer point first. For large m the upper point lies in the window of $I_{K'-1}$ and the lower in that of $I_{K'}$. At the feasible shifts of Theorem 3.23(a), the anchor is the point on the side where the dip opens.

Proof of Theorem 3.24. At the tip, $\bar{n} = 1$ and $\Lambda_1 = -2\sin(\omega/2)\sin(\Theta - \omega/2)$ is strictly negative away from $0, \omega_*$. Its second derivatives at those zeros are $-(1+2\tau_q) < 0$ and $-2\sin c'\, \Theta''(\omega_*) < 0$. The perturbation

$$T = (1-2\varepsilon)\Lambda_1 + \varepsilon\Lambda_{1+K'} + \varepsilon\Lambda_{1-K'}$$

has the same zero values and slopes. For sufficiently small $\varepsilon > 0$, strict curvature in fixed neighborhoods and compact negativity elsewhere persist. Lemma H.3 gives one open neighborhood with every claimed threshold, regularity, and extremal conclusion. On it $\sigma \le 0$ is exactly $s_* \le 0$, the closed tongue; both adjacent gaps have $s_* > 0$. □

Corollary I.1 (A neighborhood of $\partial\Omega_1$)**.** *Every point q_0 of $\partial\Omega_1$ has a neighborhood V with $\mathcal{D} \cap V = \emptyset$ if q_0 is not phase-locked, and $\mathcal{D} \cap V = \{q \in V : \Theta_q(2\pi/K') \le \pi/K'\}$ if q_0 is a tip of the K'-tongue. In particular $\operatorname{cl}\mathcal{B}_K \cap \partial\Omega_1 = \emptyset$ for every K: near $\partial\Omega_1$ the phase diagram is that of level zero.*

Proof. At a nonlocked tangency, Theorem 3.8 gives a strict finite symbol, whose sign persists uniformly in the tuning. The locked points are the tips and have the neighborhoods above. A gap contains neither the locked endpoint in its closed dip nor any point of the local closed tongue; its boundary consequently cannot accumulate on $\partial\Omega_1$. □

I.2. Junctions

For $K \ge 3$ put $t_1 = c_{K+1}$, $t_2 = c_K$ and let $q_\times(a)$ have quadratic $\Xi(c) = 1 + c - a(c-t_1)(c-t_2)$, where $a > a_0(t_2, t_1)$. Its closed dip is $[2\pi/(K+1), 2\pi/K]$, so it lies on both tongue edges and is approached from I_K.

Proposition I.2 (Junctions)**.** *Let $K \ge 3$ and $K' = K+1$. At $q_\times(a)$, with respect to $\omega_* = 2\pi/K'$, $\chi = -a(t_1 - t_2)\sin^2 c'$, and for $|\chi| < K'/2$*

$$F_q''(\omega_*) = |\chi| \cos c' \Big[4|\chi| - (K+3) + \frac{2(1-t_1)}{t_1 - t_2} + \frac{t_1}{1+t_1}\Big].$$

The interval

$$\mathcal{J}_K = \Big\{ a : \ \frac{2}{(1-t_1)(1-t_2)} < a < \frac{K+3-2(1-t_1)/(t_1-t_2)-t_1/(1+t_1)}{4(t_1-t_2)\sin^2 c'} \Big\}$$

is nonempty and contained in $\{|\chi| < K'/2\}$*, and for* $a > a_0(t_2,t_1)$ *with* $\chi \notin (K'/2)\mathbb{Z}$ *one has* $F_q''(\omega_*) < 0$ *if and only if* $a \in \mathcal{J}_K$*. Consequently, for* $a \in \mathcal{J}_K$*,* $\sigma(q_\times(a)) < 0$*, and all tunings of* I_K *near* $q_\times(a)$ *lie in the interior of* $\mathcal{D}$*.*

The proof follows Lemmas I.3 and I.4.

Lemma I.3 (The curve at a junction)**.** *Let* $m \geq 4$*,* $\omega_* = 2\pi/m$*,* $c' = \pi/m$*,* $t_1 = \cos\omega_*$*,* $t_2 < t_1$*,* $a > 0$*, and let* $\Theta = \operatorname{arccot}(\Xi(\cos\omega)/\sin\omega)$ *on* $(0,\pi)$*, where* $\Xi(c) = 1 + c - a(c-t_1)(c-t_2)$*. Then* $\Theta(\omega_*) = c'$*,* $\Theta'(\omega_*) - \frac{1}{2} = -a(t_1-t_2)\sin^2 c'$*, and*

$$\Theta''(\omega_*) = -\tfrac{1}{2}\cot c' + \tfrac{a}{2}\tan c'\,(2 - t_1^2 - t_1 t_2) + 2\Theta'(\omega_*)^2 \cot c'.$$

Proof. Differentiate $\cot\Theta = \cot(\omega/2) - a(\cos\omega - t_1)(\cos\omega - t_2)/\sin\omega$ at ω_*. The perturbation has value zero, first derivative $a(t_1 - t_2)$ and second derivative $a(t_1^2 + t_1 t_2 - 2)/\sin\omega_*$. The identities $\Theta' = -\mathsf{f}'\sin^2\Theta$ and $\Theta'' = -\mathsf{f}''\sin^2\Theta + 2\Theta'^2\cot\Theta$ for $\mathsf{f} = \cot\Theta$ give both formulas. □

Lemma I.4. *For* $m \geq 4$ *put* $u_m = 1 - \sin^2(\pi/m)/\sin^2(\pi/(m-1))$ *and* $t = \cos(2\pi/m)$*. Then* $4u_m + 2/u_m + t/(1+t) < m + 4$*.*

Proof. For $m = 4, 5$ the left sides are $22/3$ and $4\sqrt{5} - 1$. For $m \geq 6$, the alternating sine bounds through degrees five and three, together with monotonicity of $\sin x/x$, give

$$\frac{2m^2 - m - 10.57}{m^3} \leq u_m \leq \frac{2}{m}.$$

Therefore the required left side minus m is bounded above by

$$\frac{8}{m} + \frac{m^2 + 10.57m}{2m^2 - m - 10.57} + \frac{1}{2}.$$

This expression decreases for $m \geq 6$ and is less than 3.63 at $m = 6$, proving the strict inequality. □

Proof of Proposition I.2. The parameterization of Proposition E.7 gives a tuning with the stated closed dip, and $(a, t_2 + \epsilon, t_1 - \epsilon)$ approaches it from the gap. Lemma I.3, substituted in (99), gives the displayed curvature. The lower endpoint of $\mathcal{J}_K$ is exactly $a_0(t_2,t_1)$. Its being smaller than the upper endpoint is Lemma I.4, after using $1 - t_1 = 2\sin^2 c'$ and $u_m = 1 - \sin^2 c'/\sin^2(\pi/(m-1))$, $m = K+1$. On this interval $0 < |\chi| < (m+2)/4 \leq m/2$. Beyond the first slope strip, put $g = |\chi| > m/2$; the adjacent-lag curvature formula and $\vartheta(1-\vartheta) \leq 1/4$ give

$$F_q''(\omega_*) \geq \cos c'[2g(g-1) - m^2/8] > 0.$$

Thus, at every nonresonant slope, negative curvature holds precisely on $\mathcal{J}_K$. Theorem 3.23(d) gives $\sigma < 0$ there; continuity gives an open divergent neighborhood reaching into I_K. □